Mathematics

Class 9 (English Medium)

Solved Exercises — Sindh Textbook Board

16 lessons — 1063 questions

1. Real and Complex Numbers

Question: Exercise 1.1, Question 1: Identify the rational and irrational numbers among the following and write each in a separate column. (i) 1/5

Answer: It is a rational number, because it can be written in p/q form (1/5 is already in that form).

Question: Exercise 1.1, Question 1: (ii) √2/8

Answer: It is an irrational number, because √2 is irrational and dividing an irrational number by 8 still leaves it irrational.

Question: Exercise 1.1, Question 1: (iii) 5/√6

Answer: It is an irrational number, because its denominator contains √6, which is irrational.

Question: Exercise 1.1, Question 1: (iv) 2/8

Answer: It is a rational number, because 2/8 = 1/4, which is in p/q form.

Question: Exercise 1.1, Question 1: (v) 1/√3

Answer: It is an irrational number, because its denominator contains √3, which is irrational.

Question: Exercise 1.1, Question 1: (vi) √8

Answer: It is an irrational number, because √8 = √(4×2) = 2√2, and √2 is irrational.

Question: Exercise 1.1, Question 1: (vii) 0

Answer: It is a rational number, because 0 = 0/1, which is in p/q form.

Question: Exercise 1.1, Question 1: (viii) π

Answer: It is an irrational number, because the decimal form of π never terminates and never repeats: π = 3.14159...

Question: Exercise 1.1, Question 1: (ix) √5

Answer: It is an irrational number, because 5 is not a perfect square, so the decimal form of √5 is non-terminating and non-repeating.

Question: Exercise 1.1, Question 1: (x) 22/3

Answer: It is a rational number, because it is a ratio of two integers (p/q form).

Question: Exercise 1.1, Question 1: (xi) 1/π

Answer: It is an irrational number, because π is irrational and dividing 1 by an irrational number still gives an irrational number.

Question: Exercise 1.1, Question 1: (xii) 11/12

Answer: It is a rational number, because it is a ratio of two integers (p/q form).

Question: Exercise 1.1, Question 2: Convert the following fractions to decimal form and identify whether they are terminating or non-terminating. (i) 5/8

Answer: 5/8 = 0.625. The decimal form is terminating (it ends after three decimal places).

Question: Exercise 1.1, Question 2: (ii) 4/18

Answer: 4/18 =

= 2/9

= 0.222...

= $0.\bar{2}$. The decimal form is non-terminating and repeating (2 keeps repeating).

Question: Exercise 1.1, Question 2: (iii) 1/15

Answer: 1/15 =

= 0.0666...

= $0.0\bar{6}$. The decimal form is non-terminating and repeating (6 keeps repeating).

Question: Exercise 1.1, Question 2: (iv) 49/8

Answer: 49/8 = 6.125. The decimal form is terminating (it ends after three decimal places).

Question: Exercise 1.1, Question 2: (v) 207/15

Answer: 207/15 =

= 69/5

= 13.8. The decimal form is terminating (it ends after one decimal place).

Question: Exercise 1.1, Question 2: (vi) 50/76

Answer: 50/76 = 25/38 = 0.6578947368421052 63... The decimal form is non-terminating and repeating, because the denominator 38 = 2×19 and 19 is neither 2 nor 5.

Question: Exercise 1.1, Question 3: Represent the following rational numbers on the number line. (i) 8/10

Answer: 8/10 = 0.8. On the number line, divide the unit between 0 and 1 into 10 equal parts; the 8th mark from 0 is 0.8.



Question: Exercise 1.1, Question 3: (ii) -8/10

Answer: -8/10 = -0.8. On the number line, divide the unit between 0 and -1 into 10 equal parts; the 8th mark to the left of 0 is -0.8.

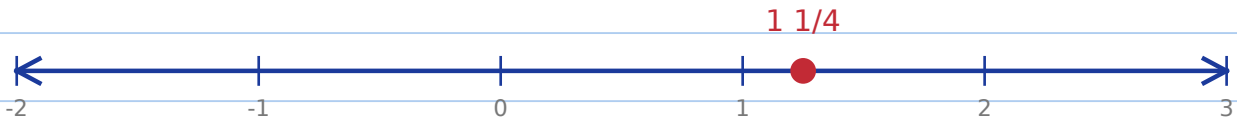


Question: Exercise 1.1, Question 3: (iii) 1 1/4

Answer: 1 1/4 =

= 5/4

= 1.25. On the number line, divide the unit between 1 and 2 into 4 equal parts; the first mark after 1 is 1.25.



Question: Exercise 1.1, Question 3: (iv) -1 1/4

Answer: -1 1/4 =

= -5/4

= -1.25. On the number line, divide the unit between -1 and -2 into 4 equal parts; the first mark to the left of -1 is -1.25.



Question: Exercise 1.1, Question 3: (v) 2/3

Answer: 2/3 ≈ 0.67. On the number line, divide the unit between 0 and 1 into 3 equal parts; the second mark after 0 is 2/3.



Question: Exercise 1.1, Question 3: (vi) -2/3

Answer: -2/3 ≈ -0.67. On the number line, divide the unit between 0 and -1 into 3 equal parts; the second mark to the left of 0 is -2/3.



Question: Exercise 1.1, Question 4: Can you list all the real numbers between 1 and 2?

Answer: No. There are infinitely many real numbers between 1 and 2, so a complete list cannot be made. For example 1.1, 1.01, 1.001, 1.0001 … never end.

Question: Exercise 1.1, Question 5: Give a reason why π is an irrational number.

Answer: The decimal form of π never terminates and never repeats (π = 3.1415926535…), and π cannot be written as a ratio p/q of two integers. Therefore π is irrational.

Question: Exercise 1.1, Question 6: Tick (✓) the correct statement. (i) 5/7 is an example of an irrational number.

Answer: Wrong (✗) — 5/7 is a rational number, because it is in p/q form.

Question: Exercise 1.1, Question 6: (ii) π is an irrational number.

Answer: Correct (✓) — the decimal form of π is non-terminating and non-repeating.

Question: Exercise 1.1, Question 6: (iii) 0.31591… is a non-terminating and non-repeating decimal.

Answer: Correct (✓) — the … in the given form means the digits never end and there is no repeating pattern.

Question: Exercise 1.1, Question 6: (iv) $0.1\bar{2}3$ is a repeating decimal.

Answer: Correct (✓) — the bar over 3 means that 3 keeps repeating: 0.12333…

Question: Exercise 1.1, Question 6: (v) There are infinitely many numbers between 0 and 1 2/3. [note: printed line faint; read as the density claim matching Q4]

Answer: Correct (✓) — between any two different real numbers there are always more real numbers, so there are infinitely many numbers between

0 and 1 2/3 as well.

Question: Exercise 1.1, Question 6: (vi) 1/√3 is an example of a rational number.

Answer: Wrong (✗) — 1/√3 is irrational, because its denominator contains √3, which is irrational.

Question: Exercise 1.2, Question 1: Identify the properties of real numbers in the following. (i) 1/2 + 2/3 = 2/3 + 1/2

Answer: Commutative property of addition — changing the order of two numbers in addition does not change the sum.

Question: Exercise 1.2, Question 1: (ii) 4/3 + (1 1/3 + 2/3) = (4/3 + 1 1/3) + 2/3

Answer: Associative property of addition — changing the position of the brackets when adding three numbers does not change the sum.

Question: Exercise 1.2, Question 1: (iii) 9 × (10/9 + 20/9) = (9 × 10/9) + (9 × 20/9)

Answer: Distributive property of multiplication over addition.

Question: Exercise 1.2, Question 1: (iv) (4/5 + 5/7) × 7/8 = (4/5 × 7/8) + (5/7 × 7/8)

Answer: Distributive property of multiplication over addition.

Question: Exercise 1.2, Question 1: (v) (7/5 − 3/5) × 10/15 = (7/5 × 10/15) − (3/5 × 10/15)

Answer: Distributive property of multiplication over subtraction.

Question: Exercise 1.2, Question 1: (vi) d/c × e/f = e/f × d/c

Answer: Commutative property of multiplication — changing the order in multiplication does not change the product.

Question: Exercise 1.2, Question 1: (vii) 11 × (15 × 21) = (11 × 15) × 21

Answer: Associative property of multiplication.

Question: Exercise 1.2, Question 1: (viii) 2/11 × 11/2 = 11/2 × 2/11 = 1

Answer: Commutative property of multiplication and multiplicative inverse — the multiplicative inverse of 2/11 is 11/2 and their product is 1.

Question: Exercise 1.2, Question 1: (ix) (3/5) + (−3/5) = (−3/5) + (3/5) = 0

Answer: Commutative property of addition and additive inverse — the additive inverse of 3/5 is −3/5 and their sum is 0.

Question: Exercise 1.2, Question 1: (x) (a/b) × (b/a) = (b/a) × (a/b) = 1

Answer: Commutative property of multiplication and multiplicative inverse — the multiplicative inverse of a/b is b/a and their product is 1.

Question: Exercise 1.2, Question 1: (xi) 15/10 × (8/5 − 4/10) = (15/10 × 8/5) − (15/10 × 4/10)

Answer: Distributive property of multiplication over subtraction.

Question: Exercise 1.2, Question 1: (xii) √2/3 × 3/√2 = 3/√2 × √2/3 = 1

Answer: Commutative property of multiplication and multiplicative inverse — the multiplicative inverse of √2/3 is 3/√2 and their product is 1.

Question: Exercise 1.2, Question 2: Fill in the blanks with the correct real

numbers to complete the following. (i) √2/5 + 3/√6 = □/√6 + √2/5

Answer: □ = 3 (commutative property of addition: the order of the two terms is changed).

Question: Exercise 1.2, Question 2: (ii) 7/10 + (70/□ + 16/33) = (7/□ + □/10) + 16/□ [note: printed RHS middle term appears shuffled/misprinted; intended associative form below]

Answer: Intended associative form: 7/10 + (70/33 + 16/33) = (7/10 + 70/33) + 16/33, i.e. the blanks in order are 33, 10, 70, 33.

Question: Exercise 1.2, Question 2: (iii) 99/50 × 50/99 = □

Answer: □ = 1, because 99/50 and 50/99 are multiplicative inverses of each other: (99×50)/(50×99) = 1.

Question: Exercise 1.2, Question 2: (iv) (59/95) × (95/59) = □

Answer: □ = 1, because 59/95 and 95/59 are multiplicative inverses of each other: (59×95)/(95×59) = 1.

Question: Exercise 1.2, Question 2: (v) (−21) + (□) = 0

Answer: □ = 21, because the additive inverse of −21 is 21: −21 + 21 = 0.

Question: Exercise 1.2, Question 2: (vi) 5/8 × (2/3 + 5/7) = (□/□ × 2/3) + (5/8 × □/□)

Answer: The blanks in order are 5/8 and 5/7 (distributive property: 5/8 is multiplied by both terms inside the brackets).

Question: Exercise 1.2, Question 3: Fill in the blanks to make the following properties correct. (i) 5 < 8 and 8 < 10 ⇒ ___ < ___

Answer: 5 < 10 (transitive property: if 5 < 8 and 8 < 10 then 5 < 10).

Question: Exercise 1.2, Question 3: (ii) 10 > 8 and 8 > 5 ⇒ ___ > ___

Answer: 10 > 5 (transitive property: if 10 > 8 and 8 > 5 then 10 > 5).

Question: Exercise 1.2, Question 3: (iii) 3 < 6 ⇒ 3 + 9 < ___ + ___

Answer: 6 + 9 (addition property: adding 9 to both sides keeps the inequality true: 12 < 15).

Question: Exercise 1.2, Question 3: (iv) 4 < 6 ⇒ 4 + 8 < ___ + ___

Answer: 6 + 8 (addition property: adding 8 to both sides keeps the inequality true: 12 < 14).

Question: Exercise 1.2, Question 3: (v) 8 > 6 ⇒ 6 + 8 > ___ + ___

Answer: 6 + 6 (addition property: adding 6 to both sides keeps the inequality true: 14 > 12).

Question: Exercise 1.2, Question 4: Fill in the blanks to make the following properties correct. (i) 5 < 7 ⇒ 5 × 12 < ___ × ___

Answer: 7 × 12 (multiplication property: multiplying by the positive number 12 keeps the inequality true: 60 < 84).

Question: Exercise 1.2, Question 4: (ii) 7 > 5 ⇒ 7 × 12 > ___ × ___

Answer: 5 × 12 (multiplication property: multiplying by the positive number 12 keeps the inequality true: 84 > 60).

Question: Exercise 1.2, Question 4: (iii) 6 > 4 ⇒ 6 × (−7) ___ 4 × (−7)

Answer: < , because multiplying by a negative number reverses the

inequality sign: 6×(−7) = −42 < −28 = 4×(−7).

Question: Exercise 1.2, Question 4: (iv) 2 < 8 ⇒ 2 × (−4) ___ 8 × (−4)

Answer: > , because multiplying by a negative number reverses the inequality sign: 2×(−4) = −8 > −32 = 8×(−4).

Question: Exercise 1.2, Question 5: Find the additive and multiplicative inverses of the following real numbers. (i) 3

Answer: Additive inverse = −3 (3 + (−3) = 0); multiplicative inverse = 1/3 (3 × 1/3 = 1).

Question: Exercise 1.2, Question 5: (ii) −7

Answer: Additive inverse = 7 (−7 + 7 = 0); multiplicative inverse = −1/7 (−7 × (−1/7) = 1).

Question: Exercise 1.2, Question 5: (iii) 0.3

Answer: Additive inverse = −0.3 (0.3 + (−0.3) = 0); multiplicative inverse = 1/0.3 = 10/3 (0.3 × 10/3 = 1).

Question: Exercise 1.2, Question 5: (iv) −√5/5

Answer: Additive inverse = √5/5 (−√5/5 + √5/5 = 0); multiplicative inverse = −5/√5 = −√5 ((−√5/5) × (−√5) = 5/5 = 1).

Question: Exercise 1.2, Question 5: (v) 9/√12

Answer: Additive inverse = −9/√12 (9/√12 + (−9/√12) = 0); multiplicative inverse = √12/9 = 2√3/9 ((9/√12) × (√12/9) = 1).

Question: Exercise 1.2, Question 5: (vi) 0

Answer: Additive inverse = 0 (0 + 0 = 0); multiplicative inverse does not exist, because 1/0 is meaningless (no number multiplied by 0 gives 1).

Question: Exercise 1.3, Question 1: Identify the index and the radicand in the following. (i) $\sqrt[3]{5}$

Answer: Index = 3, radicand = 5.

Question: Exercise 1.3, Question 1: (ii) $\sqrt[4]{(x/y)}$

Answer: Index = 4, radicand = x/y.

Question: Exercise 1.3, Question 1: (iii) $\sqrt[5]{(x^2yz)}$

Answer: Index = 5, radicand = x^2yz.

Question: Exercise 1.3, Question 2: Convert the following to exponential form. (i) √(3/4)

Answer: √(3/4) = (3/4)^(1/2), because a square root means the power 1/2.

Question: Exercise 1.3, Question 2: (ii) $\sqrt{((x/y)^5)}$

Answer: $\sqrt{((x/y)^5)}$ = (x/y)^(5/2), because $\sqrt[n]{(a^m)}$ = a^(m/n) and here n = 2, m = 5.

Question: Exercise 1.3, Question 2: (iii) $\sqrt[3]{((y/x)^{(-5)})}$

Answer: $\sqrt[3]{((y/x)^{(-5)})}$ = (y/x)^(−5/3), because $\sqrt[n]{(a^m)}$ = a^(m/n) and here n = 3, m = −5.

Question: Exercise 1.3, Question 2: (iv) $\sqrt[3]{((yz)^7)}$

Answer: $\sqrt[3]{((yz)^7)}$ = (yz)^(7/3), because $\sqrt[n]{(a^m)}$ = a^(m/n) and here n = 3, m = 7.

Question: Exercise 1.3, Question 2: (v) $\sqrt[9]{27}$

Answer: $\sqrt[9]{27}$ =

= 27^(1/9)

= (3^3)^(1/9)

= 3^(3/9)

= 3^(1/3).

Question: Exercise 1.3, Question 2: (vi) $\sqrt[3]{((-64)^2)}$

Answer: $\sqrt[3]{((-64)^2)}$ =

= (-64)^(2/3)

= $((-4)^3)$^(2/3)

= $(-4)^2$

= 16.

Question: Exercise 1.3, Question 2: (vii) $\sqrt[3]{((1/2)^m)}$

Answer: $\sqrt[3]{((1/2)^m)}$ = (1/2)^(m/3), because $\sqrt[n]{(a^m)}$ = a^(m/n) and here n = 3.

Question: Exercise 1.3, Question 2: (viii) $\sqrt[5]{((xy)^3)}$

Answer: $\sqrt[5]{((xy)^3)}$ = (xy)^(3/5), because $\sqrt[n]{(a^m)}$ = a^(m/n) and here n =

5, m = 3.

Question: Exercise 1.3, Question 2: (ix) ³√(√(4/3))

Answer: ³√(√(4/3)) = ((4/3)^(1/2))^(1/3) = (4/3)^(1/6), because (a^(1/2))^(1/3) = a^(1/6).

Question: Exercise 1.3, Question 3: Convert the following to radical form.

(i) (5³)^(1/7)

Answer: (5³)^(1/7) = ⁷√(5³), because a^(m/n) = ⁿ√(a^m).

Question: Exercise 1.3, Question 3: (ii) (ab^(−2))^(1/3)

Answer: (ab^(−2))^(1/3) = ³√(ab^(−2)) = ³√(a/b²), because b^(−2) = 1/b².

Question: Exercise 1.3, Question 3: (iii) [(5/7)³]^(5/7)

Answer: [(5/7)³]^(5/7) = (5/7)^(15/7) = ⁷√((5/7)^15), because (a^m)^(p/q) = a^(mp/q).

Question: Exercise 1.3, Question 3: (iv) (b/a)^(−m/2)

Answer: (b/a)^(−m/2) =

= 1/(b/a)^(m/2)

= 1/√((b/a)^m)

= √((a/b)^m).

Question: Exercise 1.3, Question 3: (v) [(11/13)(12/13)]^(1/5)

Answer: [(11/13)(12/13)]^(1/5) =

= ⁵√((11/13)(12/13))

= ⁵√(132/169).

Question: Exercise 1.4, Question 1: Express the following in simplest form.

(i) $3^5/3^2$

Answer: $3^5/3^2$ =

= 3^(5−2)

= 3^3

= 27 (by the quotient law of exponents).

Question: Exercise 1.4, Question 1: (ii) $2^4 \cdot 5^3/10^2$

Answer: $2^4 \cdot 5^3/10^2$ =

= (16 × 125)/100

= 2000/100

= 20.

Question: Exercise 1.4, Question 1: (iii) $(a+b)^2 \cdot (c+d)^3/((a+b) \cdot (c+d)^2)$

Answer: $(a+b)^2/(a+b) = (a+b)$ and $(c+d)^3/(c+d)^2 = (c+d)$, so the answer is $(a+b)(c+d)$.

Question: Exercise 1.4, Question 2: Simplify using the laws of exponents.

(i) $(1/3)^4 \times (1/3)^5$

Answer: $(1/3)^4 \times (1/3)^5 = (1/3)^{(4+5)} = (1/3)^9$ (product law: with the same base, exponents are added).

Question: Exercise 1.4, Question 2: (ii) $(3/4)^5 \times (3/4)^2$

Answer: $(3/4)^5 \times (3/4)^2 =$

$= (3/4)^{(5+2)}$

$= (3/4)^7$.

Question: Exercise 1.4, Question 2: (iii) $(-4/5)^3 \times (-4/5)^5$

Answer: $(-4/5)^3 \times (-4/5)^5 =$

$= (-4/5)^{(3+5)}$

$= (-4/5)^8$.

Question: Exercise 1.4, Question 2: (iv) $(-3 \times 5^2)^3$

Answer: $(-3 \times 5^2)^3 =$

$= (-3)^3 \times (5^2)^3$

$= -27 \times 5^6$

$= -27 \times 15625$

= −421875.

Question: Exercise 1.4, Question 2: (v) $[3 \times (-4)^2]^3$

Answer: $[3 \times (-4)^2]^3$ =

$= 3^3 \times ((-4)^2)^3$

$= 27 \times (-4)^6$

$= 27 \times 4096$

= 110592.

Question: Exercise 1.4, Question 2: (vi) $(-a/bc)^5 \times (-a/bc)^4$

Answer: $(-a/bc)^5 \times (-a/bc)^4$ =

= (−a/bc)^(5+4)

$= (-a/bc)^9$.

Question: Exercise 1.4, Question 2: (vii) $(-c/d)^2(-c/d)^3(-c/d)^5$

Answer: $(-c/d)^2(-c/d)^3(-c/d)^5$ =

= (−c/d)^(2+3+5)

= (−c/d)^10

= (c/d)^10.

Question: Exercise 1.4, Question 2: (viii) $mn^2t^4 \cdot n^3m^5t^7$

Answer: $mn^2t^4 \cdot n^3m^5t^7$ =

= m^(1+5)·n^(2+3)·t^(4+7)

= $m^6n^5t^{11}$.

Question: Exercise 1.4, Question 2: (ix) $a^2c^5b^2a^3c^3b^4a^4$

Answer: $a^2c^5b^2a^3c^3b^4a^4$ =

= a^(2+3+4)·b^(2+4)·c^(5+3)

= $a^9b^6c^8$.

Question: Exercise 1.4, Question 3: Simplify using the laws of exponents.

(i) $(5^2)^3$

Answer: $(5^2)^3$ =

= 5^(2×3)

= 5^6

= 15625.

Question: Exercise 1.4, Question 3: (ii) $\{(xy)^3\}^5$

Answer: $\{(xy)^3\}^5$ =

= (xy)^(3×5)

= (xy)^15

= x^15·y^15.

Question: Exercise 1.4, Question 3: (iii) $\{(-4)^2\}^5$

Answer: $\{(-4)^2\}^5$ =

= (−4)^(2×5)

= (−4)^10

= 4^10

= 1048576.

Question: Exercise 1.4, Question 3: (iv) $\{(-3)^3(-4)^2\}^3$

Answer: $\{(-3)^3(-4)^2\}^3$ =

= $(-3)^9 \cdot (-4)^6$

= (−19683) × 4096

= −80621568.

Question: Exercise 1.4, Question 3: (v) $(b^2/5)^3$

Answer: $(b^2/5)^3$ =

= $(b^2)^3/5^3$

= $b^6/125$.

Question: Exercise 1.4, Question 3: (vi) $\{(-4/9)^2\}^3$

Answer: $\{(-4/9)^2\}^3$ =

= (−4/9)^(2×3)

= $(-4/9)^6$

= $(4/9)^6$

= 4096/531441.

Question: Exercise 1.4, Question 3: (vii) $\{(z^3)^2\}^4$

Answer: $\{(z^3)^2\}^4$ =

= z^(3×2×4)

= z^24.

Question: Exercise 1.4, Question 3: (viii) $\{(mm^2m^3m^4)^2\}^5$

Answer: $mm^2m^3m^4$ = m^(1+2+3+4) = m^10; then $\{(m^{10})^2\}^5$ = m^(10×2×5) = m^100.

Question: Exercise 1.4, Question 3: (ix) $-[(-0.1)^2(-0.1)^3(-0.1)^4]^2$

Answer: Inside: (−0.1)^(2+3+4) = $(-0.1)^9 = -(0.1)^9$; squaring: $(-(0.1)^9)^2$

= (0.1)^18; with the outer minus: −(0.1)^18 = −10^(−18).

Question: Exercise 1.5, Question 1: Write the following complex numbers in a+ib form. (i) (1,2)

Answer: (1,2) = 1 + 2i (in the pair, the first number is the real part and the second is the imaginary part).

Question: Exercise 1.5, Question 1: (ii) (2,2)

Answer: (2,2) = 2 + 2i.

Question: Exercise 1.5, Question 1: (iii) (3,4)

Answer: (3,4) = 3 + 4i.

Question: Exercise 1.5, Question 1: (iv) (−1,1)

Answer: (−1,1) = −1 + i.

Question: Exercise 1.5, Question 1: (v) (−2,2)

Answer: (−2,2) = −2 + 2i.

Question: Exercise 1.5, Question 1: (vi) (−3,4)

Answer: (−3,4) = −3 + 4i.

Question: Exercise 1.5, Question 2: Identify the real and imaginary parts of the following complex numbers. (i) 1+2i

Answer: Real part = 1, imaginary part = 2.

Question: Exercise 1.5, Question 2: (ii) 9i+4

Answer: 9i+4 = 4+9i, so real part = 4, imaginary part = 9.

Question: Exercise 1.5, Question 2: (iii) (−5,6)

Answer: Real part = −5, imaginary part = 6.

Question: Exercise 1.5, Question 2: (iv) −1−i

Answer: Real part = −1, imaginary part = −1.

Question: Exercise 1.5, Question 2: (v) (−3/4) − (−4/5)i

Answer: (−3/4) − (−4/5)i = −3/4 + (4/5)i, so real part = −3/4, imaginary part = 4/5.

Question: Exercise 1.5, Question 2: (vi) 2i−1

Answer: 2i−1 = −1+2i, so real part = −1, imaginary part = 2.

Question: Exercise 1.5, Question 3: Find the conjugate of the following complex numbers. (i) 3+2i

Answer: Conjugate = 3−2i (the sign of the imaginary part is reversed).

Question: Exercise 1.5, Question 3: (ii) (4,9)

Answer: (4,9) = 4+9i, so conjugate = 4−9i.

Question: Exercise 1.5, Question 3: (iii) (−1,1)

Answer: (−1,1) = −1+i, so conjugate = −1−i.

Question: Exercise 1.5, Question 3: (iv) 1−i

Answer: Conjugate = 1+i.

Question: Exercise 1.5, Question 3: (v) (−3/4) + (−4/5)i

Answer: Conjugate =

= (−3/4) − (−4/5)i

= −3/4 + (4/5)i.

Question: Exercise 1.5, Question 3: (vi) 3i+1

Answer: 3i+1 = 1+3i, so conjugate = 1−3i.

Question: Exercise 1.5, Question 4: Verify that the conjugate of the conjugate of the following complex numbers is the number itself. (i) (4/7)+(9/10)i

Answer: Conjugate = (4/7)−(9/10)i; conjugate again = (4/7)+(9/10)i = the original number z. Verified.

Question: Exercise 1.5, Question 4: (ii) (−9/11)+(10/9)i

Answer: Conjugate = (−9/11)−(10/9)i; conjugate again = (−9/11)+(10/9)i = the original number z. Verified.

Question: Exercise 1.5, Question 4: (iii) 1/2−3i

Answer: Conjugate = 1/2+3i; conjugate again = 1/2−3i = the original number z. Verified.

Question: Exercise 1.5, Question 4: (iv) 2+3i

Answer: Conjugate = 2−3i; conjugate again = 2+3i = the original number z. Verified.

Question: Exercise 1.5, Question 4: (v) −2−3(−10/9)i

Answer: −2−3(−10/9)i = −2+(10/3)i; conjugate = −2−(10/3)i; conjugate

again = −2+(10/3)i = the original number z. Verified.

Question: Exercise 1.5, Question 4: (vi) 4x+3iy

Answer: Conjugate = 4x−3iy; conjugate again = 4x+3iy = the original number z. Verified.

Question: Exercise 1.5, Question 5: Find the values of x and y when the two sides are equal. (i) x+yi = −5+5i

Answer: Equating real parts: x = −5; equating imaginary parts: y = 5.

Question: Exercise 1.5, Question 5: (ii) $x^2+iy^2 = 16/9 + 9/25\ i$

Answer: $x^2 = 16/9 \Rightarrow x = \pm 4/3$; $y^2 = 9/25 \Rightarrow y = \pm 3/5$.

Question: Exercise 1.5, Question 5: (iii) $y^2 + (x/3)i = 121 - 9/5\ i$

Answer: $y^2 = 121 \Rightarrow y = \pm 11$; $x/3 = -9/5 \Rightarrow x = -27/5$.

Question: Exercise 1.5, Question 5: (iv) $(\sqrt{5}/3)x - (3/\sqrt{2})yi = 6\sqrt{3}/\sqrt{2} + 2\sqrt{2}/9\ i$

Answer: Real parts: $(\sqrt{5}/3)x = 6\sqrt{3}/\sqrt{2} \Rightarrow x = (6\sqrt{3}/\sqrt{2})\times(3/\sqrt{5}) = 18\sqrt{3}/\sqrt{10} = 9\sqrt{30}/5$. Imaginary parts: $-(3/\sqrt{2})y = 2\sqrt{2}/9 \Rightarrow y = -(2\sqrt{2}/9)\times(\sqrt{2}/3) = -4/27$.

Question: Exercise 1.6, Question 1: Solve the following complex numbers using the basic operation signs. (i) (3,2)+(9,3)

Answer: (3,2)+(9,3) = (3+9, 2+3) = (12,5) = 12+5i.

Question: Exercise 1.6, Question 1: (ii) (3/2, 2/3)+(2/3, 3/2)

Answer: (3/2+2/3, 2/3+3/2) = (13/6, 13/6) = 13/6 + (13/6)i, because

3/2+2/3 = 9/6+4/6 = 13/6.

Question: Exercise 1.6, Question 1: (iii) (15,12)−(10,−9)

Answer: (15,12)−(10,−9) = (15−10, 12−(−9)) = (5,21) = 5+21i.

Question: Exercise 1.6, Question 1: (iv) (4/5, 8/15)−(4/5, 6/10)

Answer: (4/5−4/5, 8/15−6/10) = (0, 16/30−18/30) = (0, −1/15) = −(1/15)i.

Question: Exercise 1.6, Question 1: (v) (1,2)(1,−2)

Answer: (1,2)(1,−2) = (1×1−2×(−2), 1×(−2)+2×1) = (5,0) = 5 (multiplication rule: (a,b)(c,d) = (ac−bd, ad+bc)).

Question: Exercise 1.6, Question 1: (vi) (4,−5)(5,−4)

Answer: (4,−5)(5,−4) = (4×5−(−5)(−4), 4×(−4)+(−5)×5) = (20−20, −16−25) = (0,−41) = −41i.

Question: Exercise 1.6, Question 1: (vii) (3,−7)÷(3,2)

Answer: (3−7i)/(3+2i) =

= (3−7i)(3−2i)/((3+2i)(3−2i))

= (9−6i−21i+14i²)/13

= (−5−27i)/13

= −5/13 − (27/13)i.

Question: Exercise 1.6, Question 1: (viii) (4,5)÷(2,−3)

Answer: (4+5i)/(2−3i) =

= (4+5i)(2+3i)/((2−3i)(2+3i))

= (8+12i+10i+15i²)/13

= (−7+22i)/13

= −7/13 + (22/13)i.

Question: Exercise 1.6, Question 2: Simplify and write your answer in a+ib form. (i) −1/(1+i)

Answer: −1/(1+i) =

= −(1−i)/((1+i)(1−i))

= −(1−i)/2

= −1/2 + (1/2)i.

Question: Exercise 1.6, Question 2: (ii) (1+i)⁴

Answer: (1+i)² = 1+2i+i² = 2i; then (1+i)⁴ = (2i)² = 4i² = −4 = −4+0i.

Question: Exercise 1.6, Question 2: (iii) (1/(1+i))²

Answer: (1/(1+i))² = 1/(1+i)² = 1/(2i) = −i/2 = 0 − (1/2)i, because 1/i = −i.

Question: Exercise 1.6, Question 2: (iv) (1+i)⁸

Answer: $(1+i)^8$ =

= $((1+i)^4)^2$

= $(-4)^2$

= 16

= 16+0i.

Question: Exercise 1.6, Question 3: If $z_1 = -4+6i$ and $z_2 = 2\ 1/2 - 2i$, verify that (i) conjugate(z_1+z_2) = conjugate(z_1) + conjugate(z_2)

Answer: $z_1+z_2 = (-4+5/2)+(6-2)i = -3/2+4i$; left side: conjugate = $-3/2-4i$. Right side: $(-4-6i)+(5/2+2i) = -3/2-4i$. Both sides are equal.

Question: Exercise 1.6, Question 3: (ii) conjugate(z_1-z_2) = conjugate(z_1) − conjugate(z_2)

Answer: $z_1-z_2 = (-4-5/2)+(6+2)i = -13/2+8i$; left side: conjugate = $-13/2-8i$. Right side: $(-4-6i)-(5/2+2i) = -13/2-8i$. Both sides are equal.

Question: Exercise 1.6, Question 4: If $z_1 = 1+i$ and $z_2 = 1-i$, verify that (i) conjugate(z_1z_2) = conjugate(z_1)·conjugate(z_2)

Answer: $z_1z_2 = (1+i)(1-i) = 1-i^2 = 2$; left side: conjugate(2) = 2. Right side: $(1-i)(1+i) = 2$. Both sides are equal.

Question: Exercise 1.6, Question 4: (ii) conjugate(z_1/z_2) = conjugate(z_1)/conjugate(z_2)

Answer: $z_1/z_2 = (1+i)/(1-i) = (1+i)^2/2 = 2i/2 = i$; left side: conjugate(i) =

$-i$. Right side: $(1-i)/(1+i) = (1-i)^2/2 = -2i/2 = -i$. Both sides are equal.

Review Exercise 1, Question 1: Fill in the following blanks. (i) The multiplicative inverse of $\sqrt{5}$ is _____.

Answer: $\times (1/\sqrt{5}) = 1$.

Review Exercise 1, Question 1: (ii) $Q \cup Q' =$ _____.

Answer: R (the set of real numbers), because every real number is either rational or irrational.

Review Exercise 1, Question 1: (iii) The additive identity element in R is _____.

Answer: 0, because for every real number a, a + 0 = 0 + a = a.

Review Exercise 1, Question 1: (iv) $5+(6+7) = (5+6)+$ _____.

Answer: 7 (associative property of addition: changing the position of the brackets does not change the sum).

Review Exercise 1, Question 1: (v) $3+(-3) =$ _____.

Answer: 0, because -3 is the additive inverse of 3.

Review Exercise 1, Question 1: (vi) π is a _____ number.

Answer: non-terminating and non-repeating.

Review Exercise 1, Question 1: (vii) 22/7 is a _____ number.

Answer: ratio of two integers in p/q form.

Review Exercise 1, Question 1: (viii) The conjugate of $-3+5i$ is _____.

Answer: reversed).

Review Exercise 1, Question 1: (ix) The real part in 2i(3−i) is _____2._____ .

Review Exercise 1, Question 1: (x) The product of two complex numbers (a,b) and (c,d), i.e. (a,b).(c,d) = _____.

Answer: (ac−bd)+(ad+bc)i.

Question: Review Exercise 1, Question 2: Read the following statements carefully, circle T for a true statement and F for a false statement. (i) R is closed under multiplication.

Answer: T — the product of two real numbers is also a real number.

Question: Review Exercise 1, Question 2: (ii) If $x<y$ and $y<z$ then $x<z$.

Answer: T — this is the transitive property of inequality.

Question: Review Exercise 1, Question 2: (iii) For all $x,y,z \in R$, $x(y-z) = xy-xz$.

Answer: T — this is the distributive property of multiplication over subtraction.

Question: Review Exercise 1, Question 2: (iv) The product of any two imaginary numbers is real.

Answer: T — the product of two purely imaginary numbers is real, e.g.(2i)(3i) =

$= 6i^2$

= −6.

Question: Review Exercise 1, Question 2: (v) The sum of two real numbers is a real number.

Answer: T — this is the closure property of R under addition.

Review Exercise 1, Question 3: Tick () the correct answer. (i) The additive inverse of √5 is:

✓ (a) −√5

(b) 1/√5

(c) √−5

(d) −5

Review Exercise 1, Question 3: (ii) (5i)·(−2i) = _____.

(a) −10

✓ (b) 10

(c) −10i

(d) 10i

Review Exercise 1, Question 3: (iii) 3(5+7) = 3·5+3·7, the name of the property is:

(a) Commutative

(b) Associative

✓ (c) Distributive

(d) Closure

Review Exercise 1, Question 3: (iv) √−2 × √−2 = _____.

(a) 2

(b) −2 ✓

(c) 2i

(d) −2i

2. Logarithms

Question: Exercise 2.1, Question 1: Express each of the following numbers in scientific notation. (i) 9700

Answer: $9700 = 9.7 \times 10^3$ (the decimal point is moved 3 places to the left).

Question: Exercise 2.1, Question 1: (ii) 4,980,000

Answer: $4{,}980{,}000 = 4.98 \times 10^6$ (the decimal point is moved 6 places to the left).

Question: Exercise 2.1, Question 1: (iii) 96,000,000

Answer: $96{,}000{,}000 = 9.6 \times 10^7$ (the decimal point is moved 7 places to the left).

Question: Exercise 2.1, Question 1: (iv) 4169

Answer: $4169 = 4.169 \times 10^3$ (the decimal point is moved 3 places to the left).

Question: Exercise 2.1, Question 1: (v) 84,000

Answer: $84{,}000 = 8.4 \times 10^4$ (the decimal point is moved 4 places to the left).

Question: Exercise 2.1, Question 1: (vi) 0.718

Answer: $0.718 = 7.18 \times 10^{-1}$ (the decimal point is moved 1 place to the right).

Question: Exercise 2.1, Question 1: (vii) 0.00643

Answer: $0.00643 = 6.43 \times 10^{-3}$ (the decimal point is moved 3 places to the right).

Question: Exercise 2.1, Question 1: (viii) 0.0074

Answer: $0.0074 = 7.4 \times 10^{-3}$ (the decimal point is moved 3 places to the right).

Question: Exercise 2.1, Question 1: (ix) 0.21005

Answer: $0.21005 = 2.1005 \times 10^{-1}$ (the decimal point is moved 1 place to the right).

Question: Exercise 2.1, Question 2: Express the following numbers in ordinary (standard) notation. (i) 7×10^4

Answer: $7 \times 10^4 = 70{,}000$.

Question: Exercise 2.1, Question 2: (ii) 8.072×10^{-10}

Answer: $8.072 \times 10^{-10} = 0.0000000008072$ (the decimal point is moved 10 places to the left).

Question: Exercise 2.1, Question 2: (iii) 6.018×10^6

Answer: $6.018 \times 10^6 = 6{,}018{,}000$.

Question: Exercise 2.1, Question 2: (iv) 7.865×10^8

Answer: $7.865 \times 10^8 = 786{,}500{,}000$.

Question: Exercise 2.1, Question 2: (v) 2.05×10^{-4}

Answer: $2.05 \times 10^{-4} = 0.000205$.

Question: Exercise 2.1, Question 2: (vi) 7.25×10^{10}

Answer: $7.25 \times 10^{10} = 72{,}500{,}000{,}000$.

Question: Exercise 2.1, Question 2: (vii) 4.502×10^{6}

Answer: $4.502 \times 10^{6} = 4{,}502{,}000$.

Question: Exercise 2.1, Question 2: (viii) 2.865×10^{-8}

Answer: $2.865 \times 10^{-8} = 0.00000002865$.

Question: Exercise 2.1, Question 2: (ix) 3.056×10^{6}

Answer: $3.056 \times 10^{6} = 3{,}056{,}000$.

Question: Exercise 2.2, Question 1: Write the following in logarithmic form.

(i) $7^{3} = 343$

Answer: $\log_{7}343 = 3$ (rule: a^x = y ⇔ $\log_{a}y = x$).

Question: Exercise 2.2, Question 1: (ii) $3^{-4} = 1/81$

Answer: $\log_{3}(1/81) = -4$.

Question: Exercise 2.2, Question 1: (iii) $10^{-3} = 0.001$

Answer: $\log_{10}(0.001) = -3$.

Question: Exercise 2.2, Question 1: (iv) $\sqrt[3]{(8^{2})} = 4$

Answer: $\sqrt[3]{(8^{2})} = 8^{(2/3)} = 4$, so $\log_{8}4 = 2/3$.

Question: Exercise 2.2, Question 2: Write the following in exponential form. (i) $\log_{27}81 = 4/3$

Answer: $27^{(4/3)} = 81$, because $(3^{3})^{(4/3)} = 3^{4} = 81$.

Question: Exercise 2.2, Question 2: (ii) $\log_2(1/8) = -3$

Answer: $2^{-3} = 1/8$.

Question: Exercise 2.2, Question 2: (iii) $\log_{10}1 = 0$

Answer: $10^0 = 1$.

Question: Exercise 2.2, Question 2: (iv) $\log_{10}(0.01) = -2$

Answer: $10^{-2} = 0.01$.

Question: Exercise 2.2, Question 3: Find the unknown in the following. (i) $\log_{32}x = 1/2$

Answer: x =

= 32^(1/2)

= √32

= √(16×2)

= 4√2.

Question: Exercise 2.2, Question 3: (ii) $\log_a 3 = 1/2$

Answer: a^(1/2) =

= 3 ⇒ a

= 3^2

= 9.

Question: Exercise 2.2, Question 3: (iii) log_√5 25 = y

Answer: (√5)^y =

= 25 ⇒ 5^(y/2)

= 5^2 ⇒ y/2

= 2 ⇒ y

= 4.

Question: Exercise 2.2, Question 3: (iv) $\log_4 x = 3/2$

Answer: x =

= 4^(3/2)

= $(\sqrt{4})^3$

= 2^3

= 8.

Question: Exercise 2.2, Question 3: (v) $\log_{10} 100 = y$

Answer: 10^y =

= 100

$= 10^2 \Rightarrow y$

$= 2.$

Question: Exercise 2.2, Question 3: (vi) $\log_a 64 = 3$

Answer: $a^3 =$

$= 64$

$= 4^3 \Rightarrow a$

$= 4.$

Question: Exercise 2.2, Question 3: (vii) $\log_a 1 = 0$

Answer: $a^0 = 1$, which holds for every valid base $a > 0, a \neq 1$.

Question: Exercise 2.2, Question 3: (viii) $\log_{55} 55 = y$

Answer: 55^y =

$= 55 \Rightarrow y$

$= 1.$

Question: Exercise 2.2, Question 3: (ix) $\log_{64} 8 = x/2$

Answer: $\log_{64} 8 = 1/2$, because 64^(1/2) = 8.So x/2 =

$= 1/2 \Rightarrow x$

= 1.

Question: Exercise 2.3, Question 1: Find the characteristic and mantissa of the logarithm of the following. (i) 8

Answer: log 8 = 0.9031; characteristic = 0, mantissa = 0.9031.

Question: Exercise 2.3, Question 1: (ii) 5054

Answer: 5054 = 5.054 × 10^3, so log 5054 = 3.7036; characteristic = 3, mantissa = 0.7036.

Question: Exercise 2.3, Question 1: (iii) 9.992

Answer: log 9.992 = 0.9997; characteristic = 0, mantissa = 0.9997.

Question: Exercise 2.3, Question 1: (iv) 765.3

Answer: 765.3 = 7.653 × 10^2, so log 765.3 = 2.8838; characteristic = 2, mantissa = 0.8838.

Question: Exercise 2.3, Question 1: (v) 0.00329

Answer: log 0.00329 =

= −2.4828

= $\bar{3}$.5172; characteristic = $\bar{3}$ (−3), mantissa = 0.5172.

Question: Exercise 2.3, Question 1: (vi) 0.0000300

Answer: 0.0000300 = 3.00 × 10^{-5}, so log 0.0000300 = −4.5229 = $\bar{5}$.4771; characteristic = $\bar{5}$ (−5), mantissa = 0.4771.

Question: Exercise 2.3, Question 2: Find the logarithm of the following numbers. (i) 9

Answer: log 9 = 0.9542 (from the table: the mantissa of 9 is 0.9542 and the characteristic is 0).

Question: Exercise 2.3, Question 2: (ii) 55.56

Answer: log 55.56 = 1.7448 (mantissa 0.7448 and characteristic 1).

Question: Exercise 2.3, Question 2: (iii) 29.592

Answer: log 29.592 = 1.4712 (mantissa 0.4712 and characteristic 1).

Question: Exercise 2.3, Question 2: (iv) 405.3

Answer: log 405.3 = 2.6078 (mantissa 0.6078 and characteristic 2).

Question: Exercise 2.3, Question 2: (v) 0.00469

Answer: log 0.00469 =

$= -2.3288$

$= \bar{3}.6712$ (mantissa 0.6712 and characteristic $\bar{3}$).

Question: Exercise 2.3, Question 2: (vi) 0.000076

Answer: log 0.000076 =

$= -4.1192$

$= \bar{5}.8808$ (mantissa 0.8808 and characteristic $\bar{5}$).

Question: Exercise 2.3, Question 3: If log 31.09 = 1.4926, find the values of the following using the logarithm table. (i) log 3.109

Answer: Same mantissa 0.4926, characteristic 0 ⇒ log 3.109 = 0.4926.

Question: Exercise 2.3, Question 3: (ii) log 310.9

Answer: Same mantissa 0.4926, characteristic 2 ⇒ log 310.9 = 2.4926.

Question: Exercise 2.3, Question 3: (iii) log 0.003109

Answer: Same mantissa 0.4926, characteristic $\bar{3}$ ⇒ log 0.003109 = $\bar{3}$.4926.

Question: Exercise 2.3, Question 3: (iv) log 3109

Answer: Same mantissa 0.4926, characteristic 3 ⇒ log 3109 = 3.4926.

Question: Exercise 2.3, Question 3: (v) log 310.942

Answer: The first four significant digits of 310.942 are 3109 (mantissa 0.4926) and adding the mean difference for the fifth digit 4 gives mantissa 0.4927; characteristic 2 ⇒ log 310.942 = 2.4927.

Question: Exercise 2.3, Question 3: (vi) log 310926

Answer: The first four significant digits of 310926 are 3109 (mantissa 0.4926) and adding the mean differences for the remaining digits gives mantissa 0.4927; characteristic 5 ⇒ log 310926 = 5.4927.

Question: Exercise 2.4, Question 1: Using the table, find the numbers whose common logarithms are given below. (i) 0.35749

Answer: For mantissa 0.35749 the antilog table gives digits 2277 and the characteristic is 0 ⇒ number = 2.2777.

Question: Exercise 2.4, Question 1: (ii) 3.56721

Answer: For mantissa 0.56721 the antilog is 3691.6 and the characteristic is 3 ⇒ number = 3691.6.

Question: Exercise 2.4, Question 1: (iii) $\bar{1}$.7427

Answer: For mantissa 0.7427 the antilog is 5530 and the characteristic is $\bar{1}$ (−1) ⇒ number = 0.5530.

Question: Exercise 2.4, Question 1: (iv) 5.8196

Answer: For mantissa 0.8196 the antilog is 660085 and the characteristic is 5 ⇒ number = 660085.

Question: Exercise 2.4, Question 1: (v) $\bar{4}$.3847

Answer: For mantissa 0.3847 the antilog is 2425 and the characteristic is $\bar{4}$ (−4) ⇒ number = 0.0002425.

Question: Exercise 2.4, Question 1: (vi) 0.9187

Answer: For mantissa 0.9187 the antilog is 8293 and the characteristic is 0 ⇒ number = 8.2928.

Question: Exercise 2.4, Question 2: Using a calculator, find the logarithms of the following numbers. (i) 900

Answer: log 900 = 2.9542.

Question: Exercise 2.4, Question 2: (ii) 45.54

Answer: log 45.54 = 1.6584.

Question: Exercise 2.4, Question 2: (iii) 36582

Answer: log 36582 = 4.5633.

Question: Exercise 2.4, Question 2: (iv) 826.3

Answer: log 826.3 = 2.9171.

Question: Exercise 2.4, Question 2: (v) 0.00851

Answer: log 0.00851 =

= −2.0701

= 3.9299.

Question: Exercise 2.4, Question 2: (vi) 0.000097

Answer: log 0.000097 =

= −4.0132

= 5.9868.

Question: Exercise 2.4, Question 3: Using a calculator, find the value of x from the following. (i) log x = 1.7505

Answer: x =

= antilog(1.7505)

= 10^1.7505

= 56.30.

Question: Exercise 2.4, Question 3: (ii) log x = 0.6609

Answer: x =

= antilog(0.6609)

= 10^0.6609

= 4.5804.

Question: Exercise 2.4, Question 3: (iii) log x = $\bar{1}$.6132

Answer: x = antilog($\bar{1}$.6132) = 10^(−1+0.6132) = 0.4104.

Question: Exercise 2.4, Question 3: (iv) log x = 3.4800

Answer: x =

= antilog(3.4800)

= 10^3.4800

= 3019.95 ≈ 3020.

Question: Exercise 2.4, Question 3: (v) log x = $\bar{7}$.0038

Answer: x = antilog($\bar{7}$.0038) = 10^(−7+0.0038) = 1.0088×10^{-7} = 0.0000001009.

Question: Exercise 2.4, Question 3: (vi) log x = 0.2665

Answer: x =

= antilog(0.2665)

= 10^0.2665

= 1.8471.

Question: Exercise 2.5, Question 1: Express the following logarithms as sums of $\log_a x$, $\log_a y$ and $\log_a z$. (i) $\log_a(x^3y/z^2)$

Answer: $\log_a(x^3y/z^2)$ =

= $\log_a x^3 + \log_a y - \log_a z^2$

= $3\log_a x + \log_a y - 2\log_a z$.

Question: Exercise 2.5, Question 1: (ii) $\log_a\sqrt{(xy^2z)}$

Answer: $\log_a\sqrt{(xy^2z)}$ =

= $(1/2)\log_a(xy^2z)$

= $(1/2)(\log_a x + 2\log_a y + \log_a z)$

= $(1/2)\log_a x + \log_a y + (1/2)\log_a z$.

Question: Exercise 2.5, Question 1: (iii) $\log_a(\sqrt[3]{(x^{-1}\sqrt{(y^3)})} \div \sqrt{(y^3\sqrt{x})})$

Answer: $\sqrt[3]{(x^{-1}\sqrt{(y^3)})}$ = x^(−1/3)·y^(1/2); $\sqrt{(y^3\sqrt{x})}$ = y^(3/2)·x^(1/4); quotient = x^(−7/12)·y^(−1). So log = $-(7/12)\log_a x - \log_a y$.

Question: Exercise 2.5, Question 1: (iv) $\log_a(x\sqrt{(y^3)}/\sqrt[3]{(z^2x^5)})$

Answer: = $\log_a x + (3/2)\log_a y - (1/3)(2\log_a z + 5\log_a x) = -(2/3)\log_a x + (3/2)\log_a y - (2/3)\log_a z$.

Question: Exercise 2.5, Question 2: Combine the following into a single logarithm. (i) $\log_a 20 - \log_a 15 + (1/2)\log_a(9/2)$

Answer: = $\log_a(20/15) + \log_a\sqrt{(9/2)} = \log_a((4/3)\times(3/\sqrt{2})) = \log_a(4/\sqrt{2}) = \log_a(2\sqrt{2})$.

Question: Exercise 2.5, Question 2: (ii) $(1/3)\log_a(x-1)^3 + (10/9)\log_a(x+1) - (1/9)\log_a(x+1)$

Answer: = $\log_a(x-1) + (10/9 - 1/9)\log_a(x+1) = \log_a(x-1) + \log_a(x+1) = \log_a(x^2-1)$.

Question: Exercise 2.5, Question 2: (iii) $\log x - 2\log x + 3\log(x+1) - \log(x^2-1)$

Answer: = $-\log x + 3\log(x+1) - \log((x-1)(x+1)) = -\log x + 2\log(x+1) - \log(x-1) = \log((x+1)^2/(x(x-1)))$.

Question: Exercise 2.5, Question 3: If log 2 = 0.3010, log 3 = 0.4771 and log 5 = 0.6990, find the values of the following without using tables. (i) log 15

Answer: log 15 =

= log(3×5)

= log 3 + log 5

= 0.4771 + 0.6990

= 1.1761.

Question: Exercise 2.5, Question 3: (ii) log 64

Answer: log 64 =

= log(2^6)

= 6log 2

= 6 × 0.3010

= 1.8060.

Question: Exercise 2.5, Question 3: (iii) log √(5×2)

Answer: log √(5×2) =

= (1/2)(log 5 + log 2)

= (1/2)(0.6990 + 0.3010)

= 0.5000.

Question: Exercise 2.5, Question 3: (iv) log 48

Answer: log 48 =

= log(16×3)

= 4log 2 + log 3

= 1.2040 + 0.4771

= 1.6811.

Question: Exercise 2.5, Question 3: (v) log √18

Answer: log √18 =

= (1/2)log(2×3²)

= (1/2)(log 2 + 2log 3)

= (1/2)(0.3010 + 0.9542)

= 0.6276.

Question: Exercise 2.5, Question 3: (vi) log 30

Answer: log 30 =

= log(2×3×5)

= log 2 + log 3 + log 5

= 0.3010 + 0.4771 + 0.6990

= 1.4771.

Question: Exercise 2.5, Question 3: (vii) log(8/3)

Answer: log(8/3) =

= log 8 − log 3

= 3log 2 − log 3

= 0.9030 − 0.4771

= 0.4259.

Question: Exercise 2.5, Question 3: (viii) log(5/√3)

Answer: log(5/√3) =

= log 5 − (1/2)log 3

= 0.6990 − 0.2386

= 0.4604.

Question: Exercise 2.5, Question 4: Prove the following. (i) log_b m × log_m a = log_b a

Answer: By the change-of-base rule: log_b m = ln m/ln b and log_m a = ln a/ln m. Their product = (ln m/ln b)×(ln a/ln m) = ln a/ln b = log_b a.

Question: Exercise 2.5, Question 4: (ii) log_a b × log_c a = log_c b

Answer: log_a b = ln b/ln a and log_c a = ln a/ln c. Their product = (ln b/ln a)×(ln a/ln c) = ln b/ln c = log_c b.

Question: Exercise 2.5, Question 4: (iii) $\log_b a \cdot \log_c b \cdot (1/\log_c a) = 1$

Answer: Left side =

$= (\ln a/\ln b)\times(\ln b/\ln c)\times(\ln c/\ln a)$

$= 1$

= right side.

Question: Exercise 2.5, Question 4: (iv) $\log_a b = 1/\log_b a$

Answer: $\log_b a = \ln a/\ln b$, so $1/\log_b a = \ln b/\ln a = \log_a b$.

Question: Exercise 2.5, Question 5: Verify the following. (i) $\log_5 7 \times \log_7 25 = 2$

Answer: $\log_7 25$ =

$= \log_7(5^2)$

$= 2\log_7 5$

$= 2/\log_5 7$. So $\log_5 7 \times (2/\log_5 7) = 2$. Verified.

Question: Exercise 2.5, Question 5: (ii) $\log_3 2 \times \log_2 81 = 4$

Answer: $\log_2 81$ =

$= \log_2(3^4)$

$= 4\log_2 3$

= $4/\log_3 2$. So $\log_3 2 \times (4/\log_3 2) = 4$. Verified.

Question: Exercise 2.5, Question 5: (iii) $\log_5 343 \times \log_7 25 = 6$

Answer: $\log_5 343 = \log_5(7^3) = 3\log_5 7$; $\log_7 25 = 2\log_7 5 = 2/\log_5 7$.Product =

= 3×2

= 6. Verified.

Question: Exercise 2.5, Question 5: (iv) $\log_6 16 \times \log_2 216 = 12$

Answer: $\log_6 16 = \log_6(2^4) = 4\log_6 2$; $\log_2 216 = \log_2(6^3) = 3\log_2 6 = 3/\log_6 2$.Product =

= 4×3

= 12. Verified.

Question: Exercise 2.6, Question 1: Find the values of the following using logarithms. (i) 57.86 × 4.385

Answer: Let x = 57.86 × 4.385.log x =

= log 57.86 + log 4.385

= 1.7624 + 0.6420

= 2.4044 ⇒ x

= antilog(2.4044)

= 253.7.

Question: Exercise 2.6, Question 1: (ii) 25.753 × 0.5341 × 490.8

Answer: log x =

= log 25.753 + log 0.5341 + log 490.8

= 1.4108 + (−0.2724) + 2.6909

= 3.8293 ⇒ x

= antilog(3.8293)

= 6750.8.

Question: Exercise 2.6, Question 1: (iii) 25.753/0.5341

Answer: log x =

= log 25.753 − log 0.5341

= 1.4108 − (−0.2724)

= 1.6832 ⇒ x

= antilog(1.6832)

= 48.22.

Question: Exercise 2.6, Question 1: (iv) (790.6 × 30.32)/25.753

Answer: log x =

= log 790.6 + log 30.32 − log 25.753

= 2.8980 + 1.4817 − 1.4108

= 2.9689 ⇒ x

= antilog(2.9689)

= 930.8.

Question: Exercise 2.6, Question 1: (v) 99.87/((8.369)×(0.785))

Answer: log x =

= log 99.87 − log 8.369 − log 0.785

= 1.9994 − 0.9227 − (−0.1051)

= 1.1818 ⇒ x

= antilog(1.1818)

= 15.20.

Question: Exercise 2.6, Question 1: (vi) $\sqrt[5]{2.709} \times \sqrt[3]{1.239}$

Answer: log x =

= (1/5)log 2.709 + (1/3)log 1.239

= 0.4328/5 + 0.0931/3

= 0.0866 + 0.0310

= 0.1176 ⇒ x

= antilog(0.1176)

= 1.3109.

Question: Exercise 2.6, Question 1: (vii) (26.62)^(1/2) × $(87.19)^3/\sqrt{69.53}$

Answer: log x = (1/2)log 26.62 + 3log 87.19 − (1/2)log 69.53 = 0.7126 + 5.8215 − 0.9211 = 5.6130 ⇒ x = antilog(5.6130) = 410,200.

Question: Exercise 2.6, Question 1: (viii) $(4308)^3 \times \sqrt{80.06}/(0.3387)^3$

Answer: log x = 3log 4308 + (1/2)log 80.06 − 3log 0.3387 = 10.9029 + 0.9517 + 1.4106 = 13.2652 ⇒ x = antilog(13.2652) = 1.842×10^{13}.

Question: Exercise 2.6, Question 2: Find the number of digits in the following. (i) 4^{12}

Answer: Number of digits =

= [12 × log 4] + 1

= [12 × 0.6021] + 1

= [7.2252] + 1

= 7 + 1

= 8.

Question: Exercise 2.6, Question 2: (ii) 7^{25}

Answer: Number of digits =

= [25 × log 7] + 1

= [25 × 0.8451] + 1

= [21.1275] + 1

= 21 + 1

= 22.

Question: Exercise 2.6, Question 2: (iii) 3^{30}

Answer: Number of digits =

= [30 × log 3] + 1

= [30 × 0.4771] + 1

= [14.3130] + 1

= 14 + 1

= 15.

Question: Exercise 2.6, Question 2: (iv) 5^{20}

Answer: Number of digits =

= [20 × log 5] + 1

= [20 × 0.6990] + 1

= [13.9800] + 1

= 13 + 1

= 14.

Question: Exercise 2.6, Question 2: (v) 9^{30}

Answer: Number of digits =

= [30 × log 9] + 1

= [30 × 0.9542] + 1

= [28.6260] + 1

= 28 + 1

= 29.

Question: Review Exercise 2, Question 1: Read the following statements carefully and circle T for a true statement and F for a false statement. (i) The scientific notation of 0.025 is 2.5×10^3.

Answer: F — the scientific notation of 0.025 is 2.5×10^{-2}.

Question: Review Exercise 2, Question 1: (ii) Al-Biruni introduced the logarithm.

Answer: F — according to the book's introduction, Al-Khwarizmi introduced the logarithm (and John Napier later developed it), not Al-Biruni.

Question: Review Exercise 2, Question 1: (iii) The whole-number part of the logarithm of a number is called its characteristic.

Answer: T — the whole (integer) part of a logarithm is the characteristic.

Question: Review Exercise 2, Question 1: (iv) The mantissa of the logarithm of a number can be non-negative.

Answer: T — the mantissa is always non-negative (0 or greater), never negative.

Question: Review Exercise 2, Question 1: (v) $\log_a x = y \Leftrightarrow a^y = x$.

Answer: T — this is the basic relationship between a logarithm and a power.

Review Exercise 2, Question 2: Fill in the following blanks. (i) A logarithm whose base is 10 is called _____.

Answer: Common logarithm.

Review Exercise 2, Question 2: (ii) log 1 = _____ 1. _____ .

Review Exercise 2, Question 2: (iii) The decimal part of a logarithm is called _____.

Answer: Mantissa.

Review Exercise 2, Question 2: (iv) log 512 = _____.

Answer: 9log 2 = 9 × 0.3010 = 2.7090.

Review Exercise 2, Question 2: (v) $\log_a m \times \log_m n$ = _____.

Answer: $\log_a n$ (by the change-of-base rule).

Review Exercise 2, Question 2: (vi) The exponential form of $x = \log_a y$ is _____.

Answer: y.

Review Exercise 2, Question 2: (vii) The logarithmic form of a^10 = y is _____.

Answer: 10.

Review Exercise 2, Question 2: (viii) log_b a × log_a b = _____ 1/log_b a. _____ .

Review Exercise 2, Question 2: (ix) $\log_a(m/n) =$ $\log_a m - \log_a n$.

Review Exercise 2, Question 2: (x) log(10 × 10) = _____.

Answer: log 10 + log 10 = 1 + 1 = 2.

Review Exercise 2, Question 3: Tick () the correct answer. (i) If $\log_{10} x = 4$ then x = _____.

(a) 500

(b) 100

(c) 1000

✓ (d) 10000

Review Exercise 2, Question 3: (ii) The characteristic of log 54.58 is _____.

(a) 0

✓ (b) 1

(c) 2

(d) 4

Review Exercise 2, Question 3: (iii) The base of the common logarithm is _____.

(a) 5

✓ (b) 10

(c) e

(d) 100

Review Exercise 2, Question 3: (iv) log xyz = _____.

(a) logx logy logz

(b) logx + logy + logz

(d) logx − logy − logz

Review Exercise 2, Question 3: (v) The scientific notation of 0.00789 is:

✓ (a) 7.89×10^{-3}

(b) 7.89×10^{3}

(c) 0.789×10^{-2}

(d) 78.9×10^{-4}

Review Exercise 2, Question 3: (vi) If log x = 2 then x = _____.

(a) 200

(b) 1000

✓ (c) 100

(d) 2/10

Review Exercise 2, Question 3: (vii) If $\log_2 8 = x$ then x = _____.

(a) 64

✓ (b) 3

(c) 3

(d) 2

Review Exercise 2, Question 3: (viii) The base of the natural logarithm is _____.

(a) 10

✓ (b) e

(c) π

(d) 5

✓

Review Exercise 2, Question 3: (ix) $3^5 = 243$ written in logarithmic form is _____.

(a) $\log_3 5 = 243$

(b) $\log_3 243 = 5$

(c) $\log_5 243 = 12$

(d) $\log_5 3 = 243$

3. Algebraic Expression and Formulas

Question: Exercise 3.1, Question 1: State whether the following algebraic expressions are polynomials or not. (i) $2xy^2 - 3x^2 + 5y^3 - 6$

Answer: It is a polynomial, because all the exponents of the variables are non-negative integers.

Question: Exercise 3.1, Question 1: (ii) $3xy^{-2}$

Answer: It is not a polynomial, because the exponent of y is −2, which is negative.

Question: Exercise 3.1, Question 1: (iii) $6x^2 - 10x + 7 - \sqrt{45}$

Answer: It is a polynomial, because √45 is a constant and all exponents of x are non-negative integers.

Question: Exercise 3.1, Question 1: (iv) $5\sqrt{x} - x + 5x^2$

Answer: It is not a polynomial, because in √x = x^(1/2) the exponent of x is 1/2, which is a fraction.

Question: Exercise 3.1, Question 1: (v) 2/(x+2)

Answer: It is not a polynomial, because the variable is in the denominator.

Question: Exercise 3.1, Question 1: (vi) $2/x + x^3 - 2$

Answer: It is not a polynomial, because in $2/x = 2x^{-1}$ the exponent of x is −1, which is negative.

Question: Exercise 3.1, Question 2: State whether the following algebraic

expressions are rational or not. (i) $(x^2+2x+3)/(x-4)$

Answer: It is rational, because it is a ratio of two polynomials.

Question: Exercise 3.1, Question 2: (ii) $(x^2+5\sqrt{x}-2x)/(3x^2+5x+4)$

Answer: It is not rational, because the numerator contains $\sqrt{x}$, which is not part of a polynomial.

Question: Exercise 3.1, Question 2: (iii) $(13x^2-9x+4)/(x^2+5x+\sqrt{7})$

Answer: It is rational, because $\sqrt{7}$ is a constant and both parts are polynomials.

Question: Exercise 3.1, Question 2: (iv) $(\sqrt{x}-\sqrt{y})/(\sqrt{x}+\sqrt{y})$

Answer: It is not rational, because $\sqrt{x}$ and $\sqrt{y}$ are not polynomials.

Question: Exercise 3.1, Question 2: (v) $7/(x+7)$

Answer: It is rational, because it is a ratio of two polynomials.

Question: Exercise 3.1, Question 2: (vi) $5\sqrt{x} - x + 5x^2$

Answer: It is not rational, because it contains $\sqrt{x}$, which is not part of a polynomial.

Question: Exercise 3.1, Question 3: Simplify the following. (i) $(p^2-100)/(p+10)$

Answer:

Formula: $a^2 - b^2 = (a + b)(a - b)$

$(p^2-100)/(p+10)$ =

= (p−10)(p+10)/(p+10)

= p−10.

Question: Exercise 3.1, Question 3: (ii) $(3a^2+3ab)/(3a^2+6ab+3b^2)$

Answer:

Formula: $(a + b)^2 = a^2 + 2ab + b^2$

= $3a(a+b)/(3(a^2+2ab+b^2))$ =

= $3a(a+b)/(3(a+b)^2)$

= a/(a+b).

Question: Exercise 3.1, Question 3: (iii) $(a−b)/(a+b) \times (a^2+ab)/(2a^2−2b^2)$

Answer:

Formula: $a^2 − b^2 = (a + b)(a − b)$

= $(a−b)/(a+b) \times a(a+b)/(2(a−b)(a+b)) = a/(2(a+b))$.

Question: Exercise 3.1, Question 3: (iv) $((x+y)^2−z^2)/(x+y+z)$

Answer:

Formula: $a^2 − b^2 = (a + b)(a − b)$ (with a = x + y, b = z)

= (x+y−z)(x+y+z)/(x+y+z) = x+y−z.

Question: Exercise 3.1, Question 3: (v) (m²−6m)(3m+15)/(2m−12)

Answer: = m(m−6)·3(m+5)/(2(m−6)) = 3m(m+5)/2.

Question: Exercise 3.1, Question 3: (vi) (x²−2x−3)/(x²−x−2)

Answer: = (x−3)(x+1)/((x−2)(x+1)) = (x−3)/(x−2).

Question: Exercise 3.1, Question 4: Simplify the following. (i) (4x−1)/(2x−2) + (4x+1)/(2x+2)

Answer: = [(4x−1)(2x+2) + (4x+1)(2x−2)]/(4x²−4) = [(8x²+6x−2) + (8x²−6x−2)]/(4x²−4) = (16x²−4)/(4x²−4) = (4x²−1)/(x²−1).

Question: Exercise 3.1, Question 4: (ii) 1/(x+2) + 2/(x+3)

Answer: = (x+3 + 2x+4)/((x+2)(x+3)) = (3x+7)/((x+2)(x+3)).

Question: Exercise 3.1, Question 4: (iii) xy/(xy+1) + (xy+1)/(xy−1)

Answer: = [xy(xy−1) + (xy+1)²]/((xy+1)(xy−1)) = (x²y²−xy + x²y²+2xy+1)/(x²y²−1) = (2x²y²+xy+1)/(x²y²−1).

Question: Exercise 3.1, Question 4: (iv) (x−2)/(x+3) − (x+1)/(x+6)

Answer: = [(x−2)(x+6) − (x+1)(x+3)]/((x+3)(x+6)) = [(x²+4x−12) − (x²+4x+3)]/((x+3)(x+6)) = −15/((x+3)(x+6)).

Question: Exercise 3.1, Question 4: (v) 1/(a+b) − 1/(a−b)

Answer: = (a−b − (a+b))/(a²−b²) = −2b/(a²−b²) = 2b/(b²−a²).

Question: Exercise 3.1, Question 4: (vi) 4y/(y²−1) − (y+1)/(y−1)

Answer: = [4y − (y+1)²]/((y−1)(y+1)) = (4y − y²−2y−1)/((y−1)(y+1)) = −(y−1)²/((y−1)(y+1)) = −(y−1)/(y+1) = (1−y)/(y+1).

Question: Exercise 3.1, Question 5: Simplify. (i) (x²/(4y²−x²) + 1) ÷ (1 − x/(2y))

Answer: First bracket: (x²+4y²−x²)/(4y²−x²) = 4y²/((2y−x)(2y+x)); second bracket: (2y−x)/(2y).Division =

= [4y²/((2y−x)(2y+x))] × [2y/(2y−x)]

= 8y³/((2y+x)(2y−x)²).

Question: Exercise 3.1, Question 5: (ii) (x+3)/(3y−2x) · (4x²−9y²)/(xy+3y)

Answer: = (x+3)/(3y−2x) × (2x−3y)(2x+3y)/(y(x+3)); 3y−2x = −(2x−3y), so = −(2x+3y)/y.

Question: Exercise 3.1, Question 5: (iii) (x²−1)/(x²+2x+1) × (x+1)/(x−1)

Answer: = (x−1)(x+1)/(x+1)² × (x+1)/(x−1) = 1.

Question: Exercise 3.1, Question 5: (iv) 8(y+3)/9 × 12(y+1)/(4(y+3)) ÷ 8(y+1)/5

Answer: First two fractions multiplied: (8×12(y+1))/(9×4) = (8/3)(y+1); then division: (8/3)(y+1) × 5/(8(y+1)) = 5/3.

Question: Exercise 3.1, Question 5: (v) (q²−25)/(q²−3q) ÷ (q²+5q)/(q²−9)

Answer: = (q−5)(q+5)/(q(q−3)) × (q−3)(q+3)/(q(q+5)) = (q−5)(q+3)/q².

Question: Exercise 3.1, Question 5: (vi) 4/(z²−4z−5) ÷ 2/(4z²−4)

Answer: = $4/((z-5)(z+1)) \times 4(z^2-1)/2 = 4\times4(z-1)(z+1)/(2(z-5)(z+1)) = 8(z-1)/(z-5)$.

Question: Exercise 3.1, Question 6: Find the value of $t + 1/t$, when $t = (x-y)/(x+y)$.

Answer: $t + 1/t =$

$= (x-y)/(x+y) + (x+y)/(x-y)$

$= [(x-y)^2 + (x+y)^2]/((x+y)(x-y))$

Formulas: $(x-y)^2 + (x+y)^2 = 2(x^2+y^2)$ and $(x+y)(x-y) = x^2-y^2$

$= 2(x^2+y^2)/(x^2-y^2)$.

Question: Exercise 3.1, Question 7: Find the values of the following. (i) $5(x+y)/(3x^2\sqrt{(y+6)})$, if $x = -4$, $y = 9$

Answer: $= 5(-4+9)/(3\times16\times\sqrt{15}) =$

$= 25/(48\sqrt{15})$

$= 25\sqrt{15}/720$

$= 5\sqrt{15}/144$.

Question: Exercise 3.1, Question 7: (ii) $42ab^2c^3/(3a^2b+1)$, if $a = 3$, $b = 2$

and c = 1

Answer: = 42×3×4×1/(3×9×2+1) = 504/55.

Question: Exercise 3.1, Question 7: (iii) $((x+y)^3-z^2)/(x^2y^2+z^2)$, if x = 2, y = −4 and z = 3

Answer: = $((-2)^3-9)/(4\times16+9)$ =

= (−8−9)/73

= −17/73.

Question: Exercise 3.1, Question 7: (iv) $3x^2y/z - bc/(x+1)$, if x = 2, y = −1, z = 3, b = 4, c = 1/3

Answer: = 3×4×(−1)/3 − (4×(1/3))/(2+1) = −4 − 4/9 = −40/9.

Question: Exercise 3.1, Question 7: (v) $(ab^2-c)/(a+cd^2) \times (c+d)/(a^2b-d)$, if a = 1, b = 3, c = −3 and d = 2

Answer: = (9−(−3))/(1+(−3)×4) × (−3+2)/(3−2) = (12/−11) × (−1/1) = 12/11.

Question: Exercise 3.2, Question 1: Find the values of a^2+b^2 and ab, when a+b = 8 and a−b = 6.

Answer:

Formulas: $a^2 + b^2 = ((a + b)^2 + (a - b)^2) / 2$ and $ab = ((a + b)^2 - (a - b)^2) / 4$

$a^2+b^2 = ((a+b)^2+(a-b)^2)/2 = (64+36)/2 = 50$; $ab = ((a+b)^2-(a-b)^2)/4$ =

$(64-36)/4 = 7$.

Question: Exercise 3.2, Question 2: Find the values of a^2+b^2 and ab, when $a+b = 5$ and $a-b = 3$.

Answer:

Formulas: $a^2 + b^2 = ((a + b)^2 + (a - b)^2) / 2$ and $ab = ((a + b)^2 - (a - b)^2) / 4$

$a^2+b^2 = (25+9)/2 = 17$; $ab = (25-9)/4 = 4$.

Question: Exercise 3.2, Question 3: Find the value of $a^2+b^2+c^2$, when $a+b+c = 9$ and $ab+bc+ac = 13$.

Answer:

Formula: $(a + b + c)^2 = a^2 + b^2 + c^2 + 2(ab + bc + ca)$

$a^2+b^2+c^2 =$

$= (a+b+c)^2 - 2(ab+bc+ac)$

$= 81 - 26$

$= 55$.

Question: Exercise 3.2, Question 4: Find the value of $a^2+b^2+c^2$, when $a+b+c = 1/3$ and $ab+bc+ac = -2/9$.

Answer:

Formula: $(a + b + c)^2 = a^2 + b^2 + c^2 + 2(ab + bc + ca)$

$a^2+b^2+c^2 =$

$= (1/3)^2 - 2(-2/9)$

$= 1/9 + 4/9$

$= 5/9.$

Question: Exercise 3.2, Question 5: Find the value of a+b+c, when $a^2+b^2+c^2 = 29$ and ab+bc+ac = 10.

Answer:

Formula: $(a + b + c)^2 = a^2 + b^2 + c^2 + 2(ab + bc + ca)$

$(a+b+c)^2 =$

$= 29 + 2\times10$

$= 49 \Rightarrow a+b+c$

$= \pm7$ (positive value 7).

Question: Exercise 3.2, Question 6: Find the value of a+b+c, when $a^2+b^2+c^2 = 0.9$ and ab+bc+ac = 0.8.

Answer:

Formula: $(a + b + c)^2 = a^2 + b^2 + c^2 + 2(ab + bc + ca)$

$(a+b+c)^2 =$

$= 0.9 + 2\times0.8$

$= 2.5 \Rightarrow a+b+c$

$= \pm\sqrt{2.5} \approx \pm1.58.$

Question: Exercise 3.2, Question 7: Find the value of ab+bc+ac, when a+b+c = 10 and $a^2+b^2+c^2 = 20$.

Answer:

Formula: $(a + b + c)^2 = a^2 + b^2 + c^2 + 2(ab + bc + ca)$

$ab+bc+ac =$

$= ((a+b+c)^2 - (a^2+b^2+c^2))/2$

$= (100-20)/2$

$= 40.$

Question: Exercise 3.2, Question 8: Find the value of a^3+b^3, when a+b = 4 and ab = 3.

Answer:

Formula: $a^3 + b^3 = (a + b)^3 - 3ab(a + b)$

$a^3+b^3 =$

$= (a+b)^3 - 3ab(a+b)$

$= 64 - 3\times3\times4$

$= 64 - 36$

$= 28.$

Question: Exercise 3.2, Question 9: Find the value of ab, when $a^3-b^3 = 16$ and $a-b = 4$.

Answer:

Formula: $a^3 - b^3 = (a - b)^3 + 3ab(a - b)$

$a^3-b^3 =$

$= (a-b)^3 + 3ab(a-b) \Rightarrow 16$

$= 64 + 12ab \Rightarrow 12ab$

$= -48 \Rightarrow ab$

$= -4.$

Question: Exercise 3.2, Question 10: Find the value of ab, when $a^3-b^3 = 5$ and $a-b = 5$.

Answer:

Formula: $a^3 - b^3 = (a - b)^3 + 3ab(a - b)$

$a^3-b^3 =$

$= (a-b)^3 + 3ab(a-b) \Rightarrow 5$

$= 125 + 15ab \Rightarrow 15ab$

$= -120 \Rightarrow ab$

$= -8.$

Question: Exercise 3.2, Question 11: Find the value of a^3-b^3, when $a-b = 5$ and $ab = 7$.

Answer:

Formula: $a^3 - b^3 = (a - b)^3 + 3ab(a - b)$

$a^3-b^3 =$

$= (a-b)^3 + 3ab(a-b)$

$= 125 + 3\times7\times5$

= 125 + 105

= 230.

Question: Exercise 3.2, Question 12: Find the value of $125x^3+y^3$, when $5x+y = 13$ and $xy = 10$.

Answer:

Formula: $a^3 + b^3 = (a + b)^3 - 3ab(a + b)$, applied to $5x$ and y

$(5x)^3+y^3 =$

$= (5x+y)^3 - 3(5x)(y)(5x+y)$

$= 2197 - 3\times50\times13$

$= 2197 - 1950$

$= 247.$

Question: Exercise 3.2, Question 13: Find the value of $216a^3-343b^3$, when $6a-7b = 11$ and $ab = 8$.

Answer:

Formula: $a^3 - b^3 = (a - b)^3 + 3ab(a - b)$, applied to $6a$ and $7b$

$(6a)^3-(7b)^3 =$

= $(6a-7b)^3 + 3(6a)(7b)(6a-7b)$

= 1331 + 3×42×8×11

= 1331 + 11088

= 12419.

Question: Exercise 3.2, Question 14: Find the value of x^3+1/x^3, when $x+1/x = 7$.

Answer:

Formula: $x^3 + 1/x^3 = (x + 1/x)^3 - 3(x + 1/x)$

x^3+1/x^3 =

= $(x+1/x)^3 - 3(x+1/x)$

= 343 − 21

= 322.

Question: Exercise 3.2, Question 15: Find the value of x^3-1/x^3, when $x-1/x = 11$.

Answer:

Formula: $x^3 - 1/x^3 = (x - 1/x)^3 + 3(x - 1/x)$

$x^3 - 1/x^3$ =

= $(x - 1/x)^3 + 3(x - 1/x)$

= 1331 + 33

= 1364.

Question: Exercise 3.2, Question 16: State whether the following algebraic expressions are polynomials or not. (i) $(3b/2 + 2/(3b))(9b^2/4 + 4/(9b^2) - 1)$

Answer: It is not a polynomial, because it contains terms in 1/b. Multiplying gives = $(3b/2)^3 + (2/(3b))^3 = 27b^3/8 + 8/(27b^3)$, which also has a negative power.

Question: Exercise 3.2, Question 16: (ii) $(7y^2/9 + 9/(7y^2))(49y^4/81 + 81/(49y^4) - 1)$

Answer:

Formula: $a^3 + b^3 = (a + b)(a^2 - ab + b^2)$

It is not a polynomial, because it contains terms in $1/y^2$. Multiplying gives = $(7y^2/9)^3 + (9/(7y^2))^3 = 343y^6/729 + 729/(343y^6)$.

Question: Exercise 3.2, Question 16: (iii) $(x^4/12 + 12/x^4)(x^8/144 + 144/x^8 + 1)$

Answer:

Formula: $a^3 + b^3 = (a + b)(a^2 - ab + b^2)$

It is not a polynomial, because it contains terms in $1/x^4$. Multiplying gives

$= (x^4/12)^3 + (12/x^4)^3 = x^{12}/1728 + 1728/x^{12}$.

Question: Exercise 3.2, Question 16: (iv) $(c^2-1/c^2)(c^4+1/c^4+1)$

Answer:

Formula: $a^3 - b^3 = (a - b)(a^2 + ab + b^2)$

It is not a polynomial, because it contains terms in $1/c^2$. Multiplying gives

$= (c^2)^3 - (1/c^2)^3 = c^6 - 1/c^6$.

Question: Exercise 3.2, Question 17: Find the continued product with the help of a suitable formula. (i) $(2x^2+3y^2)(4x^4-6x^2y^2+9y^4)$

Answer:

Formula: $(a + b)(a^2 - ab + b^2) = a^3 + b^3$

$= (2x^2)^3 + (3y^2)^3$

$= 8x^6 + 27y^6$.

Question: Exercise 3.2, Question 17: (ii) $(2x^2-3y^2)(4x^4+6x^2y^2+9y^4)$

Answer:

Formula: $(a - b)(a^2 + ab + b^2) = a^3 - b^3$

$= (2x^2)^3 - (3y^2)^3$

$= 8x^6 - 27y^6$.

Question: Exercise 3.2, Question 17: (iii)

$(x-y)(x+y)(x^2+y^2)(x^2+xy+y^2)(x^2-xy+y^2)(x^4+x^2y^2+y^4)$

Answer:

Formula: $a^2 - b^2 = (a + b)(a - b)$

$(x-y)(x^2+xy+y^2) = x^3-y^3$; $(x+y)(x^2-xy+y^2) = x^3+y^3$; $(x^2+y^2)(x^4+x^2y^2+y^4)$ $= x^6+y^6$. Product $= (x^3-y^3)(x^3+y^3)(x^6+y^6) = (x^6-y^6)(x^6+y^6) = x^{12}-y^{12}$.

Question: Exercise 3.2, Question 17: (iv)

$(2x+3y)(2x-3y)(4x^2+9y^2)(16x^4+81y^4)$

Answer:

Formula: $a^2 - b^2 = (a + b)(a - b)$

$(2x+3y)(2x-3y) = 4x^2-9y^2$; $(4x^2-9y^2)(4x^2+9y^2) = 16x^4-81y^4$; $(16x^4-81y^4)(16x^4+81y^4) = 256x^8-6561y^8$.

Question: Exercise 3.3, Question 1: Simplify. (i) $\sqrt[4]{(81x^{-8}z^4)}$

Answer: $\sqrt[4]{(81x^{-8}z^4)} =$

$= \sqrt[4]{81} \times \sqrt[4]{(x^{-8})} \times \sqrt[4]{(z^4)}$

$= 3 \times x^{-2} \times z$

= $3z/x^2$.

Question: Exercise 3.3, Question 1: (ii) $\sqrt[3]{(256a^6b^{12}c^9)}$

Answer: = $\sqrt[3]{256} \times a^2b^4c^3 = \sqrt[3]{(64\times4)} \times a^2b^4c^3 = 4\sqrt[3]{4}\cdot a^2b^4c^3$.

Question: Exercise 3.3, Question 1: (iii) $\sqrt[7]{128}$

Answer: $\sqrt[7]{128}$ =

= $\sqrt[7]{(2^7)}$

= 2.

Question: Exercise 3.3, Question 1: (iv) $\sqrt{7776}$

Answer: $\sqrt{7776} = \sqrt{(1296\times6)} = 36\sqrt{6}$, because $36^2 = 1296$.

Question: Exercise 3.3, Question 1: (v) $\sqrt[3]{((125)^2\times8)}/\sqrt{((2\times32)^2)}$

Answer: = $\sqrt[3]{(15625\times8)}/64 = \sqrt[3]{125000}/64 = 50/64 = 25/32$, because 50^3 = 125000.

Question: Exercise 3.3, Question 1: (vi) $\sqrt{21} \times \sqrt{28}/\sqrt{121}$

Answer: = $\sqrt{(21\times28)}/11 = \sqrt{588}/11 = 14\sqrt{3}/11$, because $588 = 196\times3$.

Question: Exercise 3.3, Question 1: (vii) $\sqrt{(((216)\text{^}(2/3)\times(125)^2)/(0.04)^{-3})}$

Answer: (216)^(2/3) = 36; $(0.04)^{-3} = 25^3 = 15625$. The inside = $(36\times15625)/15625 = 36$, so the square root is 6.

Question: Exercise 3.3, Question 1: (viii) $(\sqrt[6]{4} \times \sqrt[3]{27} \times \sqrt{60})/(\sqrt{180} \times \sqrt[3]{0.25} \times \sqrt[4]{9})$

Answer: Numerator = 2^(1/3)×3×2√15 = 6·2^(1/3)√15; denominator = 6√5×(1/2^(2/3))×√3 = 6√15/2^(2/3).Quotient =

= 2^(1/3+2/3)

= 2.

Question: Exercise 3.3, Question 2: Find the conjugate of the following. (i) (8−4√3)

Answer: 8+4√3 (the sign of the second term is reversed).

Question: Exercise 3.3, Question 2: (ii) (6√6+2√3)

Answer:

Conjugate of (a + b) is (a − b): only the sign of the second term changes

6√6−2√3.

Question: Exercise 3.3, Question 2: (iii) (8√12+√8)

Answer:

Conjugate of (a + b) is (a − b): only the sign of the second term changes

8√12−√8.

Question: Exercise 3.3, Question 2: (iv) (2−√3)

Answer:

Conjugate of (a + b) is (a − b): only the sign of the second term changes

2+√3.

Question: Exercise 3.3, Question 3: Simplify. (i) (6√2+4√2+7√128)

Answer: = 10√2 + 7×8√2 = 10√2 + 56√2 = 66√2.

Question: Exercise 3.3, Question 3: (ii) √5+√125+7√5

Answer: = √5 + 5√5 + 7√5 = 13√5.

Question: Exercise 3.3, Question 3: (iii) (13+15√3)+(7−6√3)

Answer: = (13+7) + (15−6)√3 = 20+9√3.

Question: Exercise 3.3, Question 3: (iv) √250+√490+3√10

Answer: = 5√10 + 7√10 + 3√10 = 15√10.

Question: Exercise 3.3, Question 3: (v) √245+√625−√45

Answer: = 7√5 + 25 − 3√5 = 25+4√5.

Question: Exercise 3.3, Question 3: (vi) 10√11−√396−3√11

Answer: = 10√11 − 6√11 − 3√11 = √11.

Question: Exercise 3.3, Question 3: (vii) √17(10√17−2√17)

Answer: = √17 × 8√17 = 8×17 = 136.

Question: Exercise 3.3, Question 3: (viii) (3/2)(√18+√32−√50)

Answer: = (3/2)(3√2+4√2−5√2) =

= (3/2)(2√2)

= 3√2.

Question: Exercise 3.3, Question 3: (ix) (√2/√3+1/√3)(√2/√3−1/√3)

Answer:

Formula: (a + b)(a − b) = a² − b²

= 2/3 − 1/3

= 1/3.

Question: Exercise 3.3, Question 3: (x) (√13+√11)(√13−√11)

Answer:

Formula: a² − b² = (a + b)(a − b)

= (√13)² − (√11)² = 13 − 11 = 2.

Question: Exercise 3.3, Question 3: (xi) (3√6−4√5)²

Answer:

Formula: (a − b)² = a² − 2ab + b²

= (3√6)² + (4√5)² − 2(3√6)(4√5) = 54 + 80 − 24√30 = 134−24√30.

Question: Exercise 3.3, Question 3: (xii) (2√3+3√2)²

Answer:

Formula: $(a + b)^2 = a^2 + 2ab + b^2$

$= (2\sqrt{3})^2 + (3\sqrt{2})^2 + 2(2\sqrt{3})(3\sqrt{2}) = 12 + 18 + 12\sqrt{6} = 30+12\sqrt{6}$.

Question: Exercise 3.4, Question 1: Rationalize the denominator of the following. (i) $1/(2+\sqrt{3})$

Answer: $= (2-\sqrt{3})/((2+\sqrt{3})(2-\sqrt{3})) =$

$= (2-\sqrt{3})/(4-3)$

$= 2-\sqrt{3}$ (multiplied by the conjugate).

Question: Exercise 3.4, Question 1: (ii) $1/(3+2\sqrt{2})$

Answer:

Formula: $(a + b)(a - b) = a^2 - b^2$ (multiply by the conjugate)

$= (3-2\sqrt{2})/((3+2\sqrt{2})(3-2\sqrt{2})) =$

$= (3-2\sqrt{2})/(9-8)$

$= 3-2\sqrt{2}$.

Question: Exercise 3.4, Question 1: (iii) $1/(4\sqrt{3}-5\sqrt{2})$

Answer:

Formula: $(a + b)(a - b) = a^2 - b^2$ (multiply by the conjugate)

$= (4\sqrt{3}+5\sqrt{2})/((4\sqrt{3})^2-(5\sqrt{2})^2) =$

$= (4\sqrt{3}+5\sqrt{2})/(48-50)$

$= -(4\sqrt{3}+5\sqrt{2})/2.$

Question: Exercise 3.4, Question 1: (iv) $16/(\sqrt{12}+\sqrt{11})$

Answer:

Formula: $(a + b)(a - b) = a^2 - b^2$ (multiply by the conjugate)

$= 16(\sqrt{12}-\sqrt{11})/(12-11) =$

$= 16(2\sqrt{3}-\sqrt{11})$

$= 32\sqrt{3}-16\sqrt{11}.$

Question: Exercise 3.4, Question 1: (v) $(9-\sqrt{2})/(9+\sqrt{2})$

Answer:

Formula: $(a + b)(a - b) = a^2 - b^2$ (multiply by the conjugate)

$= (9-\sqrt{2})^2/((9+\sqrt{2})(9-\sqrt{2})) =$

$= (81+2-18\sqrt{2})/(81-2)$

= (83−18√2)/79.

Question: Exercise 3.4, Question 1: (vi) (√13+3)/(√13−3)

Answer:

Formula: (a + b)(a − b) = a² − b² (multiply by the conjugate)

= (√13+3)²/((√13−3)(√13+3)) =

= (13+9+6√13)/(13−9)

= (22+6√13)/4

= (11+3√13)/2.

Question: Exercise 3.4, Question 2: (i) If x = 8−3√7, find (x+1/x)².

Answer:

Formula: (a + b)(a − b) = a² − b² (multiply by the conjugate)

1/x = 1/(8−3√7) = (8+3√7)/(64−63) = 8+3√7; x+1/x = 16; (x+1/x)² = 256.

Question: Exercise 3.4, Question 2: (ii) If 1/x = 2√28−11, find x.

Answer: 1/x =

= 4√7−11 ⇒ x

= $1/(4\sqrt{7}-11)$

= $(4\sqrt{7}+11)/((4\sqrt{7})^2-121)$

= $(4\sqrt{7}+11)/(112-121)$

= $-(11+4\sqrt{7})/9$.

Question: Exercise 3.4, Question 2: (iii) If $x = 3-2\sqrt{2}$, find $x+1/x$, $x-1/x$, x^2+1/x^2, x^2-1/x^2 and x^4+1/x^4.

Answer:

Formulas: $x^2 + 1/x^2 = (x + 1/x)^2 - 2$, $x^2 - 1/x^2 = (x - 1/x)(x + 1/x)$ and $x^4 + 1/x^4 = (x^2 + 1/x^2)^2 - 2$

$1/x = 3+2\sqrt{2}$. $x+1/x = 6$; $x-1/x = -4\sqrt{2}$; $x^2+1/x^2 = 6^2-2 = 34$; $x^2-1/x^2 = (-4\sqrt{2})(6) = -24\sqrt{2}$; $x^4+1/x^4 = 34^2-2 = 1154$.

Review Exercise 3, Question 1: Circle the correct answer. (i) Every polynomial is:

(a) an irrational expression

✓ (b) a rational expression

(c) a sentence

(d) none of these

Review Exercise 3, Question 1: (ii) An irrational number which is the sum of two monomial irrational expressions is called:

(a) a trinomial surd

(b) a binomial surd

(c) conjugate of surds

(d) a monomial surd

Review Exercise 3, Question 1: (iii) 3x+2y−3 is an algebraic:

✓ (a) expression

(b) equation

(c) sentence

(d) inequality

Review Exercise 3, Question 1: (iv) The degree of $3x^2y+5y^4-10$ is:

✓ (a) 4

(b) 5

(c) 3

(d) 10

Review Exercise 3, Question 1: (v) $\sqrt{7}$ is:

✓ (a) a monomial surd

(b) a trinomial surd

(c) a binomial surd

(d) a conjugate of surds

Review Exercise 3, Question 1: (vi) The quotient p(x)/q(x) of two polynomial expressions p(x) and q(x), where $q(x) \neq 0$, is called:

✓ (a) a rational expression

(b) an irrational expression

(c) a polynomial

(d) a conjugate

✓

Review Exercise 3, Question 1: (vii) $1/(x-y) - 1/(x+y)$ is equal to:

Review Exercise 3, Question 1: (viii) The conjugate of $2-\sqrt{3}$ is:

(a) $2+\sqrt{3}$

(b) $-2-\sqrt{3}$

(c) $\sqrt{2}+3$

(d) $\sqrt{3}-4$

Review Exercise 3, Question 1: (ix) $a^3-3ab(a-b)-b^3$ is equal to:

(c) a^3+b^3

(d) a^3-b^3

Review Exercise 3, Question 1: (x) If $a+b = 5$ and $a-b = 3$ then the value of ab is:

(a) 4

(b) 5

(c) 3

(d) 6

Review Exercise 3, Question 1: (xi) $(5+\sqrt{15})(5-\sqrt{15})$ is equal to:

(a) 10

(b) 15

(c) 25

(d) 30

Review Exercise 3, Question 1: (xii) $a^2+b^2+c^2+2ab+2bc+2ca$ is equal to:

Review Exercise 3, Question 2: Fill in the blanks. (i) The degree of any

polynomial is _____.

Answer: the variable.

Review Exercise 3, Question 2: (ii) The conjugate of the surd expression $2-\sqrt{3}$ is _____.

Answer: $2+\sqrt{3}$.

Review Exercise 3, Question 2: (iii) The degree of the polynomial expression $2x^3+x^2-4x^4+7x-9$ is _____.

Answer: 4 (the term $-4x^4$).

Review Exercise 3, Question 2: (iv) $\sqrt{x}/(3x+5)$ is a _____ expression, $x \neq -5/3$.

Answer: An irrational (surd)

Review Exercise 3, Question 2: (v) $(x-y)(x+y)(x^2+y^2)$ is equal to _____.

Answer: x^4-y^4, because $(x-y)(x+y) = x^2-y^2$ and $(x^2-y^2)(x^2+y^2) = x^4-y^4$.

4. Factorization

Question: Exercise 4.1, Question 1: Factorize the following. (i) $4x+16y+24z$

Answer: = $4(x+4y+6z)$ (the common factor 4 is taken out of all terms).

Question: Exercise 4.1, Question 1: (ii) $x^2+3x^2y+4x^2y^2z$

Answer: = $x^2(1+3y+4y^2z)$ (the common factor x^2 is taken out).

Question: Exercise 4.1, Question 1: (iii) $3pqr+6pqt+3pqs$

Answer: = $3pq(r+2t+s)$ (the common factor $3pq$ is taken out).

Question: Exercise 4.1, Question 1: (iv) $9qr(s^2+t^2)+18q^2r^2(s^2+t^2)$

Answer: = $9qr(s^2+t^2)(1+2qr)$ (the common factor $9qr(s^2+t^2)$ is taken out).

Question: Exercise 4.1, Question 1: (v) $z^2x/16 - x^2z^2/8 + x^2z^3/12$

Answer: = $xz^2(1/16 - x/8 + xz/12) = (xz^2/48)(3-6x+4xz)$ (the common factor xz^2 is taken out).

Question: Exercise 4.1, Question 1: (vi) $a(x-y)-a^2b(x-y)+a^2b^2(x-y)$

Answer: = $a(x-y)(1-ab+ab^2)$ (the common factor $a(x-y)$ is taken out).

Question: Exercise 4.1, Question 2: Factorize the following. (i) $7x+xz+7z+z^2$

Answer: = $x(7+z)+z(7+z) = (7+z)(x+z)$ (grouped and common factors taken out).

Question: Exercise 4.1, Question 2: (ii) $9a^2b+18ab^2-6ac-12bc$

Answer: = $9ab(a+2b)-6c(a+2b)$ =

= $(a+2b)(9ab-6c)$

= $3(a+2b)(3ab-2c)$.

Question: Exercise 4.1, Question 2: (iii) $6t-12p+4tq-8pq$

Answer: = $6(t-2p)+4q(t-2p)$ =

= $(t-2p)(6+4q)$

= $2(t-2p)(3+2q)$.

Question: Exercise 4.1, Question 2: (iv) $r^2+9rs-7rs-63s^2$

Answer: = $r^2+2rs-63s^2$ = $(r+9s)(r-7s)$ (middle term split: $9s-7s = 2s$, $9\times(-7) = -63$).

Question: Exercise 4.1, Question 2: (v) $y^2/4 - y^2z/4 - z^2t/9 + z^3t/9$

Answer: = $(y^2/4)(1-z) - (z^2t/9)(1-z) = (1-z)(y^2/4 - z^2t/9)$.

Question: Exercise 4.1, Question 2: (vi) $10xy/11 + 5xz/11 - 14y^2/11 - 7yz/11$

Answer: = $(1/11)[5x(2y+z) - 7y(2y+z)] = (2y+z)(5x-7y)/11$.

Question: Exercise 4.1, Question 3: Find the factors. (i) $4a^2+12ab+9b^2$

Answer: = $(2a+3b)^2$, because $(2a)^2+2(2a)(3b)+(3b)^2 = 4a^2+12ab+9b^2$.

Question: Exercise 4.1, Question 3: (ii) $36x^4+12x^2+1$

Answer: = $(6x^2+1)^2$, because $(6x^2)^2+2(6x^2)(1)+1^2 = 36x^4+12x^2+1$.

Question: Exercise 4.1, Question 3: (iii) $x^2+1+1/(4x^2)$

Answer: = $(x+1/(2x))^2$, because $x^2+2(x)(1/(2x))+1/(4x^2) = x^2+1+1/(4x^2)$.

Question: Exercise 4.1, Question 3: (iv) $81y^2+144yz+64z^2$

Answer: = $(9y+8z)^2$, because $(9y)^2+2(9y)(8z)+(8z)^2 = 81y^2+144yz+64z^2$.

Question: Exercise 4.1, Question 3: (v) $625+50a^2b+4a^4b^2$

Answer: = $(25+2a^2b)^2$, because $25^2+2(25)(2a^2b)+(2a^2b)^2 = 625+50a^2b+4a^4b^2$.

Question: Exercise 4.1, Question 3: (vi) $a^2+0.4a+0.04$

Answer: = $(a+0.2)^2$, because $a^2+2(a)(0.2)+(0.2)^2 = a^2+0.4a+0.04$.

Question: Exercise 4.1, Question 4: Factorize. (i) $b^4-4b^2c^2+4c^4$

Answer: = $(b^2-2c^2)^2$, because $(b^2)^2-2(b^2)(2c^2)+(2c^2)^2 = b^4-4b^2c^2+4c^4$.

Question: Exercise 4.1, Question 4: (ii) $(9/4)x^4-2+4/(9x^4)$

Answer: = $(3x^2/2 - 2/(3x^2))^2$, because $(3x^2/2)^2-2(3x^2/2)(2/(3x^2))+(2/(3x^2))^2 = (9/4)x^4-2+4/(9x^4)$.

Question: Exercise 4.1, Question 4: (iii) $2a^3b^3-16a^2b^4+32ab^5$

Answer: = $2ab^3(a^2-8ab+16b^2) = 2ab^3(a-4b)^2$.

Question: Exercise 4.1, Question 4: (iv) $9(p+q)^2-6(p+q)r^2+r^4$

Answer: = $(3(p+q)-r^2)^2$, because $(3(p+q))^2-2(3(p+q))(r^2)+(r^2)^2 =$

$9(p+q)^2-6(p+q)r^2+r^4$.

Question: Exercise 4.1, Question 4: (v) $x^2y^2-0.1xy+0.0025$

Answer: $= (xy-0.05)^2$, because $(xy)^2-2(xy)(0.05)+(0.05)^2 = x^2y^2-0.1xy+0.0025$.

Question: Exercise 4.1, Question 4: (vi) $(a-b)^2-18(a-b)+81$

Answer: $= ((a-b)-9)^2 = (a-b-9)^2$.

Question: Exercise 4.1, Question 5: Factorize. (i) $4a^2-9b^2$

Answer: $= (2a-3b)(2a+3b)$ (difference-of-squares formula: $a^2-b^2 = (a-b)(a+b)$).

Question: Exercise 4.1, Question 5: (ii) $16x^2-25y^2$

Answer:

Formula: $a^2 - b^2 = (a + b)(a - b)$

$16x^2-25y^2 = (4x)^2 - (5y)^2$

$= (4x-5y)(4x+5y)$.

Question: Exercise 4.1, Question 5: (iii) $100x^2z^2-y^4$

Answer:

Formula: $a^2 - b^2 = (a + b)(a - b)$

$100x^2z^2-y^4 = (10xz)^2 - (y^2)^2$

$= (10xz - y^2)(10xz + y^2)$.

Question: Exercise 4.1, Question 5: (iv) $(1/100)x^4 - 100y^4$

Answer:

Formula: $a^2 - b^2 = (a + b)(a - b)$

$(1/100)x^4 - 100y^4 = (x^2/10)^2 - (10y^2)^2$

$= (x^2/10 - 10y^2)(x^2/10 + 10y^2) = (1/100)(x - 10y)(x + 10y)(x^2 + 100y^2)$.

Question: Exercise 4.1, Question 5: (v) $(64/81)f^2 - (81/64)g^4$

Answer:

Formula: $a^2 - b^2 = (a + b)(a - b)$

$(64/81)f^2 - (81/64)g^4 = (8f/9)^2 - (9g^2/8)^2$

$= (8f/9 - 9g^2/8)(8f/9 + 9g^2/8)$.

Question: Exercise 4.1, Question 5: (vi) $x^4/121 - 121y^2$

Answer:

Formula: $a^2 - b^2 = (a + b)(a - b)$

$x^4/121 - 121y^2 = (x^2/11)^2 - (11y)^2$

= $(x^2/11-11y)(x^2/11+11y)$.

Question: Exercise 4.1, Question 6: Find the factors. (i) $(2x+z)^2-(2x-z)^2$

Answer: = $[(2x+z)-(2x-z)][(2x+z)+(2x-z)]$ =

= $(2z)(4x)$

= $8xz$.

Question: Exercise 4.1, Question 6: (ii) $(4a-9b)^2-(2a+5b)^2$

Answer: = $[(4a-9b)-(2a+5b)][(4a-9b)+(2a+5b)]$ =

= $(2a-14b)(6a-4b)$

= $4(a-7b)(3a-2b)$.

Question: Exercise 4.1, Question 6: (iii) $169x^4-(3t+4)^2$

Answer: = $(13x^2-(3t+4))(13x^2+(3t+4))$ = $(13x^2-3t-4)(13x^2+3t+4)$.

Question: Exercise 4.1, Question 6: (iv) $(9x^2-4y^2)^2-(4x^2-y^2)^2$

Answer: = $[(9x^2-4y^2)-(4x^2-y^2)][(9x^2-4y^2)+(4x^2-y^2)]$ =

$(5x^2-3y^2)(13x^2-5y^2)$.

Question: Exercise 4.1, Question 6: (v) $(a^2+2+1/a^2)-(b^2-2+1/b^2)$

Answer: = $(a+1/a)^2-(b-1/b)^2$ = $(a+1/a-b+1/b)(a+1/a+b-1/b)$.

Question: Exercise 4.1, Question 6: (vi) $9x^2+1/(9x^2)-4y^2-1/(4y^2)+4$

Answer: = $(3x+1/(3x))^2 - (2y+1/(2y))^2$ =

$(3x+1/(3x)-2y-1/(2y))(3x+1/(3x)+2y+1/(2y))$, because 4 was split as 2+2.

Question: Exercise 4.1, Question 7: Find the factors. (i) $(x^2+2xy+y^2)-9z^4$

Answer: $= (x+y)^2-(3z^2)^2 = (x+y-3z^2)(x+y+3z^2)$.

Question: Exercise 4.1, Question 7: (ii) $(4a^2+8ab^2+4b^4)-9c^2$

Answer: $= (2a+2b^2)^2-(3c)^2 = (2a+2b^2-3c)(2a+2b^2+3c)$.

Question: Exercise 4.1, Question 7: (iii) $16d^4-(c^4-2c^2d+d^2)$

Answer: $= (4d^2)^2-(c^2-d)^2 = (4d^2-c^2+d)(4d^2+c^2-d)$.

Question: Exercise 4.1, Question 7: (iv) $4(x^2+2xy^2+y^4)-9y^6$

Answer: $= (2x+2y^2)^2-(3y^3)^2 = (2x+2y^2-3y^3)(2x+2y^2+3y^3)$.

Question: Exercise 4.1, Question 7: (v) $x^2-y^2-4x-2y+3$

Answer: $= (x^2-4x+4)-(y^2+2y+1) =$

$= (x-2)^2-(y+1)^2$

$= (x-y-3)(x+y-1)$.

Question: Exercise 4.1, Question 7: (vi) $4x^2-y^2-2y-1$

Answer: $= (2x)^2-(y^2+2y+1) =$

$= (2x)^2-(y+1)^2$

$= (2x-y-1)(2x+y+1)$.

Question: Exercise 4.2, Question 1: Factorize the following. (i) $a^4+a^2x^2+x^4$

Answer: = $(a^4+2a^2x^2+x^4)-a^2x^2 = (a^2+x^2)^2-(ax)^2 = (a^2+ax+x^2)(a^2-ax+x^2)$.

Question: Exercise 4.2, Question 1: (ii) b^4+b^2+1

Answer: = $(b^4+2b^2+1)-b^2 =$

$= (b^2+1)^2-b^2$

$= (b^2+b+1)(b^2-b+1)$.

Question: Exercise 4.2, Question 1: (iii) $a^8+a^4x^4+x^8$

Answer: = $(a^4+x^4)^2-a^4x^4 = (a^4+a^2x^2+x^4)(a^4-a^2x^2+x^4) =$

$(a^4+a^2x^2+x^4)(a^2+ax+x^2)(a^2-ax+x^2)$.

Question: Exercise 4.2, Question 1: (iv) z^8+z^4+1

Answer: = $(z^4+1)^2-z^4 =$

$= (z^4+z^2+1)(z^4-z^2+1)$

$= (z^2+z+1)(z^2-z+1)(z^4-z^2+1)$.

Question: Exercise 4.2, Question 2: Factorize. (i) x^4+4y^4 [note: the printed x^4+4y^2 does not factor over the rationals; read as x^4+4y^4 per this section's $4a^4+b^4$ pattern]

Answer: = $(x^2+2y^2)^2-(2xy)^2 = (x^2+2xy+2y^2)(x^2-2xy+2y^2)$.

Question: Exercise 4.2, Question 2: (ii) $36x^4z^4+9y^4$

Answer: = $9(4x^4z^4+y^4) =$

$= 9[(2x^2z^2+y^2)^2-(2xyz)^2]$

$= 9(2x^2z^2+2xyz+y^2)(2x^2z^2-2xyz+y^2)$.

Question: Exercise 4.2, Question 2: (iii) $4t^4+625$

Answer: $= (2t^2+25)^2-(10t)^2 = (2t^2+10t+25)(2t^2-10t+25)$.

Question: Exercise 4.2, Question 2: (iv) $4t^4+1$

Answer: $= (2t^2+1)^2-(2t)^2 = (2t^2+2t+1)(2t^2-2t+1)$.

Question: Exercise 4.2, Question 3: Factorize. (i) $x^2+3x-10$

Answer: $= (x+5)(x-2)$ (middle term: $5x-2x = 3x$, $5\times(-2) = -10$).

Question: Exercise 4.2, Question 3: (ii) $a^2b^2-3ab-10$

Answer: $= (ab-5)(ab+2)$ (treating ab as one variable: $-5+2 = -3$, $-5\times2 = -10$).

Question: Exercise 4.2, Question 3: (iii) $y^2+7y-98$

Answer: $= (y+14)(y-7)$ ($14-7 = 7$, $14\times(-7) = -98$).

Question: Exercise 4.2, Question 3: (iv) $(xyz)^2+2xyz-24$ [note: the printed first term shows only a superscript 2 before y^2z^2; read as $(xyz)^2$ per the factorization pattern]

Answer: Treating xyz as one variable: $= (xyz+6)(xyz-4)$ ($6-4 = 2$, $6\times(-4) = -24$).

Question: Exercise 4.2, Question 4: Factorize. (i) $9y^2+21yz-8z^2$

Answer: = $9y^2+24yz-3yz-8z^2$ =

= $3y(3y+8z)-z(3y+8z)$

= $(3y+8z)(3y-z)$.

Question: Exercise 4.2, Question 4: (ii) $42x^2-8x-2$

Answer: = $2(21x^2-4x-1)$ =

= $2(21x^2-7x+3x-1)$

= $2(3x-1)(7x+1)$.

Question: Exercise 4.2, Question 4: (iii) $4x^2+12x+5$

Answer: = $4x^2+10x+2x+5$ =

= $2x(2x+5)+1(2x+5)$

= $(2x+5)(2x+1)$.

Question: Exercise 4.2, Question 4: (iv) $3x^2-38xy-13y^2$

Answer: = $3x^2-39xy+xy-13y^2$ =

= $3x(x-13y)+y(x-13y)$

= $(x-13y)(3x+y)$.

Question: Exercise 4.3, Question 1: Factorize the following. (i)

$(x^2-4x-5)(x^2-4x-12)-144$

Answer: Let $x^2-4x = t$: $(t-5)(t-12)-144 = t^2-17t-84 = (t-21)(t+4)$. $t-21 = (x-7)(x+3)$; $t+4 = (x-2)^2$. Result: $(x-7)(x+3)(x-2)^2$.

Question: Exercise 4.3, Question 1: (ii) $(x^2+5x+6)(x^2+5x+4)-3$

Answer: Let $x^2+5x = t$: $(t+6)(t+4)-3 = t^2+10t+21 = (t+7)(t+3) = (x^2+5x+7)(x^2+5x+3)$.

Question: Exercise 4.3, Question 1: (iii) $(x^2-2x+3)(x^2-2x+4)-42$

Answer: Let $x^2-2x = t$: $(t+3)(t+4)-42 = t^2+7t-30 = (t+10)(t-3) = (x^2-2x+10)(x-3)(x+1)$.

Question: Exercise 4.3, Question 1: (iv) $(x^2-8x+4)(x^2-8x-4)+15$

Answer: Let $x^2-8x = t$: $(t+4)(t-4)+15 = t^2-1 = (t-1)(t+1) = (x^2-8x-1)(x^2-8x+1)$.

Question: Exercise 4.3, Question 1: (v) $(x^2+9x-1)(x^2+9x+5)-7$

Answer: Let $x^2+9x = t$: $(t-1)(t+5)-7 = t^2+4t-12 = (t+6)(t-2) = (x^2+9x+6)(x^2+9x-2)$.

Question: Exercise 4.3, Question 1: (vi) $(x^2-5x+4)(x^2-5x+6)-120$

Answer: Let $x^2-5x = t$: $(t+4)(t+6)-120 = t^2+10t-96 = (t+16)(t-6) = (x^2-5x+16)(x-6)(x+1)$.

Question: Exercise 4.3, Question 2: Factorize. (i) $(x+1)(x+2)(x+3)(x+4)-48$

Answer: $(x+1)(x+4) = x^2+5x+4$; $(x+2)(x+3) = x^2+5x+6$. Let $x^2+5x = t$: $(t+4)(t+6)-48 = t^2+10t-24 = (t+12)(t-2) = (x^2+5x+12)(x^2+5x-2)$.

Question: Exercise 4.3, Question 2: (ii) $(x+2)(x+3)(x+4)(x+5)-24$

Answer: $(x+2)(x+5) = x^2+7x+10$; $(x+3)(x+4) = x^2+7x+12$. Let $x^2+7x = t$: $(t+10)(t+12)-24 = t^2+22t+96 = (t+16)(t+6) = (x^2+7x+16)(x+6)(x+1)$.

Question: Exercise 4.3, Question 2: (iii) $(x-1)(x-2)(x-3)(x-4)-99$

Answer: $(x-1)(x-4) = x^2-5x+4$; $(x-2)(x-3) = x^2-5x+6$. Let $x^2-5x = t$: $(t+4)(t+6)-99 = t^2+10t-75 = (t+15)(t-5) = (x^2-5x+15)(x^2-5x-5)$.

Question: Exercise 4.3, Question 2: (iv) $(x-3)(x-5)(x-7)(x-9)+15$

Answer: $(x-3)(x-9) = x^2-12x+27$; $(x-5)(x-7) = x^2-12x+35$. Let $x^2-12x = t$: $(t+27)(t+35)+15 = t^2+62t+960 = (t+30)(t+32) = (x^2-12x+30)(x-8)(x-4)$.

Question: Exercise 4.3, Question 2: (v) $(x-1)(x-2)(x-3)(x-4)-224$

Answer: With the same pairing, $x^2-5x = t$: $(t+4)(t+6)-224 = t^2+10t-200 = (t+20)(t-10) = (x^2-5x+20)(x^2-5x-10)$.

Question: Exercise 4.3, Question 2: (vi) $(x-1)(x-3)(x-4)(x-5)-255$ [note: the printed expression has no rational factors; the expanded form is $x^4-13x^3+59x^2-107x-195$ and it does not factor further over the rationals]

Answer: [flagged: the printed expression does not factor over the rationals — the expanded form $x^4-13x^3+59x^2-107x-195$ has no rational root or rational quadratic factor]

Question: Exercise 4.3, Question 3: Factorize. (i) $(x-2)(x-3)(x+2)(x+3)-2x^2$

Answer: $= (x^2-4)(x^2-9)-2x^2 =$

$= x^4-15x^2+36$

$= (x^2-12)(x^2-3).$

Question: Exercise 4.3, Question 3: (ii) $(x-1)(x+1)(x+3)(x-3)-3x^2-23$

Answer: $= (x^2-1)(x^2-9)-3x^2-23 =$

$= x^4-13x^2-14$

$= (x^2-14)(x^2+1).$

Question: Exercise 4.3, Question 3: (iii) $(x-1)(x+1)(x-3)(x+3)+4x^2$

Answer: $= (x^2-1)(x^2-9)+4x^2 =$

$= x^4-6x^2+9$

$= (x^2-3)^2.$

Question: Exercise 4.3, Question 3: (iv) $(x-2)(x+2)(x-4)(x+4)-14x^2$

Answer: $= (x^2-4)(x^2-16)-14x^2 =$

$= x^4-34x^2+64$

$= (x^2-32)(x^2-2).$

Question: Exercise 4.3, Question 3: (v) $(x+5)(x+2)(x-5)(x-2)+4x^2$

Answer: $= (x^2-25)(x^2-4)+4x^2 =$

$= x^4-25x^2+100$

$= (x^2-20)(x^2-5)$.

Question: Exercise 4.3, Question 3: (vi) $(x^2-x-12)(x^2-x-12)-x^2$

Answer: $= (x^2-x-12)^2-x^2 =$

$= [(x^2-x-12)-x][(x^2-x-12)+x]$

$= (x^2-2x-12)(x^2-12)$.

Question: Exercise 4.4, Question 1: Factorize the following. (i) $b^3+3b^2c+3bc^2+c^3$

Answer: $= (b+c)^3$ (formula $a^3+3a^2b+3ab^2+b^3 = (a+b)^3$).

Question: Exercise 4.4, Question 1: (ii) $8x^3+12x^2y+6xy^2+y^3$

Answer: $= (2x+y)^3$, because $(2x)^3+3(2x)^2(y)+3(2x)(y)^2+y^3 = 8x^3+12x^2y+6xy^2+y^3$.

Question: Exercise 4.4, Question 1: (iii) $64x^3+12x^2+(3/4)x+1/64$ [note: the printed third term is 3x; the $(4x+1/4)^3$ pattern gives 3x/4]

Answer: $= (4x+1/4)^3$, because $(4x)^3+3(4x)^2(1/4)+3(4x)(1/4)^2+(1/4)^3 = 64x^3+12x^2+(3/4)x+1/64$.

Question: Exercise 4.4, Question 1: (iv) $8x^3+36x^2+54x+27$

Answer: $= (2x+3)^3$, because $(2x)^3+3(2x)^2(3)+3(2x)(3)^2+3^3 = 8x^3+36x^2+54x+27$.

Question: Exercise 4.4, Question 1: (v) $1/27+(1/3)y^2+y^4+y^6$

Answer: = $(1/3+y^2)^3$, because $(1/3)^3+3(1/3)^2(y^2)+3(1/3)(y^2)^2+(y^2)^3$ = $1/27+(1/3)y^2+y^4+y^6$.

Question: Exercise 4.4, Question 1: (vi) $(8/27)x^3+2x^2y+(9/2)xy^2+(27/8)y^3$ [note: the printed third term is 3xy; the $(2x/3+3y/2)^3$ pattern gives $(9/2)xy^2$]

Answer: = $(2x/3+3y/2)^3$, because $(2x/3)^3+3(2x/3)^2(3y/2)+3(2x/3)(3y/2)^2+(3y/2)^3$ = $(8/27)x^3+2x^2y+(9/2)xy^2+(27/8)y^3$.

Question: Exercise 4.4, Question 1: (vii) $x^3+4x^2+(16/3)x+64/27$ [note: the printed coefficients appear shuffled; the $(x+4/3)^3$ pattern gives this order]

Answer: = $(x+4/3)^3$, because $x^3+3x^2(4/3)+3x(4/3)^2+(4/3)^3$ = $x^3+4x^2+(16/3)x+64/27$.

Question: Exercise 4.4, Question 1: (viii) $z^3/8+z^2y/4+zy^2/6+y^3/27$

Answer: = $(z/2+y/3)^3$, because $(z/2)^3+3(z/2)^2(y/3)+3(z/2)(y/3)^2+(y/3)^3$ = $z^3/8+z^2y/4+zy^2/6+y^3/27$.

Question: Exercise 4.4, Question 2: Factorize. (i) $d^3-6d^2c+12dc^2-8c^3$

Answer: = $(d-2c)^3$, because $d^3-3d^2(2c)+3d(2c)^2-(2c)^3$ = $d^3-6d^2c+12dc^2-8c^3$.

Question: Exercise 4.4, Question 2: (ii) $x^6-4x^4+(16/3)x^2-64/27$ [note: the printed coefficients appear shuffled; the $(x^2-4/3)^3$ pattern gives this order]

Answer: = $(x^2-4/3)^3$, because $(x^2)^3-3(x^2)^2(4/3)+3(x^2)(4/3)^2-(4/3)^3$ = $x^6-4x^4+(16/3)x^2-64/27$.

Question: Exercise 4.4, Question 2: (iii) $x^3/125-(3/25)x^2y+(3/5)xy^2-y^3$

Answer: $= (x/5-y)^3$, because $(x/5)^3-3(x/5)^2(y)+3(x/5)(y)^2-y^3 = x^3/125-(3/25)x^2y+(3/5)xy^2-y^3$.

Question: Exercise 4.4, Question 2: (iv) $125z^3-75z^2y^2+15zy^4-y^6$

Answer: $= (5z-y^2)^3$, because $(5z)^3-3(5z)^2(y^2)+3(5z)(y^2)^2-(y^2)^3 = 125z^3-75z^2y^2+15zy^4-y^6$.

Question: Exercise 4.4, Question 2: (v) $z^3/27-2z^2y+36zy^2-216y^3$ [note: the printed second term is $-z^2y/3$; the $(z/3-6y)^3$ pattern gives $-2z^2y$]

Answer: $= (z/3-6y)^3$, because $(z/3)^3-3(z/3)^2(6y)+3(z/3)(6y)^2-(6y)^3 = z^3/27-2z^2y+36zy^2-216y^3$.

Question: Exercise 4.4, Question 2: (vi) $b^6/27-b^4c^2/6+b^2c^4/4-c^6/8$ [note: the printed middle coefficients appear swapped; the $(b^2/3-c^2/2)^3$ pattern gives this order]

Answer: $= (b^2/3-c^2/2)^3$, because $(b^2/3)^3-3(b^2/3)^2(c^2/2)+3(b^2/3)(c^2/2)^2-(c^2/2)^3 = b^6/27-b^4c^2/6+b^2c^4/4-c^6/8$.

Question: Exercise 4.4, Question 2: (vii) $216+(9/2)z^2-54z-z^3/8$

Answer: $= (6-z/2)^3$, because $6^3-3(6)^2(z/2)+3(6)(z/2)^2-(z/2)^3 = 216-54z+(9/2)z^2-z^3/8$.

Question: Exercise 4.4, Question 2: (viii) $(8/27)x^3-2x^2y+(9/2)xy^2-(27/8)y^3$ [note: the printed third term is $3xy$; the $(2x/3-3y/2)^3$ pattern gives $(9/2)xy^2$]

Answer: $= (2x/3-3y/2)^3$, because $(2x/3)^3-3(2x/3)^2(3y/2)+3(2x/3)(3y/2)^2-(3y/2)^3 =$

$(8/27)x^3-2x^2y+(9/2)xy^2-(27/8)y^3$.

Question: Exercise 4.5, Question 1: Factorize the following. (i) x^3+8y^3

Answer: = $(x+2y)(x^2-2xy+4y^2)$ (formula $a^3+b^3 = (a+b)(a^2-ab+b^2)$).

Question: Exercise 4.5, Question 1: (ii) $a^{11}+a^2b^9$

Answer: = $a^2(a^9+b^9) = a^2(a^3+b^3)(a^6-a^3b^3+b^6) =$

$a^2(a+b)(a^2-ab+b^2)(a^6-a^3b^3+b^6)$.

Question: Exercise 4.5, Question 1: (iii) a^6+1

Answer: = $(a^2)^3+1^3 = (a^2+1)(a^4-a^2+1)$.

Question: Exercise 4.5, Question 1: (iv) a^3b^3+512

Answer: = $(ab)^3+8^3 = (ab+8)(a^2b^2-8ab+64)$.

Question: Exercise 4.5, Question 1: (v) $a^3b^3+27b^6$

Answer: = $b^3(a^3+27b^3) = b^3(a+3b)(a^2-3ab+9b^2)$.

Question: Exercise 4.5, Question 1: (vi) $x^3/125+125/x^3$

Answer: = $(x/5)^3+(5/x)^3 = (x/5+5/x)(x^2/25-1+25/x^2)$.

Question: Exercise 4.5, Question 1: (vii) $x^9+x^3y^6z^9$

Answer: = $x^3(x^6+y^6z^9) = x^3((x^2)^3+(y^2z^3)^3) = x^3(x^2+y^2z^3)(x^4-x^2y^2z^3+y^4z^6)$.

Question: Exercise 4.5, Question 1: (viii) $x^6/27+8/x^3$

Answer: = $(x^2/3)^3+(2/x)^3 = (x^2/3+2/x)(x^4/9-2x/3+4/x^2)$.

Question: Exercise 4.5, Question 2: Factorize. (i) x^3-8y^3

Answer: = $(x-2y)(x^2+2xy+4y^2)$ (formula $a^3-b^3 = (a-b)(a^2+ab+b^2)$).

Question: Exercise 4.5, Question 2: (ii) x^9-8y^9

Answer: = $(x^3)^3-(2y^3)^3 = (x^3-2y^3)(x^6+2x^3y^3+4y^6)$.

Question: Exercise 4.5, Question 2: (iii) $1000-x^3y^3/125$

Answer: = $10^3-(xy/5)^3 = (10-xy/5)(100+2xy+x^2y^2/25)$.

Question: Exercise 4.5, Question 2: (iv) a^6-b^6

Answer:

Formula: $a^2 - b^2 = (a + b)(a - b)$

$a^6-b^6 = (a^3)^2 - (b^3)^2$

$= (a^3-b^3)(a^3+b^3) = (a-b)(a^2+ab+b^2)(a+b)(a^2-ab+b^2)$.

Question: Exercise 4.5, Question 2: (v) $x^6/64-64/x^{12}$

Answer: = $(x^2/4)^3-(4/x^4)^3 = (x^2/4-4/x^4)(x^4/16+1/x^2+16/x^8)$.

Question: Exercise 4.5, Question 2: (vi) $x^{12}-y^{12}$

Answer:

Formula: $a^2 - b^2 = (a + b)(a - b)$

$x^{12}-y^{12} = (x^6)^2 - (y^6)^2$

$= (x^6-y^6)(x^6+y^6) = (x-y)(x+y)(x^2+y^2)(x^4+x^2y^2+y^4)(x^4-x^2y^2+y^4)$.

Question: Exercise 4.5, Question 2: (vii) $27/x^3-8y^6$

Answer: $= (3/x)^3-(2y^2)^3 = (3/x-2y^2)(9/x^2+6y^2/x+4y^4)$.

Question: Exercise 4.5, Question 2: (viii) $8x^6-1/729$

Answer: $= (2x^2)^3-(1/9)^3 = (2x^2-1/9)(4x^4+2x^2/9+1/81)$.

Question: Exercise 4.6, Question 1: Use the remainder theorem to find the remainder when the following are divided by the given divisor. (i) $x^3-6x^2+11x-8$ divided by $(x-1)$

Answer: R =

$= p(1)$

$= 1-6+11-8$

$= -2.$

Question: Exercise 4.6, Question 1: (ii) $x^3+6x^2+11x+8$ divided by $(x+1)$

Answer: R =

$= p(-1)$

$= -1+6-11+8$

$= 2.$

Question: Exercise 4.6, Question 1: (iii) x^3-x^2+14 divided by $(x-2)$

Answer: R =

= p(2)

= 8−4+14

= 18.

Question: Exercise 4.6, Question 1: (iv) $x^3-3x^2+4x-14$ divided by (x+2)

Answer: R =

= p(−2)

= −8−12−8−14

= −42.

Question: Exercise 4.6, Question 1: (v) $(2y-1)^3+6(3+4y)-9$ divided by (2y+1)

Answer: The divisor is zero when y = −1/2: R = p(−1/2) = $(-2)^3+6(3-2)-9 = -8+6-9 = -11$.

Question: Exercise 4.6, Question 1: (vi) $4y^3-4y+3$ divided by (2y−1)

Answer: The divisor is zero when y = 1/2: R = p(1/2) = 4(1/8)−2+3 = 3/2.

Question: Exercise 4.6, Question 1: (vii) $(2y+1)^3-6(3-4y)-10$ divided by (2y−1)

Answer: The divisor is zero when y = 1/2: R = p(1/2) = $2^3-6(3-2)-10$ = 8−6−10 = −8.

Question: Exercise 4.6, Question 1: (viii) $x^4+x^2y^2+y^4$ divided by $(x-y)$

Answer: Treating x as the variable, putting x = y: R = p(y) = $y^4+y^4+y^4$ = $3y^4$.

Question: Exercise 4.6, Question 4: Find the value of m, if p(y) = my^3+4y^2+3y-4 and q(y) = y^3-4y+m leave the same remainder when divided by $(y-3)$. [note: questions 2 and 3 are not printed in the book; the numbering goes straight from 1 to 4]

Answer: p(3) = $27m+36+9-4 = 27m+41$; q(3) = $27-12+m = 15+m$. Equating: $27m+41 = 15+m \Rightarrow 26m = -26 \Rightarrow m = -1$.

Question: Exercise 4.6, Question 5: If the polynomial $4x^3-7x^2+6x-3k$ is exactly divisible by $(x+2)$, find the value of k.

Answer: For exact division p(−2) = 0: $4(-8)-7(4)+6(-2)-3k = -72-3k = 0 \Rightarrow k = -24$.

Question: Exercise 4.6, Question 6: If $(y+2)$ is a factor of the polynomial $3y^2-4ry-4r^2$, find the value of r.

Answer: For a factor p(−2) = 0: $3(4)-4r(-2)-4r^2 = 12+8r-4r^2 = 0 \Rightarrow r^2-2r-3 = 0 \Rightarrow (r-3)(r+1) = 0 \Rightarrow r = 3$ or $r = -1$.

Question: Exercise 4.7, Question 1: Divide the following polynomials by the synthetic division method and find their quotients and remainders. (i) p(x) = x^3-x^2+x-1 by $x-1$

Answer: Synthetic division (1): coefficients 1, −1, 1, −1 → rows: 1; 0; 1; 0. Quotient = x^2+1, remainder = 0.

Question: Exercise 4.7, Question 1: (ii) p(x) = x^3-x^2-x-1 by $x+1$

Answer: Synthetic division (−1): coefficients 1, −1, −1, −1 → 1; −2; 1; −2. Quotient = x^2-2x+1, remainder = −2.

Question: Exercise 4.7, Question 1: (iii) $p(x) = x^3-6x^2+11x-6$ by $x+2$

Answer: Synthetic division (−2): 1; −8; 27; −60. Quotient = $x^2-8x+27$, remainder = −60.

Question: Exercise 4.7, Question 1: (iv) $p(x) = x^3+6x^2-11x-6$ by $x-2$

Answer: Synthetic division (2): 1; 8; 5; 4. Quotient = x^2+8x+5, remainder = 4.

Question: Exercise 4.7, Question 1: (v) $p(x) = x^4-x^3+x^2-x-1$ by $x+2$

Answer: Synthetic division (−2): 1; −3; 7; −15; 29. Quotient = $x^3-3x^2+7x-15$, remainder = 29.

Question: Exercise 4.7, Question 1: (vi) $p(x) = x^4+x^3-x^2+x-1$ by $x-1$

Answer: Synthetic division (1): 1; 2; 1; 2; 1. Quotient = x^3+2x^2+x+2, remainder = 1.

Question: Exercise 4.7, Question 1: (vii) $p(x) = x^5+x^3-2x^2-3$ by $x+3$

Answer: Coefficients 1, 0, 1, −2, 0, −3; synthetic division (−3): 1; −3; 10; −32; 96; −291. Quotient = $x^4-3x^3+10x^2-32x+96$, remainder = −291.

Question: Exercise 4.7, Question 1: (viii) $p(x) = x^5-x^4+x^3-3x^2+6x-6$ by $x-3$

Answer: Synthetic division (3): 1; 2; 7; 18; 60; 174. Quotient = $x^4+2x^3+7x^2+18x+60$, remainder = 174.

Question: Exercise 4.7, Question 1: (ix) $p(x) = 2x^4-2x^3+100x^2-168x+95$ by $x-2$

Answer: Synthetic division (2): 2; 2; 104; 40; 175. Quotient = $2x^3+2x^2+104x+40$, remainder = 175.

Question: Exercise 4.7, Question 1: (x) $p(x) = 6x^4-72x^3+300x^2-564x+270$ by $x-5$

Answer: Synthetic division (5): 6; −42; 90; −114; −300. Quotient = $6x^3-42x^2+90x-114$, remainder = −300.

Question: Exercise 4.7, Question 2: For which value of k is −2 a root of the polynomial $p(x) = x^3+x^2-14x-k$?

Answer: $p(-2) =$

$= 0 \Rightarrow -8+4+28-k$

$= 24-k$

$= 0 \Rightarrow k$

$= 24.$

Question: Exercise 4.7, Question 3: For which value of m will −2 be a root of $x^3+mx^2-7x-10$?

Answer: $p(-2) =$

$= 0 \Rightarrow -8+4m+14-10$

= 4m−4

= 0 ⇒ m

= 1.

Question: Exercise 4.7, Question 4: For which value of m is x+2 a factor of $x^3-7x^2+6x-3m$?

Answer: p(−2) =

= 0 ⇒ −8−28−12−3m

= −48−3m

= 0 ⇒ m

= −16.

Question: Exercise 4.7, Question 5: For which value of m is −1 a root of $P(x) = 2x^3-4mx^2+x-1$? [note: the question prints k but the polynomial contains m; the value of m is given]

Answer: P(−1) =

= 0 ⇒ −2−4m−1−1

= −4−4m

= 0 ⇒ m

= −1.

Question: Exercise 4.8: Find the factors by the factor theorem method. 1. x^3-x^2+x-1

Answer: p(1) = 1−1+1−1 = 0, so (x−1) is a factor. By synthetic division: $(x-1)(x^2+1)$.

Question: Exercise 4.8: 2. x^3+x^2-x-1

Answer: p(−1) = −1+1+1−1 = 0, so (x+1) is a factor. By synthetic division: $(x+1)(x^2-1) = (x-1)(x+1)^2$.

Question: Exercise 4.8: 3. $x^3-6x^2+11x-6$

Answer: p(1) = 1−6+11−6 = 0, so (x−1) is a factor. By synthetic division: $(x-1)(x^2-5x+6) = (x-1)(x-2)(x-3)$.

Question: Exercise 4.8: 4. $x^3+6x^2-11x-6$ [note: the printed expression has no rational root]

Answer: [flagged: the printed expression does not factor over the rationals — none of the possible rational roots ±1, ±2, ±3, ±6 makes p(x) = 0 (e.g. p(1) = −10, p(−1) = 10, p(2) = 4, p(−2) = 32, p(3) = 42, p(−3) = 54, p(6) = 360, p(−6) = 60)]

Question: Exercise 4.8: 5. x^3-2x^2+x-1 [note: the printed expression has no rational root]

Answer: [flagged: the printed expression does not factor over the rationals — neither of the possible rational roots ±1 makes p(x) = 0 (p(1) = −1, p(−1) = −5)]

Question: Exercise 4.8: 6. $6x^3+7x^2-x-2$

Answer: $p(1/2) = 6/8+7/4-1/2-2 = 0$, so $(2x-1)$ is a factor. By synthetic division: $(2x-1)(3x^2+5x+2) = (2x-1)(3x+2)(x+1)$.

Question: Exercise 4.8: 7. $x^3+8x^2+19x+12$

Answer: $p(-1) = -1+8-19+12 = 0$, so $(x+1)$ is a factor. By synthetic division: $(x+1)(x^2+7x+12) = (x+1)(x+3)(x+4)$.

Question: Exercise 4.8: 8. $2x^3+9x^2+10x+3$

Answer: $p(-1) = -2+9-10+3 = 0$, so $(x+1)$ is a factor. By synthetic division: $(x+1)(2x^2+7x+3) = (x+1)(2x+1)(x+3)$.

Question: Exercise 4.8: 9. $x^3+12x^2+44x+48$

Answer: $p(-2) = -8+48-88+48 = 0$, so $(x+2)$ is a factor. By synthetic division: $(x+2)(x^2+10x+24) = (x+2)(x+4)(x+6)$.

Question: Review Exercise 4, Question 1: True and false questions. Read the following sentences carefully and circle T for a true statement and F for a false statement. (i) $x^2+x-6 = (x+3)(x-2)$

Answer: T, because $(x+3)(x-2) = x^2+x-6$.

Question: Review Exercise 4, Question 1: (ii) $a^3+27 = (a+3)(a^2-3a+9)$

Answer: T (the a^3+b^3 formula, with $b = 3$).

Question: Review Exercise 4, Question 1: (iii) $b^3-8 = (b-2)(b^2+2b+4)$

Answer: T (the a^3-b^3 formula, with $b = 2$).

Question: Review Exercise 4, Question 1: (iv) $a^4-b^4 = (a-b)(a+b)(a+b)^2$

Answer: F, because $a^4-b^4 = (a-b)(a+b)(a^2+b^2)$, not $(a+b)^2$.

Question: Review Exercise 4, Question 1: (v) $a^6+b^6 = (a^3+b^3)(a^3-b^3)$

Answer: F, because $(a^3+b^3)(a^3-b^3) = a^6-b^6$, not a^6+b^6.

Question: Review Exercise 4, Question 1: (vi) $a^5+b^5 = (a+b)^5$

Answer: F, because the expansion of $(a+b)^5$ also has middle terms $(5a^4b+10a^3b^2+10a^2b^3+5ab^4)$.

Review Exercise 4, Question 2: Complete the following sentences. (i) $16x^2-y^4 = (4x-y^2)$ _____

Answer: $(4x-y^2)(4x+y^2)$.

Review Exercise 4, Question 2: (ii) $x^3-64y^3 = (x-4y)$ _____

Answer: $(x-4y)(x^2+4xy+16y^2)$.

Review Exercise 4, Question 2: (iii) $x^2+5x+6 = (x+2)$ $(x+2)(x+3)$.

Review Exercise 4, Question 2: (iv) $x^2+y^2 = (x-y)^2$ _____

Answer: $x^2-2xy+y^2+2xy = x^2+y^2$.

Review Exercise 4, Question 2: (v) $a^3+27b^3 = (a+3b)$ _____

Answer: $(a+3b)(a^2-3ab+9b^2)$.

Review Exercise 4, Question 3: Tick the correct answer. (i) The factors of $a^2+2a-24$ are:

(a) $a+4, a-6$

(b) $a-4$, $a+6$

(c) $a+3$, $a-8$

(d) $a+8$, $a-3$

Review Exercise 4, Question 3: (ii) The factors of $a^2+2ab+b^2-c^2$ are:

Review Exercise 4, Question 3: (v) The simplified form of $1/(x+y) + y/(x^2-y^2)$ is:

Review Exercise 4, Question 3: (vi) Find the value of m, when x^2+4x+m is a perfect square:

(a) 8

(b) −8

(c) 4 ✓

(d) −4

✓

5. Algebraic Manipulation

Question: Exercise 5.1, Question 1: Find the HCF of the following expressions by factorization. (i) $72x^4y^5z^2$ and $120x^3y^6z^8$

Answer: HCF = $24x^3y^5z^2$ (HCF of coefficients is 24; lowest powers of variables taken).

Question: Exercise 5.1, Question 1: (ii) $18r^3s^4t^5$, $120r^4s^3t^8$ and $210r^7s^7t^3$

Answer:

HCF rule: take the common factors with their smallest powers

$18r^3s^4t^5 = 2 \times 3^2 \times r^3 \times s^4 \times t^5$

$120r^4s^3t^8 = 2^3 \times 3 \times 5 \times r^4 \times s^3 \times t^8$

$210r^7s^7t^3 = 2 \times 3 \times 5 \times 7 \times r^7 \times s^7 \times t^3$

HCF = $6r^3s^3t^3$.

Question: Exercise 5.1, Question 1: (iii) $x^2-3x-18$ and x^2+5x+6

Answer: = $(x-6)(x+3)$ and $(x+2)(x+3)$ ⇒ HCF = $x+3$.

Question: Exercise 5.1, Question 1: (iv) $4x^2-9$ and $2x^2-5x+3$

Answer: = $(2x-3)(2x+3)$ and $(2x-3)(x-1)$ ⇒ HCF = $2x-3$.

Question: Exercise 5.1, Question 1: (v) $(2a^2-8b^2)$, $(4a^2+4ab-24b^2)$ and

$(2a^2-12ab+16b^2)$

Answer: = $2(a-2b)(a+2b)$, $4(a+3b)(a-2b)$, $2(a-2b)(a-4b)$ ⇒ HCF = $2(a-2b)$.

Question: Exercise 5.1, Question 1: (vi) x^3+x^2+x+1 and x^3+3x^2+3x+1

Answer: = $(x+1)(x^2+1)$ and $(x+1)^3$ ⇒ HCF = $x+1$.

Question: Exercise 5.1, Question 2: Find the LCM of the following expressions by factorization. (i) x^2+3x+2 and $3x^2-3x-6$

Answer: = $(x+1)(x+2)$ and $3(x-2)(x+1)$ ⇒ LCM = $3(x+1)(x+2)(x-2)$.

Question: Exercise 5.1, Question 2: (ii) $2x^3+15x^2+31x+12$ and $6x^3+46x^2+100x+8$

Answer: = $(x+3)(x+4)(2x+1)$ and $2(3x^3+23x^2+50x+4)$. The two share no common factor, so LCM = $2(x+3)(x+4)(2x+1)(3x^3+23x^2+50x+4)$.

Question: Exercise 5.1, Question 2: (iii) $x^3-5x^2+10x-8$ and x^3-4x^2-7x-6

Answer: = $(x-2)(x^2-3x+4)$ and (x^3-4x^2-7x-6). The two share no common factor, so LCM = $(x-2)(x^2-3x+4)(x^3-4x^2-7x-6)$.

Question: Exercise 5.1, Question 2: (iv) $x^4+3x^3+2x^2+3x+1$, x^3+4x^2+4x+1 and x^3+5x^2+7x+2

Answer: = $(x^2+1)(x^2+3x+1)$, $(x+1)(x^2+3x+1)$, $(x+2)(x^2+3x+1)$ ⇒ LCM = $(x^2+1)(x^2+3x+1)(x+1)(x+2)$.

Question: Exercise 5.1, Question 3: Find the HCF of the following expressions by factorization. (i) $27a^4b^5c^2$ and $81ab^2c^8$

Answer:

HCF rule: take the common factors with their smallest powers

$27a^4b^5c^2 = 3^3 \times a^4 \times b^5 \times c^2$

$81ab^2c^8 = 3^4 \times a \times b^2 \times c^8$

HCF = $27ab^2c^2$.

Question: Exercise 5.1, Question 3: (ii) $24p^2q^3r^4$, $100p^5q^4r^5$ and $300p^3qr^8$

Answer:

HCF rule: take the common factors with their smallest powers

$24p^2q^3r^4 = 2^3 \times 3 \times p^2 \times q^3 \times r^4$

$100p^5q^4r^5 = 2^2 \times 5^2 \times p^5 \times q^4 \times r^5$

$300p^3qr^8 = 2^2 \times 3 \times 5^2 \times p^3 \times q \times r^8$

HCF = $4p^2qr^4$.

Question: Exercise 5.1, Question 3: (iii) $21x^2-14x$ and x^2-5x+2

Answer: = $7x(3x-2)$ and (x^2-5x+2) (which has no rational factor). No common factor ⇒ HCF = 1.

Question: Exercise 5.1, Question 3: (iv) $x^2+11x+28$ and x^2+x-12

Answer: = $(x+4)(x+7)$ and $(x+4)(x-3)$ ⇒ HCF = $x+4$.

Question: Exercise 5.1, Question 3: (v) $6x^2+11x+3$, $3x^2-2x-1$ and $3x^2-11x-4$

Answer: = $(3x+1)(2x+3)$, $(3x+1)(x-1)$, $(3x+1)(x-4)$ ⇒ HCF = $3x+1$.

Question: Exercise 5.1, Question 3: (vi) x^2-y^2, x^3-y^3 and $x^4+x^2y^2+y^4$

Answer: = $(x-y)(x+y)$, $(x-y)(x^2+xy+y^2)$, $(x^2+xy+y^2)(x^2-xy+y^2)$. No factor is common to all three ⇒ HCF = 1.

Question: Exercise 5.1, Question 4: Find the LCM. (i) $x^2-25x+100$ and x^2-x-20

Answer: = $(x-20)(x-5)$ and $(x-5)(x+4)$ ⇒ LCM = $(x-5)(x-20)(x+4)$.

Question: Exercise 5.1, Question 4: (ii) $3x^2+14x+8$ and $6x^2+x-2$

Answer: = $(3x+2)(x+4)$ and $(3x+2)(2x-1)$ ⇒ LCM = $(3x+2)(x+4)(2x-1)$.

Question: Exercise 5.1, Question 4: (iii) $x^2-y^2-z^2-2yz$ and $y^2-z^2-x^2-2xz$

Answer: = $(x-y-z)(x+y+z)$ and $(y-z-x)(y+z+x)$ ⇒ LCM = $(x+y+z)(x-y-z)(y-x-z)$.

Question: Exercise 5.1, Question 4: (iv) $3x^3+9x^2-84x$ and $4x^4-24x^3+32x^2$

Answer: = $3x(x+7)(x-4)$ and $4x^2(x-2)(x-4)$ ⇒ LCM = $12x^2(x+7)(x-4)(x-2)$.

Question: Exercise 5.1, Question 5: If the HCF of $x^2-11x+24$ and $x+6$ is $(x-3)$, find their LCM. [note: the actual HCF of the two printed expressions is 1, not $(x-3)$; the given HCF is used as stated]

Answer:

Formula: LCM = (product of the two expressions) / HCF

LCM = (product)/(HCF) = $(x^2-11x+24)(x+6)/(x-3) = (x-8)(x+6) = x^2-2x-48$, because $(x^2-11x+24)/(x-3) = x-8$.

Question: Exercise 5.1, Question 6: The HCF and LCM of two expressions are $(x+3)$ and $(x^3+7x^2+7x-15)$; if one expression is $x^2+8x+15$, find the other expression.

Answer:

Formula: (first expression) × (second expression) = HCF × LCM

Other expression =

= (HCF×LCM)/(first expression)

$= (x+3)(x^3+7x^2+7x-15)/(x^2+8x+15)$. $x^3+7x^2+7x-15 = (x-1)(x+3)(x+5)$ and $x^2+8x+15 = (x+3)(x+5)$ ⇒ other expression $= (x-1)(x+3) = x^2+2x-3$.

Question: Exercise 5.1, Question 7: The HCF and LCM of two expressions are respectively $3x-2$ and x^3+7x^2-4; find the product of the two expressions.

Answer: Product of the two expressions = HCF×LCM $= (3x-2)(x^3+7x^2-4) = 3x^4+19x^3-14x^2-12x+8$.

Question: Exercise 5.1, Question 8: Verify the relation between HCF and LCM, when $p(x) = x^2-8x-20$ and $q(x) = x^2-15x+50$.

Answer:

Formula: HCF × LCM = p(x) × q(x)

p(x) = (x−10)(x+2); q(x) = (x−10)(x−5). HCF = x−10; LCM = (x−10)(x+2)(x−5).HCF×LCM =

= $(x-10)^2(x+2)(x-5)$

= p(x)q(x). Verified.

Question: Exercise 5.1, Question 9: A carpenter has some planks, some 12 m long and some 18 m long. He wants to cut them into equal pieces so that no wood is wasted. What size pieces should he cut?

Answer: The largest equal size = HCF(12, 18) = 6 m. So he should cut 6 m pieces.

Question: Exercise 5.1, Question 10: Train A stops at Hyderabad every 12 minutes and train B every 14 minutes; both stopped together at 10:30 AM. Tell when both will stop together again.

Answer: The together-stopping interval = LCM(12, 14) = 84 minutes = 1 hour 24 minutes. 10:30 + 1:24 = 11:54 AM.

Question: Exercise 5.2, Question 1: Simplify the following. (i) $4x/(x^2+2x+1) + 3/(x+1)$

Answer: $= 4x/(x+1)^2 + 3/(x+1) = (4x+3(x+1))/(x+1)^2 = (7x+3)/(x+1)^2$.

Question: Exercise 5.2, Question 1: (ii) $3/(x(2x+1)) + (6x+7)/(3x(x+1))$

Answer: = (9(x+1)+(6x+7)(2x+1))/(3x(2x+1)(x+1)) = (12x²+29x+16)/(3x(2x+1)(x+1)).

Question: Exercise 5.2, Question 1: (iii) (3x−1)/(x²+2x+1) − (4x²−1)/(x²−2x−3)

Answer: = (3x−1)/(x+1)² − (2x−1)(2x+1)/((x−3)(x+1)) = [(3x−1)(x−3) − (4x²−1)(x+1)]/((x+1)²(x−3)) = (−4x³−x²−9x+4)/((x+1)²(x−3)).

Question: Exercise 5.2, Question 1: (iv) 1/(x+1) − 2/(x+2) + 3/(x+3)

Answer: = [(x+2)(x+3) − 2(x+1)(x+3) + 3(x+1)(x+2)]/((x+1)(x+2)(x+3)) = (2x²+6x+6)/((x+1)(x+2)(x+3)) = 2(x²+3x+3)/((x+1)(x+2)(x+3)).

Question: Exercise 5.2, Question 1: (v) (x²+4x+3)/5 × 10/(x+1)

Answer: = (x+1)(x+3)/5 × 10/(x+1) = 2(x+3).

Question: Exercise 5.2, Question 1: (vi) (x²−4x−21)/(x²−6x−7) ÷ (x+9)/(x+1)

Answer: = ((x−7)(x+3)/((x−7)(x+1))) × ((x+1)/(x+9)) = (x+3)/(x+9).

Question: Exercise 5.2, Question 1: (vii) [3/(x+1) + 1/(x+2)] ÷ [2/(x+3) − 1]

Answer: First bracket: (4x+7)/((x+1)(x+2)); second bracket: (2−(x+3))/(x+3) = (−x−1)/(x+3).Division =

= (4x+7)(x+3)/((x+1)(x+2)(−x−1))

= −(4x+7)(x+3)/((x+1)²(x+2)).

Question: Exercise 5.2, Question 1: (viii) $(1/(x^2-9)) \div (1/(x+3)) - 3/(x-2)$

Answer: $= 1/(x-3) - 3/(x-2) = ((x-2)-3(x-3))/((x-3)(x-2)) = (7-2x)/((x-3)(x-2))$.

Question: Exercise 5.2, Question 1: (ix) $1/(x^2+8x+15) + 1/(x^2+7x+12) - 1/(x^2+x-12)$

Answer: First two: $(2x+9)/((x+3)(x+4)(x+5))$; then with common denominator: $[(2x+9)(x-3) - (x+3)(x+5)]/((x+3)(x+4)(x+5)(x-3)) = (x^2-5x-42)/((x+3)(x+4)(x+5)(x-3)) = (x-7)(x+6)/((x+3)(x+4)(x+5)(x-3))$.

Question: Exercise 5.2, Question 1: (x) $2((x^2+7x+12)/(x^2-16) + (x^2+x-2)/(x^2+4x+4)) \times (x^2-2x-8)/(8x^2+2x+4)$

Answer: Inside the bracket: $(x+3)/(x-4) + (x-1)/(x+2) = 2(x^2+5)/((x-4)(x+2))$. Multiplying by 2 and then by $(x-4)(x+2)/(2(4x^2+x+2)): = 2(x^2+5)/(4x^2+x+2)$.

Question: Exercise 5.3, Question 1: Find the square root of the following algebraic expressions. (i) $36x^2-60xy+25y^2$

Answer: $= (6x-5y)^2$, so the square root is $6x-5y$.

Question: Exercise 5.3, Question 1: (ii) $9x^2+1/x^2+6$

Answer: $= (3x+1/x)^2$, so the square root is $3x+1/x$.

Question: Exercise 5.3, Question 1: (iii) $4x^4y^4-(12x^3y^3)/z^2+(9x^2y^2)/z^4$

Answer: $= (2x^2y^2-3xy/z^2)^2$, so the square root is $2x^2y^2-3xy/z^2$.

Question: Exercise 5.3, Question 1: (iv) $36(3-2x)^2-48(3-2x)y+16y^2$

Answer: = $(6(3-2x)-4y)^2$, so the square root is $6(3-2x)-4y = 18-12x-4y$.

Question: Exercise 5.3, Question 1: (v) $(x^2+1/x^2)+2(x-1/x)+3$ [note: the printed expression is not a perfect square]

Answer: [flagged: the printed expression is not a perfect square — putting $x-1/x = t$ it becomes $t^2+2t+5 = (t+1)^2+4$, so it has no simple square root]

Question: Exercise 5.3, Question 1: (vi) $(4x^2-4x+1)(9x^2-54x+81)$

Answer: = $(2x-1)^2 \cdot 9(x-3)^2 = [3(2x-1)(x-3)]^2$, so the square root is $3(2x-1)(x-3) = 3(2x^2-7x+3)$.

Question: Exercise 5.3, Question 1: (vii) $(x^2-2x+1)(x^2-6x+9)$

Answer: = $(x-1)^2(x-3)^2$, so the square root is $(x-1)(x-3) = x^2-4x+3$.

Question: Exercise 5.3, Question 1: (viii) $(x^2+8x+15)(x^2+7x+10)(x^2+5x+6)$

Answer: = $(x+3)(x+5)\cdot(x+2)(x+5)\cdot(x+2)(x+3) = [(x+2)(x+3)(x+5)]^2$, so the square root is $(x+2)(x+3)(x+5)$.

Question: Exercise 5.3, Question 2: Find the square root of the following algebraic expressions by division. (i) $x^4+2x^3+3x^2+2x+1$

Answer: By the division method: square root = x^2+x+1, because $(x^2+x+1)^2 = x^4+2x^3+3x^2+2x+1$.

Question: Exercise 5.3, Question 2: (ii) $25x^4+40x^3+26x^2+8x+1$

Answer: Square root = $5x^2+4x+1$, because $(5x^2+4x+1)^2 = 25x^4+40x^3+26x^2+8x+1$.

Question: Exercise 5.3, Question 2: (iii) $4x^4+8x^3+20x^2+16x+16$

Answer: Square root = $2x^2+2x+4$, because $(2x^2+2x+4)^2 = 4x^4+8x^3+20x^2+16x+16$.

Question: Exercise 5.3, Question 2: (iv) $x^2/y^2+y^2/x^2+47-14y/x+14x/y$

Answer: Putting $x/y-y/x = t$: $(x/y)^2+(y/x)^2 = t^2+2$ and $-14y/x+14x/y = 14t$. Expression = $t^2+14t+49 = (t+7)^2$, so square root = $x/y-y/x+7$.

Question: Exercise 5.3, Question 2: (v) $x^2-2x+3-2/x+1/x^2$

Answer: $= (x-1-1/x)^2$, because $(x-1-1/x)^2 = x^2+1/x^2-2x-2/x+3$. So square root = $x-1-1/x$.

Question: Exercise 5.3, Question 2: (vi) $x^2+y^2/9+9z^2+2xy/3+2yz+6xz$

Answer: $= (x+y/3+3z)^2$, so square root = $x+y/3+3z$.

Question: Exercise 5.3, Question 2: (vii) $(x^2+1/x^2)^2-8(x^2+1/x^2)+16$

Answer: $= ((x^2+1/x^2)-4)^2$, so square root = x^2+1/x^2-4.

Question: Exercise 5.3, Question 2: (viii) $x^6+1/x^6-4(x^3+1/x^3)+6$, $x \neq 0$

Answer: Putting $x^3+1/x^3 = t$: $x^6+1/x^6 = t^2-2$. Expression = $t^2-4t+4 = (t-2)^2$, so square root = x^3+1/x^3-2.

Question: Exercise 5.3, Question 3: What should be added to $4x^4+4x^3+17x^2+8x+9$ to make it a perfect square?

Answer: $(2x^2+x+4)^2 = 4x^4+4x^3+17x^2+8x+16$, which is 7 more than the given expression. So 7 should be added.

Question: Exercise 5.3, Question 4: What should be subtracted from

$9x^6-12x^5+4x^4-18x^3-12x^2+18$ to make it a perfect square?

Answer: $(3x^3-2x^2-3)^2 = 9x^6-12x^5+4x^4-18x^3+12x^2+9$. The difference between the given expression and this square is $-24x^2+9$. So $(-24x^2+9)$ should be subtracted, i.e. $24x^2-9$ should be added.

Question: Exercise 5.3, Question 5: Find the value of m that makes $9x^4+12x^3+34x^2+mx+25$ a perfect square.

Answer: $(3x^2+2x+5)^2 = 9x^4+12x^3+34x^2+20x+25$. Comparing gives m = 20.

Question: Exercise 5.3, Question 6: For which values of p and q will the expression $x^4+8x^3+30x^2+px+q$ be a perfect square?

Answer: $(x^2+4x+a)^2 = x^4+8x^3+(16+2a)x^2+8ax+a^2.16+2a =$

$= 30 \Rightarrow a$

$= 7$. So $p = 8\times7 = 56$ and $q = 49$.

Question: Review Exercise 5, Question 1: True and false questions. Read the following sentences carefully and circle "T" for a true statement and "F" for a false statement. (i) $y-2$ is a factor of y^2-4.

Answer: T, because $y^2-4 = (y-2)(y+2)$.

Question: Review Exercise 5, Question 1: (ii) The HCF of a^2-1 and a^3-1 is $a+1$.

Answer: F, because HCF = $a-1$ ($a^2-1 = (a-1)(a+1)$; $a^3-1 = (a-1)(a^2+a+1)$).

Question: Review Exercise 5, Question 1: (iii) The LCM of $x+1$ and x^3+1 is x^3+1.

Answer: T, because $x^3+1 = (x+1)(x^2-x+1)$.

Question: Review Exercise 5, Question 1: (iv) The LCM of x^2-y^2 and x^2+y^2 is x^4-y^4.

Answer: T, because the two share no common factor, so LCM = $(x^2-y^2)(x^2+y^2) = x^4-y^4$.

Question: Review Exercise 5, Question 1: (v) The HCF of $a+3$ and a^2+5a+6 is a^2+4a+3.

Answer: F, because HCF = $a+3$ ($a^2+5a+6 = (a+2)(a+3)$).

Review Exercise 5, Question 2: Fill in the blanks. (i) There are _____ methods for finding the HCF and LCM of polynomials.

Answer: division method).

Review Exercise 5, Question 2: (ii) LCM × HCF of two expressions = _____.

Answer: expressions.

Review Exercise 5, Question 2: (iii) The HCF of $y+6$ and y^2-5y is _____.

Answer: no common factor.

Review Exercise 5, Question 2: (iv) The LCM of y^2+5y+6 and y^2+3y+2 is _____.

Answer: $(y+1)(y+2)$.

Review Exercise 5, Question 2: (v) The HCF of y^2-1/y^2 and $y+1/y$ is _____.

Answer: = $(y-1/y)(y+1/y)$.

Review Exercise 5, Question 3: Tick () the correct answer. (i) The HCF of x^3-8y^3 and $x^2-4xy+4y^2$ is:

(a) $x-4y$

(b) $x^2+2xy+y^2$

(c) $x+2y$

✓ (d) $x-2y$

Review Exercise 5, Question 3: (ii) The LCM of $(2y+3z)^5$ and $(2y+3z)^3$ is:

(a) $2y+3z$

Review Exercise 5, Question 3: (iii) The HCF of x^3-y^3 and x^2+xy+y^2 is:

(a) $x+y$

✓ (b) x^2+xy+y^2

(c) $x-y$

Review Exercise 5, Question 3: (iv) The LCM of $(x-y)^4$ and $(x-y)^3$ is:

Review Exercise 5, Question 3: (v) The simplified form of $1/(x+y) + y/(x^2-y^2)$ is:

Review Exercise 5, Question 3: (vi) The simplified form of $y/(25x^2-y^2) - 1/(5x-y)$ is:

Review Exercise 5, Question 3: (vii) _____ = $(a^3x^3+a^3y^3)/(a^2(x+y))$:

(a) ax^2+ay^2

(b) x^2+y^2

Review Exercise 5, Question 3: (viii) _____ = a/(a−b) − b/(a+b): (a) $(a^2+b^2)/(a-b)$ (b) $(a^2-b^2)/(a^2-b^2)$ (c) $(a+b)/(a^2-b^2)$ (d) $(a-b)/(a^2-b^2)$ [note: the correct answer $(a^2+b^2)/(a^2-b^2)$ is not among the options]

Answer: [flagged:

Review Exercise 5, Question 3: (ix) LCM = _____:

(c) P/HCF

(d) Q/HCF

Review Exercise 5, Question 3: (x) The LCM of x^2-x+1 and x^3+1 is:

(a) $x+1$

(b) x^2-x+1

✓ (c) x^3+1

(d) x^2+x+1

6. Linear Equation and Inequalities

Question: Exercise 6.1, Question 1: Solve the following equations. (i) (1/4)x = 5

Answer: x =

= 5×4

= 20.

Question: Exercise 6.1, Question 1: (ii) x/4 = −3

Answer: x =

= −3×4

= −12.

Question: Exercise 6.1, Question 1: (iii) −5 = −x/6

Answer: x =

= 5×6

= 30.

Question: Exercise 6.1, Question 1: (iv) −x/8 = −5

Answer: x =

= 5×8

= 40.

Question: Exercise 6.1, Question 1: (v) y − 2/5 = −1/3

Answer: y =

= 2/5 − 1/3

= 6/15 − 5/15

= 1/15.

Question: Exercise 6.1, Question 1: (vi) 2y − 3/5 = 1/2

Answer: 2y =

= 1/2 + 3/5

= 5/10 + 6/10

= 11/10 ⇒ y

= 11/20.

Question: Exercise 6.1, Question 1: (vii) (2x−4)/5 = (5x−12)/4

Answer: 4(2x−4) =

= 5(5x−12) ⇒ 8x−16

= 25x−60 ⇒ −17x

= −44 ⇒ x

= 44/17.

Question: Exercise 6.1, Question 1: (viii) 3x/5 + 7 = 2x/3

Answer: Multiplying by 15: 9x+105 = 10x ⇒ x = 105.

Question: Exercise 6.1, Question 1: (ix) 3x/5 + 7 = 2x/3 + 4x/5

Answer: Multiplying by 15: 9x+105 = 10x+12x ⇒ 105 = 13x ⇒ x = 105/13.

Question: Exercise 6.1, Question 1: (x) 6/(2x−5) − 4/(x−3) = 0

Answer: 6(x−3) =

= 4(2x−5) ⇒ 6x−18

= 8x−20 ⇒ 2

= 2x ⇒ x

= 1.

Question: Exercise 6.1, Question 1: (xi) (7x−4)/15 = (7x+4)/10

Answer: 2(7x−4) =

= 3(7x+4) ⇒ 14x−8

= 21x+12 ⇒ −20

= 7x ⇒ x

= −20/7.

Question: Exercise 6.1, Question 1: (xii) (3x−2)/10 = (7x−3)/15 − 2

Answer: Multiplying by 30: 3(3x−2) = 2(7x−3) − 60 ⇒ 9x−6 = 14x−66 ⇒ 60 = 5x ⇒ x = 12.

Question: Exercise 6.1, Question 1: (xiii) (12x−3)/12 = (12x+3)/8

Answer: 2(12x−3) =

= 3(12x+3) ⇒ 24x−6

= 36x+9 ⇒ −15

= 12x ⇒ x

= −5/4.

Question: Exercise 6.1, Question 1: (xiv) (1/4)x + x = −3 + (1/2)x

Answer: (5/4)x − (1/2)x =

= −3 ⇒ (3/4)x

= −3 ⇒ x

= −4.

Question: Exercise 6.1, Question 1: (xv) 1/3 + 2m = m − 3/2

Answer: 2m−m =

= −3/2 − 1/3 ⇒ m

= −9/6 − 2/6

= −11/6.

Question: Exercise 6.1, Question 2: When 25 is added to a number and the result is halved, the answer is three times the original number. Tell which number it is.

Answer: Let the number be x: (x+25)/2 = 3x ⇒ x+25 = 6x ⇒ 25 = 5x ⇒ x = 5.

Question: Exercise 6.1, Question 3: Adding 4 to a number equals 10 less than three times the number. Tell what the number is.

Answer: Let the number be x: x+4 = 3x−10 ⇒ 14 = 2x ⇒ x = 7.

Question: Exercise 6.1, Question 4: Bilal is 6 years older than Ali. If after 5 years the sum of their ages will be 40, what are their ages?

Answer: Let Ali's age be x, Bilal's age x+6. After 5 years: (x+5)+(x+11) = 40 ⇒ 2x+16 = 40 ⇒ x = 12. So Ali is 12 years old and Bilal is 18 years old.

Question: Exercise 6.1, Question 5: Find the solution set of the following

equations and also check the answers. (i) 6+√x = 7

Answer: √x =

= 1 ⇒ x

= 1. Check: 6+√1 = 7 ✓. Solution set = {1}.

Question: Exercise 6.1, Question 5: (ii) √(x−9) = 1

Answer: x−9 =

= 1 ⇒ x

= 10. Check: √1 = 1 ✓. Solution set = {10}.

Question: Exercise 6.1, Question 5: (iii) √(y/4) − 2 = 3

Answer: √(y/4) =

= 5 ⇒ y/4

= 25 ⇒ y

= 100. Check: √25−2 = 3 ✓. Solution set = {100}.

Question: Exercise 6.1, Question 5: (iv) √(4x+5) = √(3x−7)

Answer: Squaring both sides: 4x+5 = 3x−7 ⇒ x = −12. Check: at x = −12 the expressions under the roots are negative (√(−43)), so this is an extraneous root. Solution set = empty set.

Question: Exercise 6.1, Question 5: (v) √(3y+12)/7 = 3

Answer: √(3y+12) =

= 21 ⇒ 3y+12

= 441 ⇒ 3y

= 429 ⇒ y

= 143. Check: √441/7 = 21/7 = 3 ✓. Solution set = {143}.

Question: Exercise 6.1, Question 5: (vi) √(x+9) = 7

Answer: x+9 =

= 49 ⇒ x

= 40. Check: √49 = 7 ✓. Solution set = {40}.

Question: Exercise 6.1, Question 5: (vii) √(25y−50) = 10√(y+3)

Answer: Squaring both sides: 25y−50 = 100(y+3) ⇒ −350 = 75y ⇒ y = −14/3. Check: at y = −14/3 the expressions under the roots are negative, so this is an extraneous root. Solution set = empty set.

Question: Exercise 6.1, Question 5: (viii) √x − 8 = 1

Answer: √x =

= 9 ⇒ x

= 81. Check: 9−8 = 1 ✓. Solution set = {81}.

Question: Exercise 6.1, Question 5: (ix) 10√(x+20) = 100

Answer: √(x+20) =

= 10 ⇒ x+20

= 100 ⇒ x

= 80. Check: 10×10 = 100 ✓. Solution set = {80}.

Question: Exercise 6.2: Find the solution set of the following equations. 1. |2x+1| = 6

Answer: 2x+1 =

= 6 or 2x+1

= −6 ⇒ x

= 5/2 or x

= −7/2. Solution set = {5/2, −7/2}.

Question: Exercise 6.2: 2. |5x−12| = 7, where x ∈ W

Answer: 5x−12 =

= 7 or 5x−12

= −7 ⇒ x

= 19/5 or x

= 1. Only 1 is a whole number. Solution set = {1}.

Question: Exercise 6.2: 3. |2x/7| = 12

Answer: 2x/7 =

= 12 or 2x/7

= −12 ⇒ x

= 42 or x

= −42. Solution set = {42, −42}.

Question: Exercise 6.2: 4. |(2x+1)/3| = 8

Answer: (2x+1)/3 =

= 8 or (2x+1)/3

= −8 ⇒ 2x+1

= 24 or 2x+1

= −24 ⇒ x

= 23/2 or x

= −25/2. Solution set = {23/2, −25/2}.

Question: Exercise 6.2: 5. |5x−3|−8 = 4, where x ∈ N

Answer: |5x−3| =

= 12 ⇒ 5x−3

= 12 or 5x−3

= −12 ⇒ x

= 3 or x

= −9/5. Only 3 is a natural number. Solution set = {3}.

Question: Exercise 6.2: 6. |(5x+1)/7|−3 = 8

Answer: |(5x+1)/7| =

= 11 ⇒ (5x+1)/7

= 11 or (5x+1)/7

= −11 ⇒ 5x+1

= 77 or 5x+1

= −77 ⇒ x

= 76/5 or x

= −78/5. Solution set = {76/5, −78/5}.

Question: Exercise 6.2: 7. |(2x+3)/4|+2 = 7

Answer: |(2x+3)/4| =

= 5 ⇒ (2x+3)/4

= 5 or (2x+3)/4

= −5 ⇒ 2x+3

= 20 or 2x+3

= −20 ⇒ x

= 17/2 or x

= −23/2. Solution set = {17/2, −23/2}.

Question: Exercise 6.2: 8. |(3x+6)/12|+1 = 3, where x ∈ Z

Answer: |(3x+6)/12| =

= 2 ⇒ (3x+6)/12

= 2 or (3x+6)/12

= −2 ⇒ 3x+6

= 24 or 3x+6

= −24 ⇒ x

= 6 or x

= −10. Both are integers. Solution set = {6, −10}.

Question: Exercise 6.2: 9. 3/2 = |7x+8|

Answer: 7x+8 =

= 3/2 or 7x+8

= −3/2 ⇒ x

= −13/14 or x

= −19/14. Solution set = {−13/14, −19/14}.

Question: Exercise 6.2: 10. |(2x−3)/5|−12 = 5

Answer: |(2x−3)/5| =

= 17 ⇒ (2x−3)/5

= 17 or (2x−3)/5

= −17 ⇒ 2x−3

= 85 or 2x−3

= −85 ⇒ x

= 44 or x

= −41. Solution set = {44, −41}.

Question: Exercise 6.3, Question 1: Find the solution set of the following inequalities and show it on the number line. (i) 2x−7 > 6+x, ∀x ∈ N

Answer: x > 13 ⇒ solution set = {14, 15, 16, ...}. On the number line: open circle at 13 and a ray to the right.



Question: Exercise 6.3, Question 1: (ii) 7x−6 > 3x+10, ∀x ∈ R

Answer: 4x > 16 ⇒ x > 4. On the number line: open circle at 4 and a ray to the right.



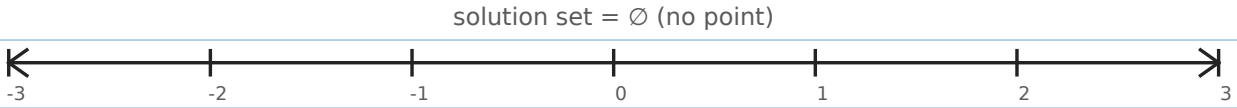
Question: Exercise 6.3, Question 1: (iii) (y+5)/20 < (25−4y)/10, ∀y ∈ N

Answer: Multiplying by 20: y+5 < 50−8y ⇒ 9y < 45 ⇒ y < 5 ⇒ solution set = {1, 2, 3, 4}. On the number line: dots at 1, 2, 3, 4.



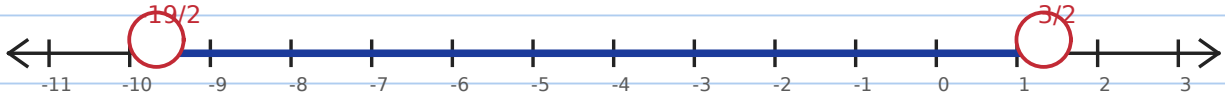
Question: Exercise 6.3, Question 1: (iv) $|2x+3| < x+2$, $\forall x \in Z$ [note: the question prints $\forall y \in Z$ but the expression is in x; $x \in Z$ is understood]

Answer: $x+2 > 0$ and $-(x+2) < 2x+3 < x+2 \Rightarrow -5/3 < x < -1$. There is no integer in this interval. Solution set = empty set. On the number line: no point.



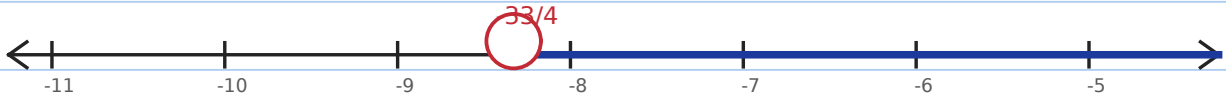
Question: Exercise 6.3, Question 1: (v) $|2y+8| < 11$, $\forall y \in R$

Answer: $-11 < 2y+8 < 11 \Rightarrow -19 < 2y < 3 \Rightarrow -19/2 < y < 3/2$. On the number line: open circles at $-19/2$ and $3/2$ and the joined segment between them.



Question: Exercise 6.3, Question 1: (vi) $5(2y-3) > 6(y-8)$, $\forall y \in R$

Answer: $10y-15 > 6y-48 \Rightarrow 4y > -33 \Rightarrow y > -33/4$. On the number line: open circle at $-33/4$ and a ray to the right.



Question: Exercise 6.3, Question 2: Ali scored 66 and 72 marks respectively in his two tests. How many marks must he score in the third test, if at least 75 average marks are required to qualify for an extra prize?

Answer: Let the third test marks be x: (66+72+x)/3 ≥ 75 ⇒ 138+x ≥ 225 ⇒ x ≥ 87. So at least 87 marks are required.

Question: Exercise 6.3, Question 3: 7 less than three times a number plus 5 is at least 10. Find all numbers satisfying this condition.

Answer: Let the number be x: 3x+5−7 ≥ 10 ⇒ 3x−2 ≥ 10 ⇒ 3x ≥ 12 ⇒ x ≥ 4. So all numbers not less than 4.

Question: Review Exercise 6, Question 1: True and false questions. Read the following sentences carefully and circle "T" for a true statement and "F" for a false one. (i) ay+b = 0 where a = 0 is a first-degree equation.

Answer: F, because at a = 0 no first-degree term of the variable remains in the expression.

Question: Review Exercise 6, Question 1: (ii) The solution set of 3y−2 < 7 ∧ y ∈ N is {4, 5, 6, ...}.

Answer: F, because 3y < 9 ⇒ y < 3 ⇒ solution set = {1, 2}.

Question: Review Exercise 6, Question 1: (iii) The solution set of √y+1 = 3 is {4}.

Answer: T, because √y = 2 ⇒ y = 4.

Question: Review Exercise 6, Question 1: (iv) The solution set of |4y| = 8 is {2, −2}.

Answer: T, because 4y = ±8 ⇒ y = ±2.

Question: Review Exercise 6, Question 1: (v) The solution set of −2 ≤ x ≤ 2, x ∈ R is {−2, 0, 2}.

Answer: F, because all real numbers between −2 and 2 are solutions, not just three points.

Review Exercise 6, Question 2: Fill in the blanks. (i) The solution set of 2y = −y is _____.

Answer: 0 ⇒ y = 0.

Review Exercise 6, Question 2: (ii) The solution set of √(y+5) = 5 is _____.

Answer: 25 ⇒ y = 20.

Review Exercise 6, Question 2: (iii) The solution set of |x|−4 = 0 is _____.

Answer: 4.

Review Exercise 6, Question 2: (iv) The solution set of √(x+5)+2 = 4 is _____.

Answer: ⇒ x = −1.

Review Exercise 6, Question 2: (v) The solution set of 0 < y+2 < 5 is _____.

Answer: gives −2 < y < 3.

Review Exercise 6, Question 3: Tick () the correct answers. (i) The solution set of a first-degree equation in one variable (Variable) is:

✓ (a) one element

(b) two elements

(c) three elements

(d) more than one element

Review Exercise 6, Question 3: (ii) |−20|:

✓ (a) = 20

(b) < 20

(c) = −20

(d) > 20

Review Exercise 6, Question 3: (iii) x ≤ 4 means:

(a) x < 4

(b) x = 4

✓ (c) x < 4 or x = 4

(d) x > 4 or x = 4

Review Exercise 6, Question 3: (iv) The solution set of √y = 10 is:

✓ (a) {100}

(b) {10}

(c) {−10}

(d) {−10, 10}

Review Exercise 6, Question 3: (v) √(y+4)+2 = 8 is a:

(a) first-degree equation

✓ (b) radical equation

(c) cubic equation

(d) second-degree equation

Review Exercise 6, Question 3: (vi) The solution set of 5−3y = −7 is:

(a) {−4}

(b) {1, 4}

✓ (c) {4}

(d) {12}

Review Exercise 6, Question 3: (vii) The solution set of $\sqrt{(5y+5)}+5 = 10$ is:

(a) {±4}

(b) {5}

✓ (c) {4}

(d) {−4}

Review Exercise 6, Question 3: (viii) The solution set of $|5y/3| = 5$ is:

(a) {3}

(b) {−5, 5}

✓ (c) {3, −3}

(d) {−3}

Review Exercise 6, Question 3: (ix) The solution set of $|-y| = 0$ is:

(a) {1}

(b) {−1}

✓ (c) {0}

(d) {}

Review Exercise 6, Question 3: (x) If $x > 0$, $y > 0$ and $x-y < 0$ then:

✓ (a) $x < y$

(b) $x+y < 0$

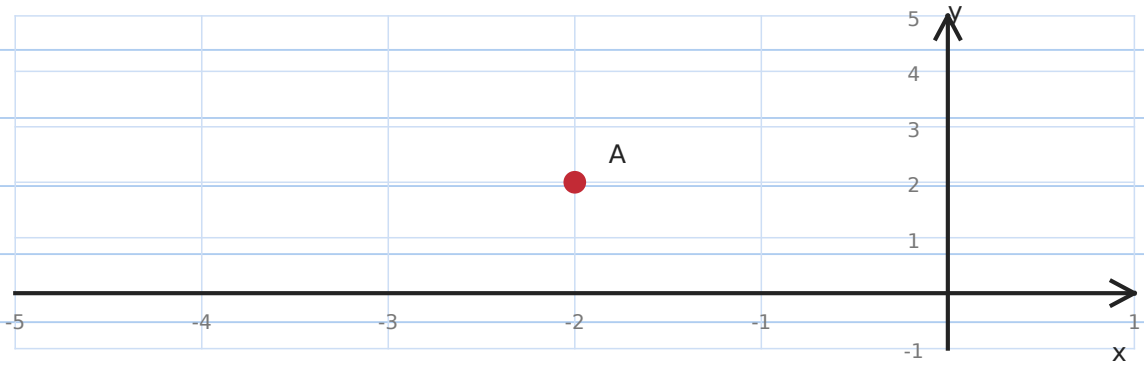
(c) $x > y$

(d) $y-x < 0$

7. Linear Graphs and their Applications

Question: Exercise 7.1, Question 1: State the x-coordinate and y-coordinate of the following points. i. A(−2, 2)

Answer: x-coordinate = −2, y-coordinate = 2.



Question: Exercise 7.1, Question 1: ii. B(5, −1)

Answer: x-coordinate = 5, y-coordinate = −1.

Question: Exercise 7.1, Question 1: iii. C(4, 0)

Answer: x-coordinate = 4, y-coordinate = 0.

Question: Exercise 7.1, Question 1: iv. D(−5, −6)

Answer: x-coordinate = −5, y-coordinate = −6.

Question: Exercise 7.1, Question 1: v. E(3, 4)

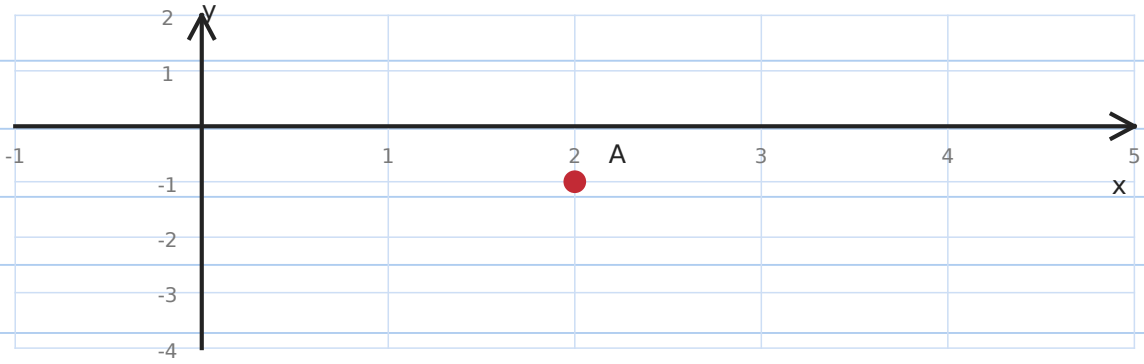
Answer: x-coordinate = 3, y-coordinate = 4.

Question: Exercise 7.1, Question 1: vi. F(−√8, √8)

Answer: x-coordinate = −√8, y-coordinate = √8.

Question: Exercise 7.1, Question 2: State in which quadrant the following points lie. i. A(2, −1)

Answer: Fourth quadrant (x positive, y negative).



Question: Exercise 7.1, Question 2: ii. B(−3, 3)

Answer: Second quadrant (x negative, y positive).

Question: Exercise 7.1, Question 2: iii. C(2√2, −2√2)

Answer: Fourth quadrant (x positive, y negative).

Question: Exercise 7.1, Question 2: iv. D(−2, −4)

Answer: Third quadrant (x negative, y negative).

Question: Exercise 7.1, Question 2: v. E(5, 4)

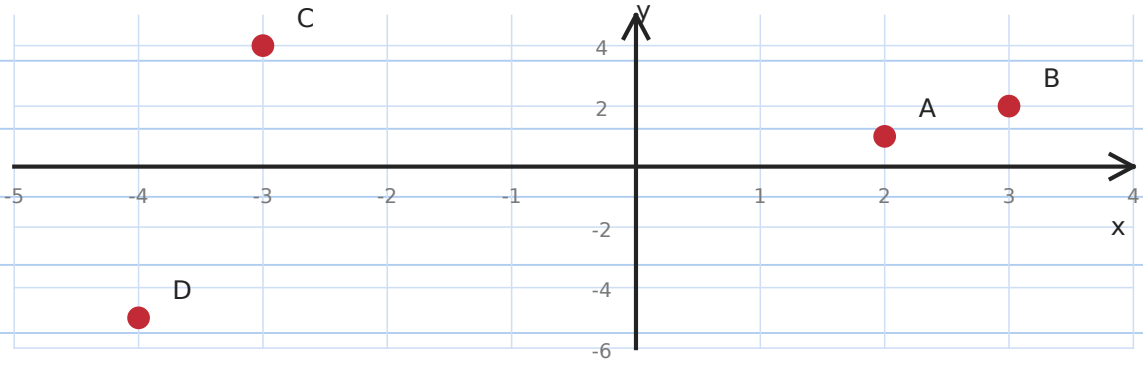
Answer: First quadrant (x positive, y positive).

Question: Exercise 7.1, Question 2: vi. F(3/2, 5/2)

Answer: First quadrant (x positive, y positive).

Question: Exercise 7.1, Question 3: Plot the following points A, B, C and D on the x-y plane. i. A(2, 1), B(3, 2), C(−3, 4), D(−4, −5)

Answer: A is in the first quadrant, B in the first quadrant, C in the second quadrant, D in the third quadrant. Plot each point on graph paper according to its x and y coordinates.



Question: Exercise 7.1, Question 3: ii. A(2, 0), B(0, 2), C(3, −3), D(−3, 3)

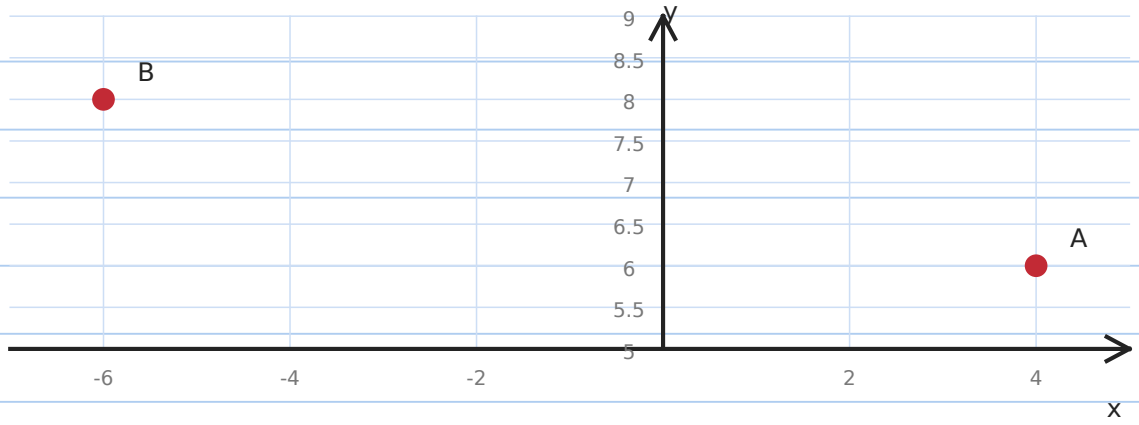
Answer: A is on the x-axis, B on the y-axis, C in the fourth quadrant, D in the second quadrant. Plot each point on graph paper according to its x and y coordinates.

Question: Exercise 7.1, Question 3: iii. A(0, 0), B(−3, −3), C(5, −6), D(−6, 5)

Answer: A is the origin, B in the third quadrant, C in the fourth quadrant, D in the second quadrant. Plot each point on graph paper according to its x and y coordinates.

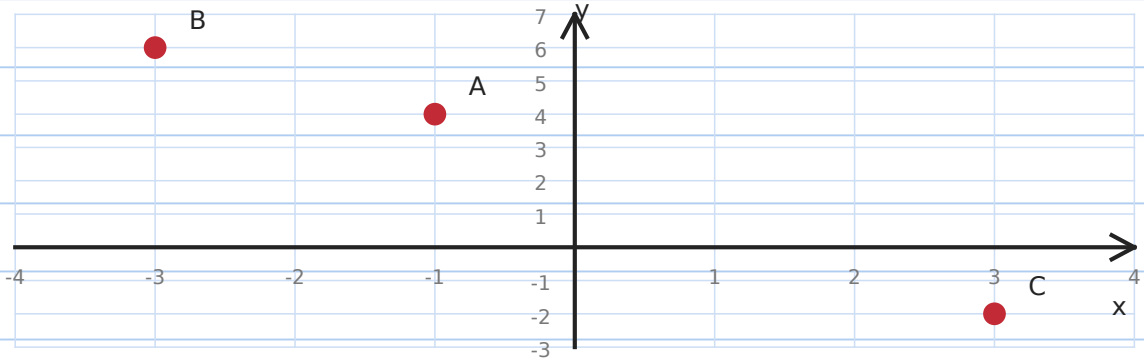
Question: Exercise 7.1, Question 4: Join the points A(4, 6) and B(−6, 8) and name the line segment AB.

Answer: Plot A in the first quadrant and B in the second quadrant, join them with a straight line. The joining part is the line segment AB (or BA).



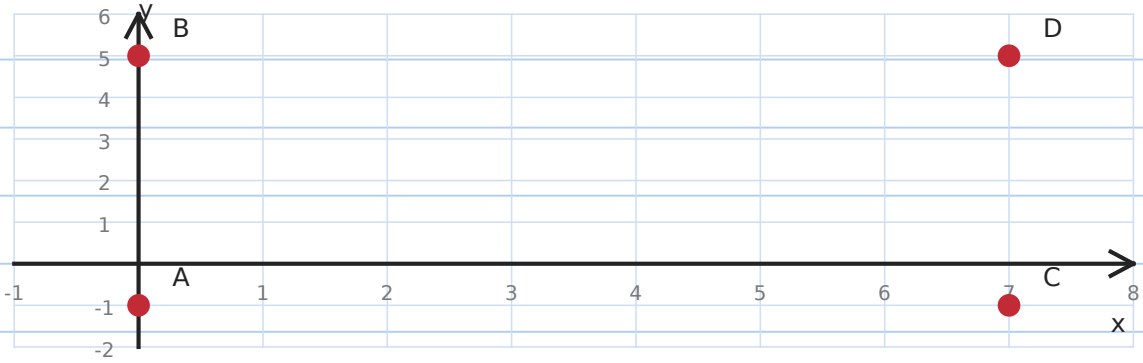
Question: Exercise 7.1, Question 5: Join the points A(−1, 4), B(−3, 6) and C(3, −2) and name the triangle ABC.

Answer: Plot the three points on graph paper, then join AB, BC and CA. The figure formed is triangle ABC.



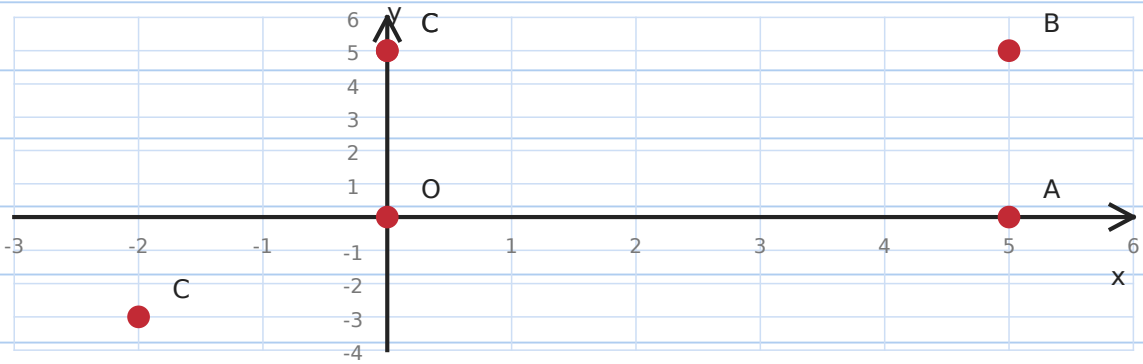
Question: Exercise 7.1, Question 6: Join the points A(0, −1), B(0, 5), C(7, −1) and D(7, 5) and name the rectangle ABCD.

Answer: Plot and join the four points. AB = CD = 6 and AC = BD = 7, opposite sides are equal and parallel and all angles are 90°, so the figure is rectangle ABCD.



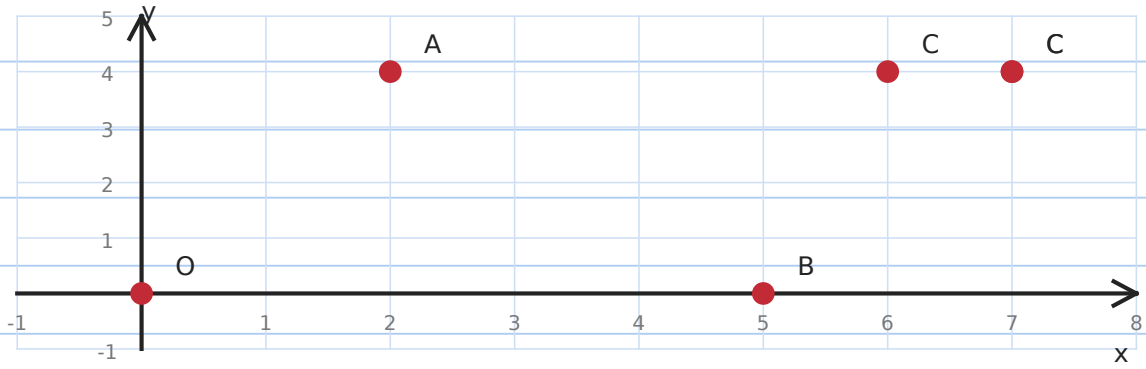
Question: Exercise 7.1, Question 7: Join the points O(0, 0), A(5, 0), B(5, 5) and C(0, 5) and name the square OABC. [note: the printed C(−2, −3) does not form a square; C(0, 5) is needed for a square]

Answer: Plot and join the four points. OA = AB = BC = CO = 5 and all angles are 90°, so the figure is square OABC.



Question: Exercise 7.1, Question 8: Join the points O(0, 0), A(2, 4), B(5, 0) and C(7, 4) and name OABC a parallelogram. [note: the printed C(6, 4) does not form a parallelogram; C(7, 4) is needed]

Answer: Plot and join the four points. OA and BC are parallel and equal (both have slope 2) and OB and AC are parallel and equal (both have slope 0), so the figure is parallelogram OABC.



Question: Exercise 7.1, Question 9: Make a table of some values of x and y for the following first-degree equations. i. x+y−2 = 0

Answer: y = 2−x. Table: x: −1, 0, 1, 2, 3; y: 3, 2, 1, 0, −1.

Question: Exercise 7.1, Question 9: ii. 2x−y−2 = 4

Answer: y = 2x−6. Table: x: 0, 1, 2, 3, 4; y: −6, −4, −2, 0, 2.

Question: Exercise 7.1, Question 9: iii. (1/2)(x+2y)−6 = 0

Answer: x+2y =

= 12 ⇒ y

= (12−x)/2. Table: x: 0, 2, 4, 6, 8; y: 6, 5, 4, 3, 2.

Question: Exercise 7.1, Question 9: iv. (2/3)(x−2y) = −2

Answer: x−2y =

= −3 ⇒ y

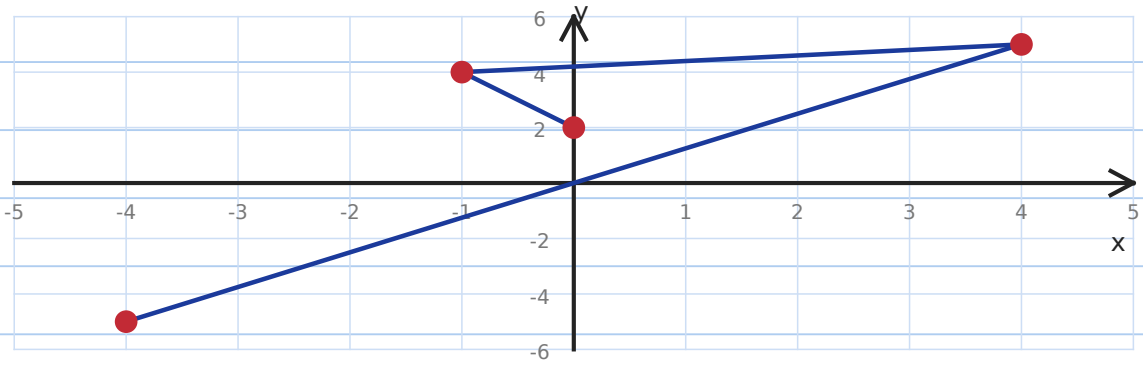
= (x+3)/2. Table: x: −3, −1, 1, 3, 5; y: 0, 1, 2, 3, 4.

Question: Exercise 7.2, Question 1: Make a table of the following equation that satisfies the pair of values x+y = 6.

Answer: Table: x: 0, 1, 2, 3, 4, 5, 6; y: 6, 5, 4, 3, 2, 1, 0.

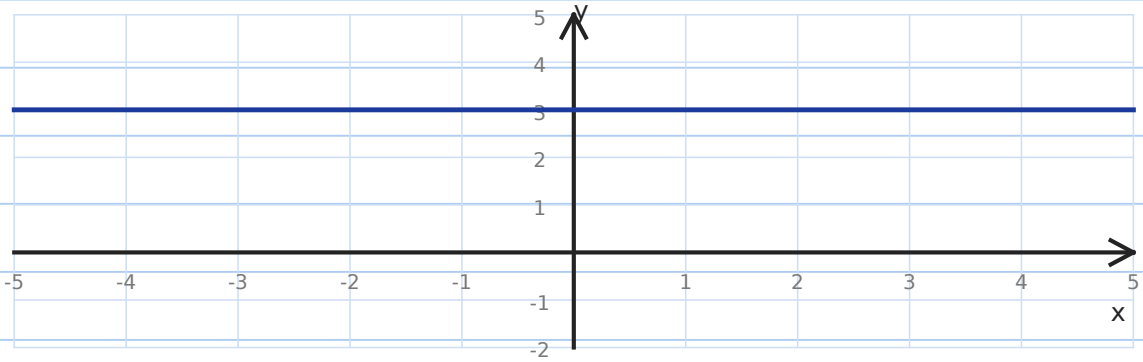
Question: Exercise 7.2, Question 2: Draw the graph from the following table taking a suitable scale. Table: x: 0, −1, 4, −4; y: 2, 4, 5, −5.

Answer: Plot the points (0, 2), (−1, 4), (4, 5), (−4, −5) on graph paper with a suitable scale (e.g. 1 square = 1 unit).



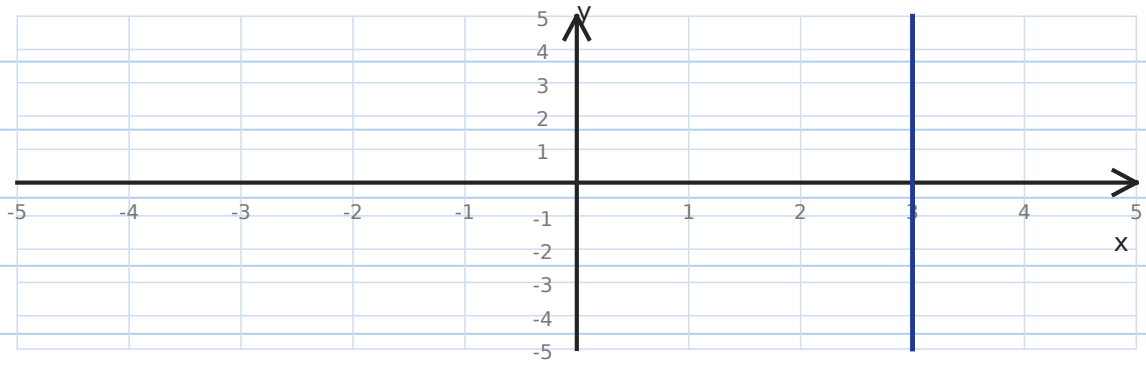
Question: Exercise 7.2, Question 3: Draw the graph of the following. i. y = 3

Answer: Horizontal line parallel to the x-axis, cutting the y-axis at (0, 3).



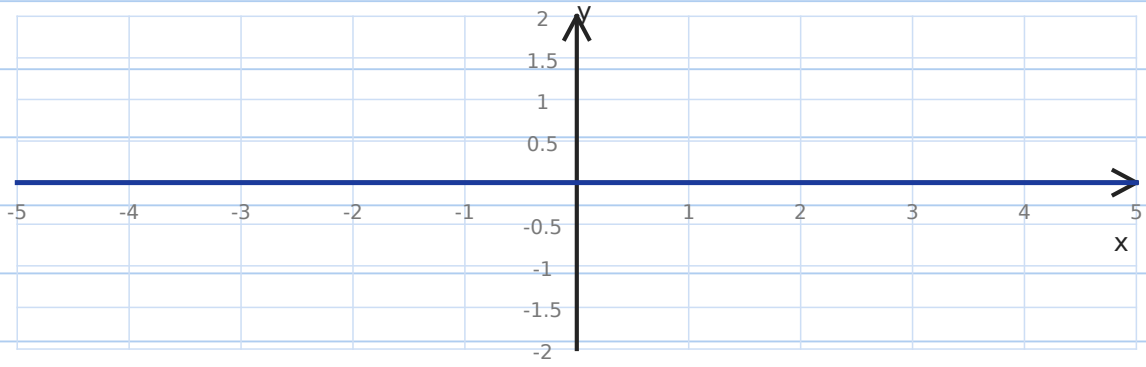
Question: Exercise 7.2, Question 3: ii. x = 3

Answer: Vertical line parallel to the y-axis, cutting the x-axis at (3, 0).



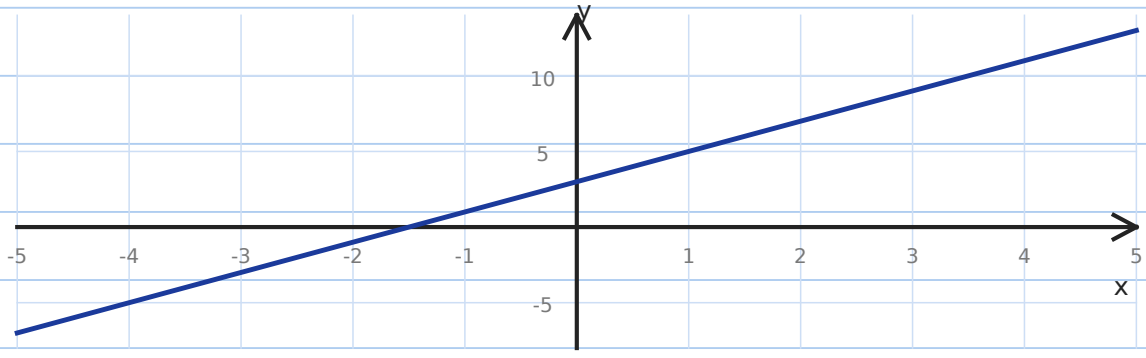
Question: Exercise 7.2, Question 3: iii. y = 0

Answer: The x-axis itself (the line y = 0).



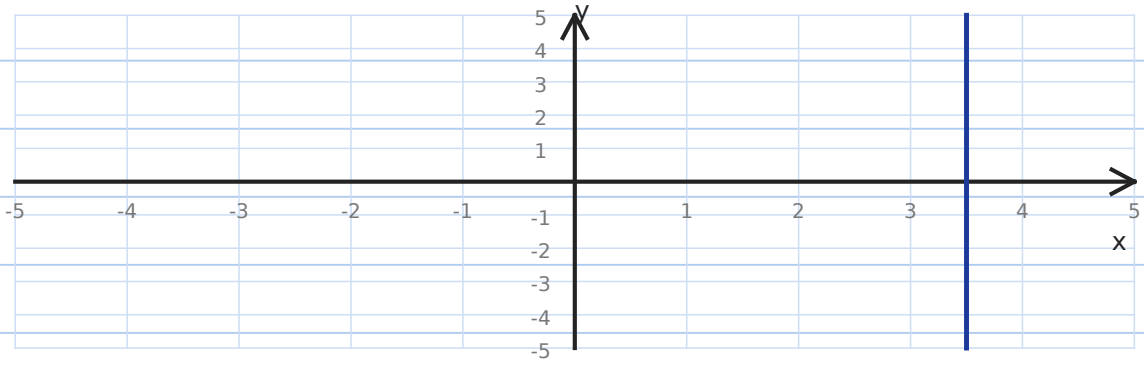
Question: Exercise 7.2, Question 3: iv. y = 2x+3

Answer: Straight line passing through the points (0, 3), (1, 5), (−1, 1).



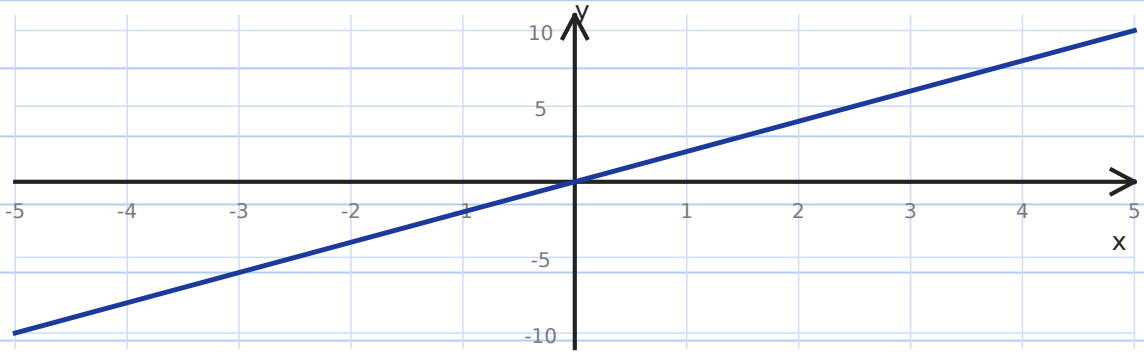
Question: Exercise 7.2, Question 3: v. x = 3.5

Answer: Vertical line parallel to the y-axis, cutting the x-axis at (3.5, 0).



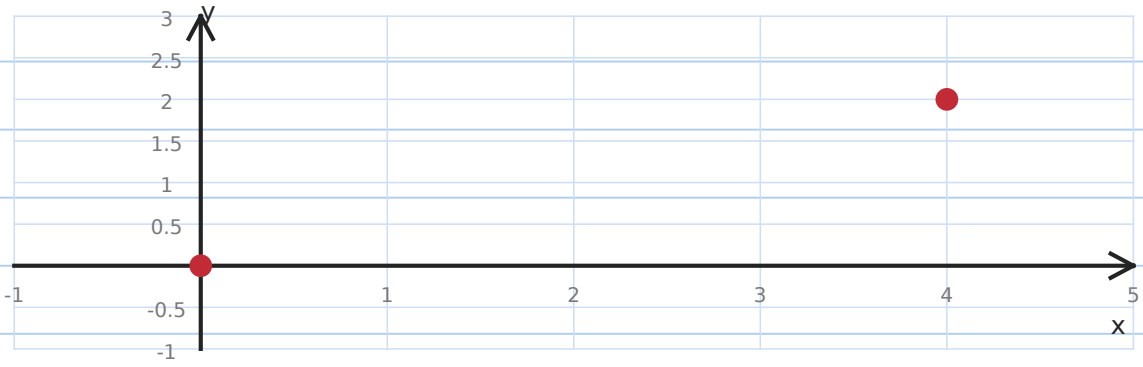
Question: Exercise 7.2, Question 3: vi. −y = 2x

Answer: y = −2x. Straight line passing through the origin (0, 0) and the points (1, −2), (−1, 2).



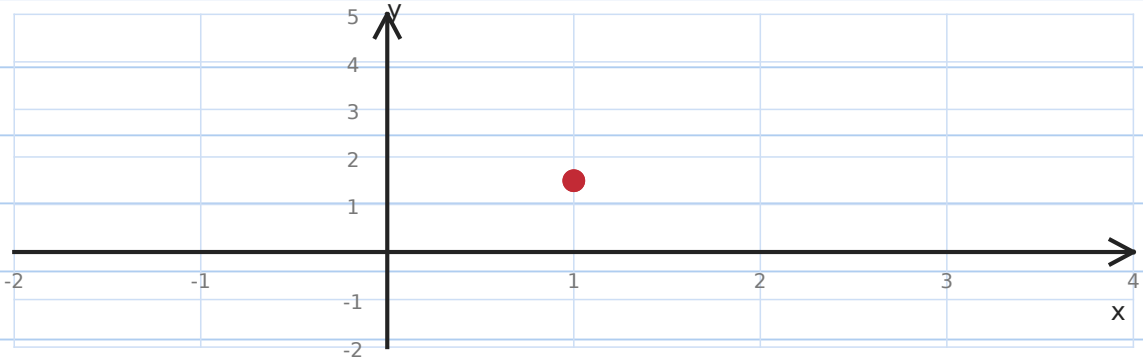
Question: Exercise 7.2, Question 4: Find the missing coordinates from the following table. (i) y = (1/2)x; first row: y-coordinate = 0; second row: x-coordinate = 4

Answer: First: y = 0 ⇒ x = 0 ⇒ point (0, 0). Second: x = 4 ⇒ y = 2 ⇒ point (4, 2).



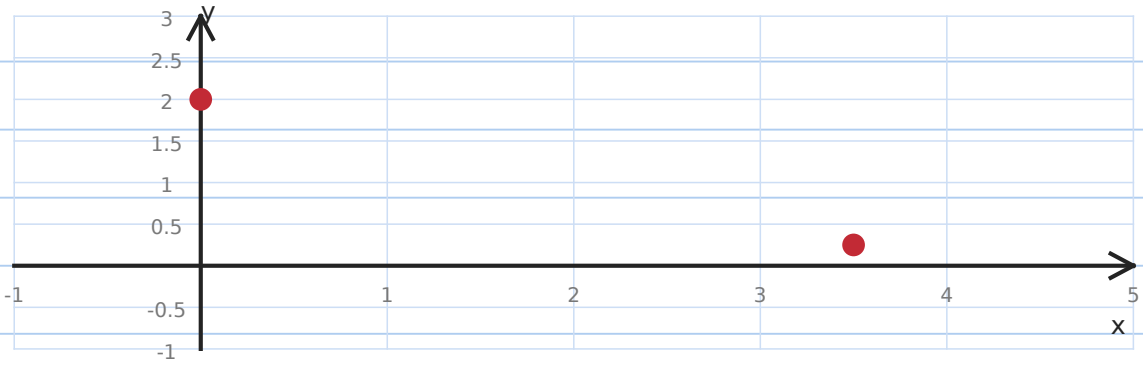
Question: Exercise 7.2, Question 4: (ii) x = (2/3)y; first row: x-coordinate = 1; second row: y-coordinate = 3/2

Answer: First: x = 1 ⇒ y = 3/2 ⇒ point (1, 3/2). Second: y = 3/2 ⇒ x = 1 ⇒ point (1, 3/2).



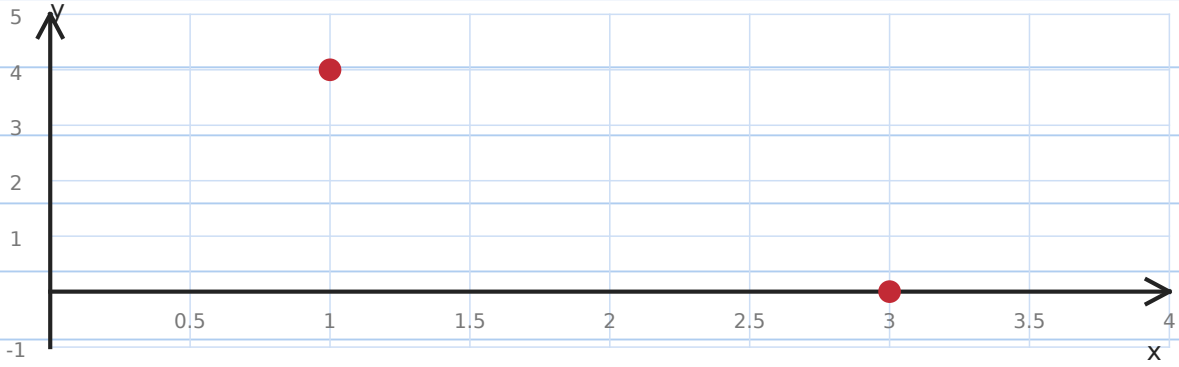
Question: Exercise 7.2, Question 4: (iii) 2x+4y = 8; first row: x-coordinate = 0; second row: y-coordinate = 1/4

Answer: First: x = 0 ⇒ 4y = 8 ⇒ y = 2 ⇒ point (0, 2). Second: y = 1/4 ⇒ 2x+1 = 8 ⇒ x = 7/2 ⇒ point (7/2, 1/4).



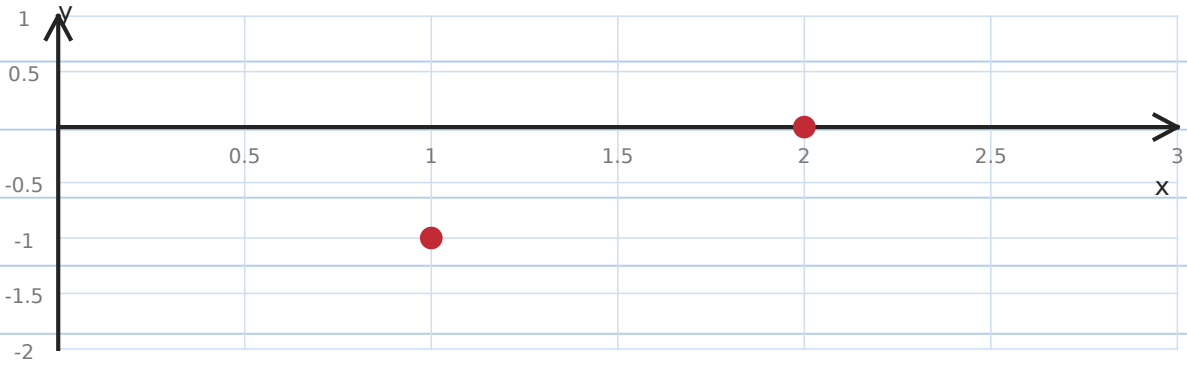
Question: Exercise 7.2, Question 4: (iv) 2x+y = 6; first row: x-coordinate = 1; second row: y-coordinate = 0

Answer: First: x = 1 ⇒ y = 4 ⇒ point (1, 4). Second: y = 0 ⇒ x = 3 ⇒ point (3, 0).



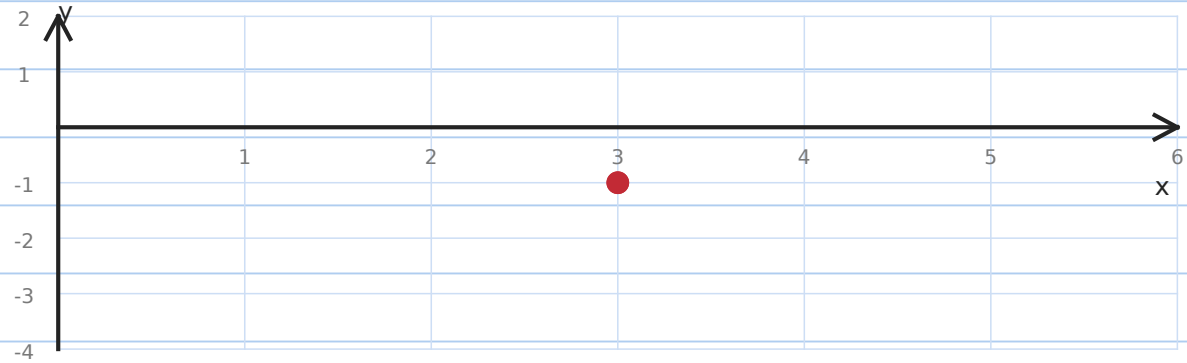
Question: Exercise 7.2, Question 4: (v) x−y = 2; first row: y-coordinate = 0; second row: x-coordinate = 1

Answer: First: y = 0 ⇒ x = 2 ⇒ point (2, 0). Second: x = 1 ⇒ y = −1 ⇒ point (1, −1).



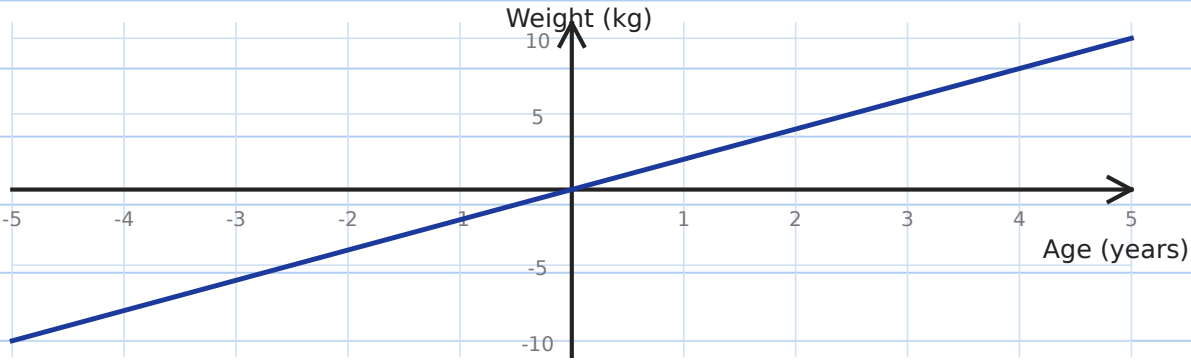
Question: Exercise 7.2, Question 4: (vi) x−3y = 6; first row: x-coordinate = 3; second row: y-coordinate = −1

Answer: First: x = 3 ⇒ 3−3y = 6 ⇒ y = −1 ⇒ point (3, −1). Second: y = −1 ⇒ x+3 = 6 ⇒ x = 3 ⇒ point (3, −1).



Question: Exercise 7.2, Question 5: A person's weight y kg is expressed in terms of age x years by the equation y = 2x. Draw the age–weight graph. [note: no table is printed; the table is made from the equation]

Answer: Table: x (age): 0, 1, 2, 3, 4, 5, 10; y (weight): 0, 2, 4, 6, 8, 10, 20. Graph: straight line through the origin, slope 2.

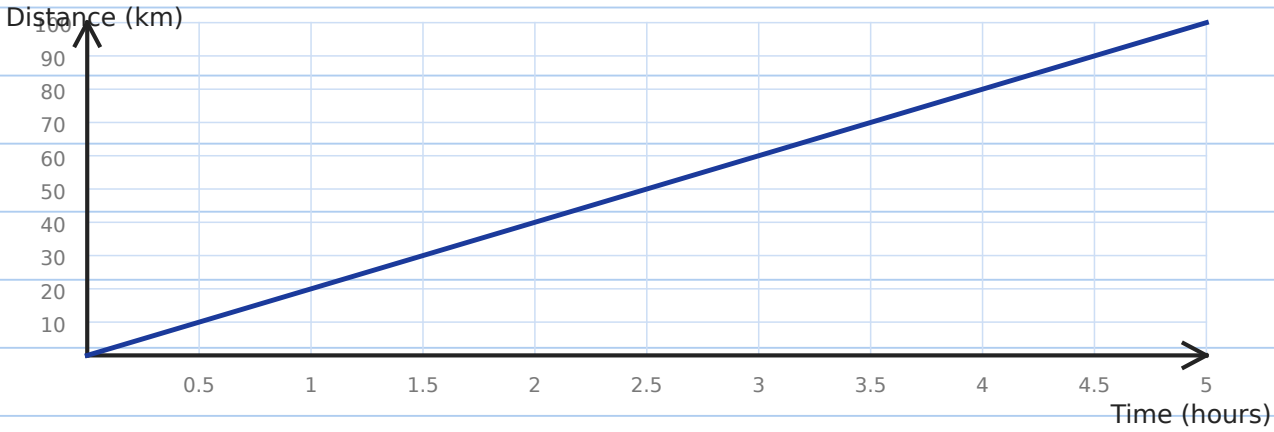


Question: Exercise 7.2, Question 6: Ayesha can drive a non-stop car continuously at 20 km per hour. Draw the distance–time graph for this situation. (a) The time needed for Ayesha to travel 100 km.

Answer: Distance = 20×time. Graph: straight line through the origin, slope 20.(a) Time =

= 100/20

= 5 hours.



Question: Exercise 7.2, Question 6: (b) The distance covered by Ayesha in a total of 3 hours.

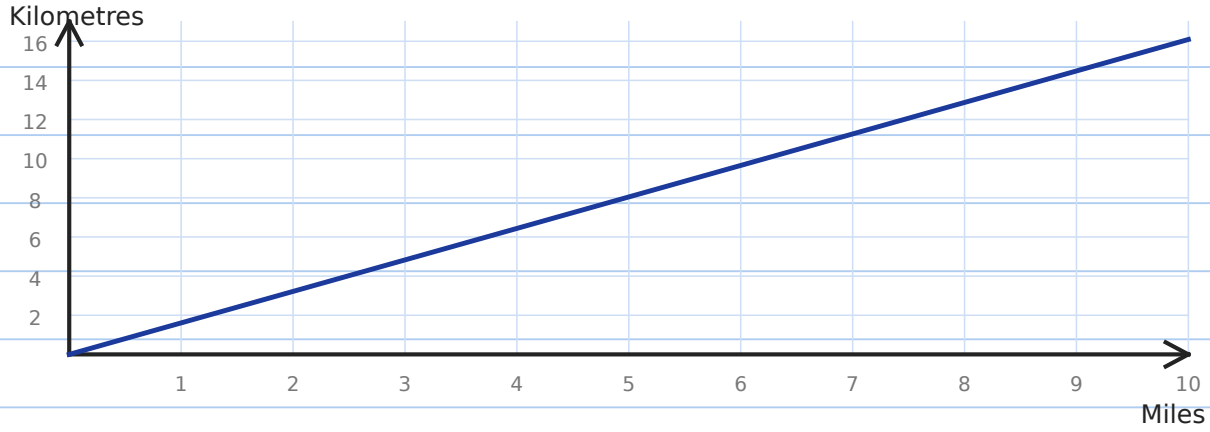
Answer: Distance =

= 20×3

= 60 km.

Question: Exercise 7.3, Question 1: Read the miles and kilometres conversion graph. (i) Convert 1 mile into kilometres.

Answer: From the graph approximately: 1 mile = 1.6 km.



Question: Exercise 7.3, Question 1: (ii) Convert 3 miles into kilometres.

Answer: From the graph approximately: 3 miles = 4.8 km.

Question: Exercise 7.3, Question 1: (iii) Convert 2 kilometres into miles.

Answer: From the graph approximately: 2 km = 1.25 miles.

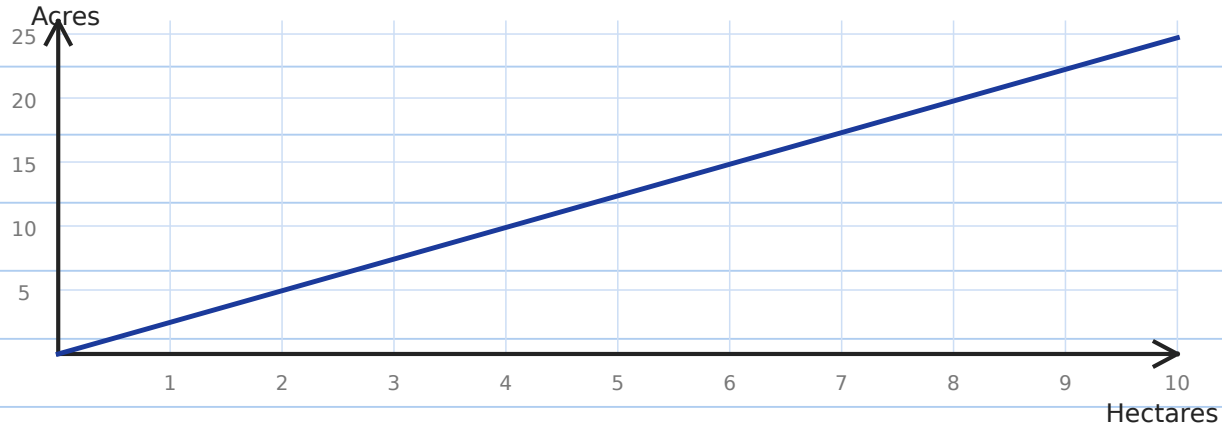
Question: Exercise 7.3, Question 1: (iv) Convert 8 kilometres into miles.

Answer: From the graph approximately: 8 km = 5 miles.

Question: Exercise 7.3, Question 2: Read the hectares and acres conversion graph. (i) Convert 2 hectares into acres.

Answer: From the graph approximately (1 hectare = 2.471 acres): 2

hectares = 4.94 acres ≈ 5 acres.



Question: Exercise 7.3, Question 2: (ii) Convert 5 hectares into acres.

Answer: From the graph approximately: 5 hectares = 12.36 acres ≈ 12.4 acres.

Question: Exercise 7.3, Question 2: (iii) Convert 5 acres into hectares.

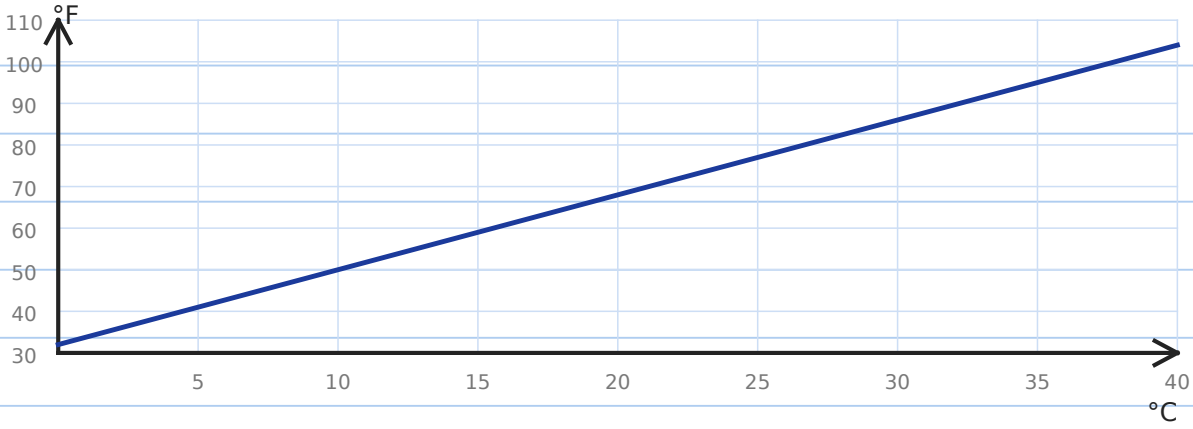
Answer: From the graph approximately: 5 acres = 2.02 hectares ≈ 2 hectares.

Question: Exercise 7.3, Question 2: (iv) Convert 15 acres into hectares.

Answer: From the graph approximately: 15 acres = 6.07 hectares ≈ 6.1 hectares.

Question: Exercise 7.3, Question 3: Read the centigrade and Fahrenheit conversion graph. (i) Convert 2° centigrade into Fahrenheit.

Answer: F = (9/5)C+32: 2°C = 35.6°F.



Question: Exercise 7.3, Question 3: (ii) Convert 80° centigrade into Fahrenheit.

Answer: 80°C =

= (9/5)(80)+32

= 176°F.

Question: Exercise 7.3, Question 3: (iii) Convert 32° Fahrenheit into centigrade.

Answer: C = (5/9)(F−32): 32°F = 0°C.

Question: Exercise 7.3, Question 3: (iv) Convert 36.4° Fahrenheit into centigrade.

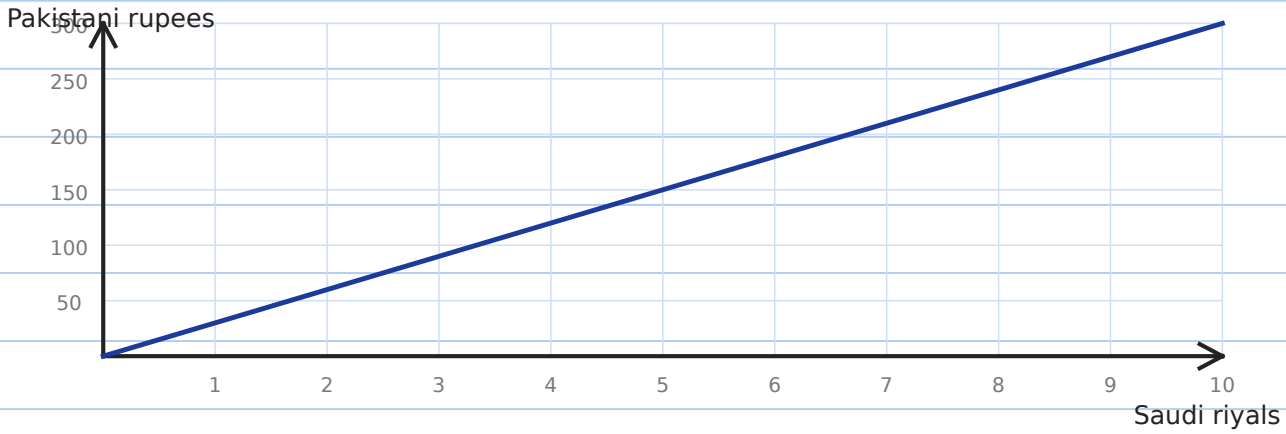
Answer: 36.4°F =

= (5/9)(36.4−32)

= (5/9)(4.4) ≈ 2.44°C ≈ 2.4°C.

Question: Exercise 7.3, Question 4: Read the Pakistani rupees and Saudi riyals conversion graph. (i) Convert 3 Saudi riyals into rupees.

Answer: From the graph approximately (1 riyal = 30 rupees): 3 riyals = 90 rupees.



Question: Exercise 7.3, Question 4: (ii) Convert 5.2 Saudi riyals into rupees.

Answer: From the graph approximately: 5.2 riyals = 156 rupees.

Question: Exercise 7.3, Question 4: (iii) Convert 150 rupees into Saudi riyals.

Answer: From the graph approximately: 150 rupees = 5 riyals.

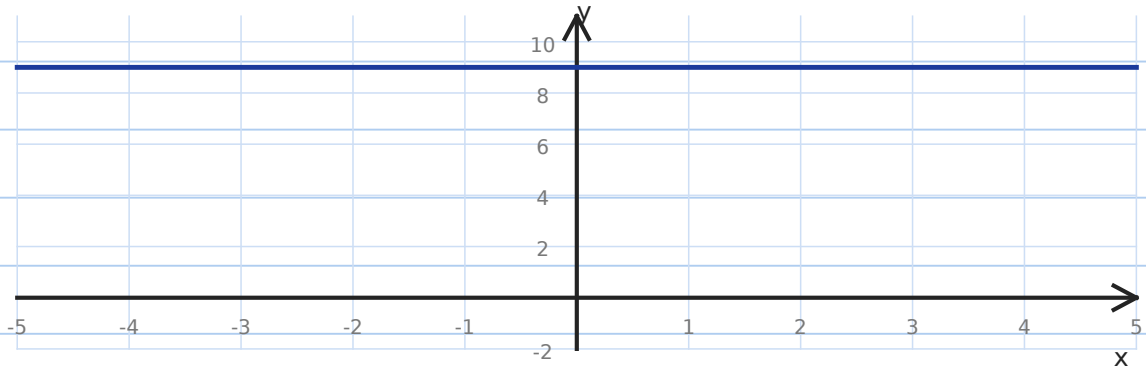
Question: Exercise 7.3, Question 4: (iv) Convert 78 rupees into Saudi riyals.

Answer: From the graph approximately: 78 rupees = 2.6 riyals.

Question: Exercise 7.3, Question 5: Solve the following simultaneous equations by the graphical method. i. $3x-11 = y$; $x-3y = 9$

Answer: Draw both lines on the graph; the intersection point is $(3, -2)$.

Check: $y = 3(3)-11 = -2$ and $3-3(-2) = 9$ ✓.



Question: Exercise 7.3, Question 5: ii. x+y = 4; 2x−1 = 5y

Answer: The intersection point is (3, 1). Check: 3+1 = 4 and 2(3)−1 = 5(1) ✓.

Question: Exercise 7.3, Question 5: iii. 2x = y+5; x = 2y+1

Answer: The intersection point is (3, 1). Check: 2(3) = 1+5 and 3 = 2(1)+1 ✓.

Question: Exercise 7.3, Question 5: iv. y = 3x−5; x+y = 11

Answer: The intersection point is (4, 7). Check: 7 = 3(4)−5 and 4+7 = 11 ✓.

Question: Exercise 7.3, Question 5: v. 2x+y = 3; x−y = 0

Answer: The intersection point is (1, 1). Check: 2(1)+1 = 3 and 1−1 = 0 ✓.

Question: Exercise 7.3, Question 5: vi. 2x+2 = y; y = x−1

Answer: The intersection point is (−3, −4). Check: 2(−3)+2 = −4 and −4 = −3−1 ✓.

Question: Exercise 7.3, Question 5: vii. 5 = x+4y; 2x+3y = 0

Answer: The intersection point is (−3, 2). Check: −3+4(2) = 5 and 2(−3)+3(2) = 0 ✓.

Question: Exercise 7.3, Question 5: viii. 3x = 5y−2; 3x+5y = 8

Answer: The intersection point is (1, 1). Check: 3(1) = 5(1)−2 and 3(1)+5(1) = 8 ✓.

Question: Exercise 7.3, Question 5: ix. (x+2)/5 + y = 6; y = 2x−12

Answer: First equation: x+5y = 28. The intersection point is (8, 4). Check: (8+2)/5+4 = 6 and 4 = 2(8)−12 ✓.

Question: Exercise 7.3, Question 5: x. 3x−2y = 13; 2x+3y = 13

Answer: The intersection point is (5, 1). Check: 3(5)−2(1) = 13 and 2(5)+3(1) = 13 ✓.

Question: Review Exercise 7, Question 1: True and false answers. Read the following sentences carefully and circle T or F, whichever applies. (i) The Cartesian plane is also called the xy-plane.

Answer: T.

Question: Review Exercise 7, Question 1: (ii) In the second quadrant both x and y coordinates are positive.

Answer: F, because in the second quadrant x is negative and y is positive.

Question: Review Exercise 7, Question 1: (iii) The point (1, −2) lies in the first quadrant.

Answer: F, because (1, −2) is in the fourth quadrant.

Question: Review Exercise 7, Question 1: (iv) The point (−3, −4) lies in the fourth quadrant.

Answer: F, because (−3, −4) is in the third quadrant.

Review Exercise 7, Question 2: Tick () the correct answer from the following. (i) The point (−3, −4) lies in:

(a) the first quadrant

(b) the second quadrant

✓ (c) the third quadrant

(d) the fourth quadrant

Review Exercise 7, Question 2: (ii) The two coordinate axes meet at an angle of:

(a) 45°

✓ (b) 90°

(c) 180°

(d) 270°

Review Exercise 7, Question 2: (iii) The line y = 4 is parallel to:

✓ (a) the x-axis

(b) the y-axis

(c) both axes

(d) none

Review Exercise 7, Question 2: (iv) The line x = 5 is parallel to:

(a) the x-axis

✓ (b) the y-axis

(c) both axes

(d) none

Review Exercise 7, Question 2: (v) The point of the line x = 5 on the x-axis is:

Review Exercise 7, Question 2: (vi) The solution set of the lines x = 2 and x = 5 is:

✓ (d) {}

Review Exercise 7, Question 2: (vii) The coordinate axes are _____ to each other:

✓ (a) perpendicular

(b) parallel

(c) meeting at 30°

(d) meeting at 45°

8. Quadratic Equations

Question: Exercise 8.1, Question 1: Solve the following quadratic equations by the factorization method. (i) $x^2+5x+6 = 0$

Answer: $(x+2)(x+3) =$

$= 0 \Rightarrow x$

$= -2$ or x

$= -3.$

Question: Exercise 8.1, Question 1: (ii) $6x^2-x-1 = 0$

Answer: $(3x+1)(2x-1) =$

$= 0 \Rightarrow x$

$= -1/3$ or x

$= 1/2.$

Question: Exercise 8.1, Question 1: (iii) $x^2-11x+30 = 0$

Answer: $(x-5)(x-6) =$

$= 0 \Rightarrow x$

$= 5$ or x

= 6.

Question: Exercise 8.1, Question 1: (iv) $x^2-2x = 0$

Answer: $x(x-2) =$

= 0 ⇒ x

= 0 or x

= 2.

Question: Exercise 8.1, Question 1: (v) $x^2-2x-15 = 0$

Answer: $(x-5)(x+3) =$

= 0 ⇒ x

= 5 or x

= −3.

Question: Exercise 8.1, Question 1: (vi) $12x^2-41x+24 = 0$

Answer: $(3x-8)(4x-3) =$

= 0 ⇒ x

= 8/3 or x

= 3/4.

Question: Exercise 8.1, Question 1: (vii) $(x-5)^2-9 = 0$

Answer: $(x-5)^2 =$

$= 9 \Rightarrow x-5$

$= \pm 3 \Rightarrow x$

= 8 or x

= 2.

Question: Exercise 8.1, Question 1: (viii) $(3x+4)^2-16 = 0$

Answer: $(3x+4)^2 =$

$= 16 \Rightarrow 3x+4$

$= \pm 4 \Rightarrow x$

= 0 or x

$= -8/3$.

Question: Exercise 8.1, Question 2: Solve each of the following by the completing the square method. (i) $x^2+6x+1 = 0$

Answer: $x^2+6x+9 =$

$= 8 \Rightarrow (x+3)^2$

$= 8 \Rightarrow x$

$= -3 \pm 2\sqrt{2}.$

Question: Exercise 8.1, Question 2: (ii) $(3x+2)(x+2) = 6-2(x+1)$

Answer: $3x^2+8x+4 =$

$= 4-2x \Rightarrow 3x^2+10x$

$= 0 \Rightarrow x^2+(10/3)x$

$= 0 \Rightarrow (x+5/3)^2$

$= 25/9 \Rightarrow x$

$= 0$ or x

$= -10/3.$

Question: Exercise 8.1, Question 2: (iii) $3x^2-8x = -1$

Answer: $x^2-(8/3)x =$

$= -1/3 \Rightarrow (x-4/3)^2$

$= 16/9-1/3$

= 13/9 ⇒ x

= (4±√13)/3.

Question: Exercise 8.1, Question 2: (iv) $24x^2 = -10x+21$

Answer: $x^2+(5/12)x =$

= 7/8 ⇒ $(x+5/24)^2$

= 7/8+25/576

= 529/576 ⇒ x

= −5/24±23/24 ⇒ x

= 3/4 or x

= −7/6.

Question: Exercise 8.1, Question 2: (v) $2(x^2-3)-3x = 2(x+3)$

Answer: $2x^2-5x-12 =$

= 0 ⇒ $x^2-(5/2)x$

= 6 ⇒ $(x-5/4)^2$

= 6+25/16

= 121/16 ⇒ x

= 5/4±11/4 ⇒ x

= 4 or x

= −3/2.

Question: Exercise 8.1, Question 2: (vi) $2x^2+4x-1 = 0$

Answer: $x^2+2x =$

= 1/2 ⇒ $(x+1)^2$

= 3/2 ⇒ x

= −1±√6/2.

Question: Exercise 8.1, Question 3: In the equation $3x^2+bx-8 = 0$, 2 is one of its roots. (i) What is the value of b?

Answer: Putting x = 2: $3(4)+2b-8 = 0 \Rightarrow 4+2b = 0 \Rightarrow b = -2$.

Question: Exercise 8.1, Question 3: (ii) What is the other root of the equation?

Answer: With b = −2: $3x^2-2x-8 = (3x+4)(x-2) = 0$ ⇒ the other root is x = −4/3.

Question: Exercise 8.2: Solve the following equations using the quadratic formula. (i) $x^2-2x = 15$

Answer:

Quadratic formula: $x = (-b \pm \sqrt{(b^2 - 4ac)}) / 2a$, with $D = b^2 - 4ac$

$x^2-2x-15 = 0.D = b^2 - 4ac$

$= (-2)^2 - 4(1)(-15)$

$= 4+60$

$= 64.x =$

$= (2\pm8)/2 \Rightarrow x$

$= 5$ or x

$= -3.$

Question: Exercise 8.2: (ii) $10x^2+19x-15 = 0$

Answer:

Quadratic formula: $x = (-b \pm \sqrt{(b^2 - 4ac)}) / 2a$, with $D = b^2 - 4ac$

$D = b^2 - 4ac$

$= (19)^2 - 4(10)(-15)$

$= 361+600$

= 961

= 31^2.x =

= (−19±31)/20 ⇒ x

= 3/5 or x

= −5/2.

Question: Exercise 8.2: (iii) $x^2 = -x+1$

Answer:

Quadratic formula: $x = (-b \pm \sqrt{(b^2 - 4ac)}) / 2a$, with $D = b^2 - 4ac$

$x^2+x-1 = 0$.

$D = b^2 - 4ac$

$= (1)^2 - 4(1)(-1)$

= 1 + 4

= 5.

$x = (-1\pm\sqrt{5})/2$.

Question: Exercise 8.2: (iv) $2x = 9-3x^2$

Answer:

Quadratic formula: $x = (-b \pm \sqrt{(b^2 - 4ac)}) / 2a$, with $D = b^2 - 4ac$

$3x^2+2x-9 = 0.D = b^2 - 4ac$

$= (2)^2 - 4(3)(-9)$

$= 4+108$

$= 112.x =$

$= (-2\pm4\sqrt{7})/6$

$= (-1\pm2\sqrt{7})/3.$

Question: Exercise 8.2: (v) $9x^2 = 12x-49$

Answer:

Quadratic formula: $x = (-b \pm \sqrt{(b^2 - 4ac)}) / 2a$, with $D = b^2 - 4ac$

$9x^2-12x+49 = 0.D = b^2 - 4ac$

$= (-12)^2 - 4(9)(49)$

$= 144-1764$

= −1620.x =

= (12±18i√5)/18

= (2±3i√5)/3.

Question: Exercise 8.2: (vi) (1/2)x²+(3/4)x−1 = 0

Answer:

Quadratic formula: x = (−b ± √(b² − 4ac)) / 2a, with D = b² − 4ac

Multiplying by 4: 2x²+3x−4 = 0.D = b² − 4ac

= (3)² − 4(2)(−4)

= 9+32

= 41. x = (−3±√41)/4.

Question: Exercise 8.2: (vii) 3x²−2x+2 = 0

Answer:

Quadratic formula: x = (−b ± √(b² − 4ac)) / 2a, with D = b² − 4ac

D = b² − 4ac

= (−2)² − 4(3)(2)

= 4−24

= −20.x =

= (2±2i√5)/6

= (1±i√5)/3.

Question: Exercise 8.2: (viii) $6x^2-x-1 = 0$

Answer:

Quadratic formula: $x = (-b \pm \sqrt{(b^2 - 4ac)}) / 2a$, with $D = b^2 - 4ac$

$D = b^2 - 4ac$

$= (-1)^2 - 4(6)(-1)$

= 1+24

= 25.x =

= (1±5)/12 ⇒ x

= 1/2 or x

= −1/3.

Question: Exercise 8.2: (ix) $4x^2-10x = 0$

Answer:

Quadratic formula: $x = (-b \pm \sqrt{(b^2 - 4ac)}) / 2a$, with $D = b^2 - 4ac$

$D = b^2 - 4ac$

$= (-10)^2 - 4(4)(0)$

$= 100.$

$x =$

$= (10 \pm 10)/8 \Rightarrow x$

$= 5/2$ or x

$= 0.$

Question: Exercise 8.2: (x) $x^2 - 1 = 0$

Answer:

Quadratic formula: $x = (-b \pm \sqrt{(b^2 - 4ac)}) / 2a$, with $D = b^2 - 4ac$

$x^2 =$

$= 1 \Rightarrow x$

$= \pm 1.$

Question: Exercise 8.2: (xi) $x^2-6x+9 = 0$

Answer:

Quadratic formula: $x = (-b \pm \sqrt{(b^2 - 4ac)}) / 2a$, with $D = b^2 - 4ac$

$(x-3)^2 =$

$= 0 \Rightarrow x$

$= 3$ (double root).

Question: Exercise 8.2: (xii) $1/(x+4) - 1/(x-4) = 4$

Answer:

Quadratic formula: $x = (-b \pm \sqrt{(b^2 - 4ac)}) / 2a$, with $D = b^2 - 4ac$

$-8/(x^2-16) =$

$= 4 \Rightarrow x^2-16$

$= -2 \Rightarrow x^2$

$= 14 \Rightarrow x$

$= \pm\sqrt{14}$.

Question: Exercise 8.3: Solve the following equations. 1. $x^4-8x^2-9 = 0$

Answer: Put $x^2 = t$: $(t-9)(t+1) = 0 \Rightarrow t = 9$ or $t = -1 \Rightarrow x = \pm 3$ or $x = \pm i$.

Question: Exercise 8.3: 2. $x^4-3x^2-4 = 0$

Answer: Put $x^2 = t$: $(t-4)(t+1) = 0 \Rightarrow t = 4$ or $t = -1 \Rightarrow x = \pm 2$ or $x = \pm i$.

Question: Exercise 8.3: 3. $12x^4-11x^2+2 = 0$

Answer: Put $x^2 = t$: $12t^2-11t+2 = (3t-2)(4t-1) = 0 \Rightarrow t = 2/3$ or $t = 1/4 \Rightarrow x = \pm\sqrt{(2/3)}$ or $x = \pm 1/2$.

Question: Exercise 8.3: 4. $(2x+3)/(x+1) + 6((x+1)/(2x+3)) = 7$

Answer: Put $t = (2x+3)/(x+1)$: $t+6/t = 7 \Rightarrow (t-1)(t-6) = 0 \Rightarrow t = 1$ or $t =$ 6.t =

$= 1 \Rightarrow x$

$= -2$.t =

$= 6 \Rightarrow x$

$= -3/4$.

Question: Exercise 8.3: 5. $\sqrt{((2x^2+1)/(x^2+1))} + 6\sqrt{((x^2+1)/(2x^2+1))} = 5$

Answer: Put $t = \sqrt{((2x^2+1)/(x^2+1))}$: $t+6/t = 5 \Rightarrow (t-2)(t-3) = 0 \Rightarrow t = 2$ or $t =$ 3.t =

$= 2 \Rightarrow x^2$

$= -3/2 \Rightarrow x$

$= \pm i\sqrt{(3/2)}$.t =

= 3 ⇒ x^2

= −8/7 ⇒ x

= ±2i√(2/7).

Question: Exercise 8.3: 6. 5^(x+1) + 5^(2−x) = 5^3+1

Answer: 5^3+1 = 126. Put t = 5^x: 5t+25/t = 126 ⇒ (5t−1)(t−25) = 0 ⇒ t = 1/5 or t = 25 ⇒ x = −1 or x = 2.

Question: Exercise 8.3: 7. 2^x + 16/2^x = 8

Answer: Put t = 2^x: t+16/t = 8 ⇒ $(t-4)^2$ = 0 ⇒ t = 4 ⇒ x = 2.

Question: Exercise 8.3: 8. 2(x/(x+1))² − 5(x/(x+1)) + 2 = 0

Answer: Put t = x/(x+1): $2t^2$−5t+2 = (2t−1)(t−2) = 0 ⇒ t = 1/2 or t = 2.t =

= 1/2 ⇒ x

= 1.t =

= 2 ⇒ x

= −2.

Question: Exercise 8.3: 9. 4(x/(x−1))² − 4(x/(x−1)) + 4 = 0

Answer: Put t = x/(x−1): t^2−t+1 = 0 ⇒ t = (1±i√3)/2. Then x/(x−1) =

(1±i√3)/2 gives x = (1±i√3)/2.

Question: Exercise 8.3: 10. 9^(x+2) − 6·3^(x+1) + 1 = 0

Answer: Put t = 3^x: 81t²−18t+1 = (9t−1)² = 0 ⇒ t = 1/9 ⇒ x = −2.

Question: Exercise 8.3: 11. 2^x + 2^(−x+6) − 20 = 0 [note: the printed first term is 2x; it is solved as 2^x, because with 2x the equation does not reduce to a quadratic]

Answer: Put t = 2^x: t+64/t−20 = 0 ⇒ t²−20t+64 = (t−4)(t−16) = 0 ⇒ t = 4 or t = 16 ⇒ x = 2 or x = 4.

Question: Exercise 8.3: 12. (x−1)(x+5)(x+8)(x+2) = 880

Answer: (x²+7x−8)(x²+7x+10) = 880. Put t = x²+7x: (t−8)(t+10) = 880 ⇒ (t+32)(t−30) = 0 ⇒ t = 30 or t = −32.t =

= 30 ⇒ x

= −10 or x

= 3.t =

= −32 ⇒ x

= (−7±i√79)/2.

Question: Exercise 8.3: 13. (x+1)(x+2)(x+3)(x+4) = 120

Answer: (x²+5x+4)(x²+5x+6) = 120. Put t = x²+5x: (t+4)(t+6) = 120 ⇒ (t+16)(t−6) = 0 ⇒ t = 6 or t = −16.t =

= 6 ⇒ x

= −6 or x

= 1.t =

= −16 ⇒ x

= (−5±i√39)/2.

Question: Exercise 8.3: 14. (x−2)(x+1)(x+3)(x−4) = 24

Answer: (x²−x−2)(x²−x−12) = 24. Put t = x²−x: (t−2)(t−12) = 24 ⇒ t(t−14) = 0 ⇒ t = 0 or t = 14.t =

= 0 ⇒ x

= 0 or x

= 1.t =

= 14 ⇒ x

= (1±√57)/2.

Question: Exercise 8.4: Solve the following equations. 1. x+√(x+5) = 7

Answer: √(x+5) =

= 7−x ⇒ x+5

= 49−14x+x² ⇒ (x−11)(x−4)

= 0 ⇒ x

= 11 or x

= 4. Check: x = 11 is extraneous (11+4 = 15 ≠ 7); x = 4 is correct. Solution set = {4}.

Question: Exercise 8.4: 2. √(x−2) = 8−x

Answer: x−2 =

= 64−16x+x² ⇒ (x−11)(x−6)

= 0 ⇒ x

= 11 or x

= 6. Check: x = 11 is extraneous (3 ≠ −3); x = 6 is correct. Solution set = {6}.

Question: Exercise 8.4: 3. √(7−5x) + √(13−5x) = 3√(4−2x)

Answer: Squaring: 20−10x+2√(25x²−100x+91) = 36−18x ⇒ √(25x²−100x+91) = 8−4x ⇒ 9x²−36x+27 = 0 ⇒ (x−1)(x−3) = 0 ⇒ x = 1 or x = 3. Check: x = 3 is outside the domain; x = 1 is correct (√2+√8 = 3√2). Solution set = {1}.

Question: Exercise 8.4: 4. $\sqrt{(x+2)} + \sqrt{(x+7)} = \sqrt{(6x+13)}$

Answer: Squaring: $2x+9+2\sqrt{(x^2+9x+14)} = 6x+13 \Rightarrow \sqrt{(x^2+9x+14)} = 2x+2$ $\Rightarrow 3x^2-x-10 = (3x+5)(x-2) = 0 \Rightarrow x = -5/3$ or $x = 2$. Check: $x = -5/3$ is extraneous; $x = 2$ is correct ($2+3 = 5$). Solution set = {2}.

Question: Exercise 8.4: 5. $\sqrt{(2x^2+3x+4)} + \sqrt{(2x^2+3x+9)} = 5$

Answer: Put $t = 2x^2+3x+4$: $\sqrt{t}+\sqrt{(t+5)} = 5 \Rightarrow \sqrt{(t+5)} = 5-\sqrt{t} \Rightarrow 10\sqrt{t} = 20 \Rightarrow$ $t = 4 \Rightarrow 2x^2+3x = 0 \Rightarrow x = 0$ or $x = -3/2$. Both check out. Solution set = $\{0, -3/2\}$.

Question: Exercise 8.4: 6. $\sqrt{(y^2-3y+9)} - \sqrt{(y^2-3y+36)} = 0$

Answer: $\sqrt{(y^2-3y+9)} = \sqrt{(y^2-3y+36)}$ gives $9 = 36$, which is impossible. Solution set = empty set.

Review Exercise 8, Question 1: Fill in the blanks. (i) A polynomial equation in which the degree of the variable is _____ is called a quadratic equation.

Answer: 2.

Review Exercise 8, Question 1: (ii) _____ is the standard form of a quadratic equation.

Answer: $ax^2+bx+c = 0$,

Review Exercise 8, Question 1: (iii) 3^x + 3^(2x) = 1 is called a _____ equation.

Answer: Exponential

Review Exercise 8, Question 1: (iv) The solution of 3^x = 9 is

2.

Review Exercise 8, Question 1: (v) The solution of $ax^2+bx+c = 0$ is _____.

Answer: $(-b\pm\sqrt{(b^2-4ac)})/(2a)$.

Review Exercise 8, Question 2: Tick () the correct answer. (i) The degree of a quadratic equation is:

(a) 1

✓ (b) 2

(c) 3

(d) 4

Review Exercise 8, Question 2: (ii) The standard form of a quadratic equation is:

✓ (a) $ax^2+bx+c = 0, a \neq 0$

(b) $ax^2+c = 0, a \neq 0$

(c) $ax^2+bx = 0, a \neq 0$

(d) $ax^3+bx^2+c = 0, a \neq 0$

Review Exercise 8, Question 2: (iii) The quadratic formula for $ax^2+bx+c = 0$ is:

Review Exercise 8, Question 2: (iv) The solution set of $x^2+10x+24 = 0$ is:

✓ (a) $\{-6, -4\}$

(b) $\{-6, 4\}$

(c) $\{6, 4\}$

(d) $\{6, -4\}$

Review Exercise 8, Question 2: (v) The maximum number of roots of a

quadratic equation is:

✓ (a) 2

(b) 3

(c) 1

(d) 4

Review Exercise 8, Question 2: (vi) The first-degree factors of $x^2-15x+56$ are:

Review Exercise 8, Question 2: (vii) A polynomial equation which does not change when x is replaced by 1/x is called:

(a) exponential equation

✓ (b) reciprocal equation

(c) radical equation

(d) none of these

Review Exercise 8, Question 2: (viii) The equation 3^x+3^(2−x)+6 = 0 is a model of _____:

✓ (a) exponential equation

(b) radical equation

(c) reciprocal equation

(d) none of these

Review Exercise 8, Question 2: (ix) The solution set of $4x^2-16 = 0$ is:

(a) {±4}

(b) {4}

✓ (c) {±2}

(d) none of these

Review Exercise 8, Question 2: (x) The equation $2x^4-3x^3+7x^2-3x+2 = 0$ is called a model of _____:

(a) reciprocal equation

(b) radical equation

(c) exponential equation

(d) none of these

Question: Review Exercise 8, Question 3: True and false answers. Read the following sentences carefully and circle "T" for a true statement and "F" for a false one. (i) Every quadratic equation can be solved by factorization.

Answer: F, because some equations do not factor over the rational numbers.

Question: Review Exercise 8, Question 3: (ii) Every quadratic equation has two roots.

Answer: T (including double roots and complex roots).

Question: Review Exercise 8, Question 3: (iii) A quadratic equation has no solution.

Answer: F, because every quadratic equation has (real or complex) roots.

Question: Review Exercise 8, Question 3: (iv) $ax^2+bx+c = 0$ is a quadratic equation in x if $a = 0$ and b, c are real numbers.

Answer: F, because a quadratic equation requires $a \neq 0$.

Question: Review Exercise 8, Question 3: (v) Extraneous roots satisfy the equation.

Answer: F, because extraneous roots do not satisfy the original equation.

Question: Review Exercise 8, Question 3: (vi) Extraneous roots do not satisfy the equation.

Answer: T.

Question: Review Exercise 8, Question 3: (vii) In a quadratic equation the maximum power of the variable is 2.

Answer: T.

9. Congruent Triangles

Question: Exercise 9.1, Question 1: In the given figure mAB = mCB and ∠A ≅ ∠C, prove that ΔABD ≅ ΔCBE.

Answer: ∠A ≅ ∠C (given), AB ≅ CB (given), ∠ABD ≅ ∠CBE (vertical angles). Therefore by ASA, ΔABD ≅ ΔCBE.

Question: Exercise 9.1, Question 2: From any point on the angle bisector line, perpendiculars are drawn to the two sides of the angle. Prove that these perpendiculars are equal in measure.

Answer: Let V be the vertex of the angle and P the point; PQ and PR are perpendiculars to the two sides. In ΔPQV and ΔPRV: ∠PQV = ∠PRV = 90°, ∠QVP = ∠RVP (VP bisects the angle), VP is common. Therefore by AAS the two triangles are congruent, so PQ = PR.

Question: Exercise 9.1, Question 3: In the figure we have ΔACE ≅ ΔBDE. Given mAC = mBD = 3 cm, ∠A = (3x+1)°, m∠E = (3y−2)° and m∠B = (x+35)°, find the values of x and y. [flagged: the value of y cannot be determined from the given data; only x is found]

Answer: From the correspondence A ↔ B, ∠A = ∠B: 3x+1 = x+35 ⇒ x = 17 (so ∠A = ∠B = 52°). The given data is not enough for y, because ∠E is the same vertical angle in both triangles and no other independent condition is given.

Question: Exercise 9.1, Question 4: In the figure ΔDEF ≅ ΔPQR, so mDE = (6x+1) cm, mEF = 8 cm, mRQ = (5y−7) cm and mPQ = (10x−19) cm. Find the values of x and y.

Answer: From the correspondence D ↔ P, E ↔ Q, F ↔ R: DE = PQ ⇒ 6x+1 = 10x−19 ⇒ x = 5.EF =

= QR ⇒ 8

= 5y−7 ⇒ y

= 3.

Question: Exercise 9.2, Question 1: ABC is a triangle in which m∠A = 35° and m∠B = 100°, BD ⊥ AC. Prove that ΔBDC is an isosceles triangle.

Answer: In ΔABC, ∠C = 180° − 35° − 100° = 45°. In ΔBDC, ∠BDC = 90° (BD ⊥ AC) and ∠C = 45°, so ∠DBC = 180° − 90° − 45° = 45°. Since ∠DBC = ∠C, ΔBDC is an isosceles triangle (DC = DB).

Question: Exercise 9.2, Question 2: If the bisector of any angle of a triangle is perpendicular to the opposite side, then that triangle is an isosceles triangle.

Answer: Let AD be the bisector of ∠A meeting BC at D, with AD ⊥ BC. In ΔABD and ΔACD: ∠BAD = ∠CAD (AD bisects), AD is common, ∠ADB = ∠ADC = 90°. By ASA the two triangles are congruent, so AB = AC, i.e. ΔABC is an isosceles triangle.

Question: Exercise 9.2, Question 3: ABC is a triangle in which m∠B = 45° and CD ⊥ AB. Prove that ΔDBC is an isosceles triangle.

Answer: In ΔDBC, ∠CDB = 90° (CD ⊥ AB) and ∠B = 45°, so ∠DCB = 180° − 90° − 45° = 45°. Since ∠DCB = ∠B, ΔDBC is an isosceles triangle (DB = DC).

Question: Exercise 9.3, Question 1: ABC is an isosceles triangle. D is the midpoint of its base BC. Prove that AD is the bisector of ∠A and AD ⊥ BC.

Answer: In ΔABD and ΔACD: AB = AC (isosceles), BD = DC (D is the midpoint), AD is common. By SSS the two triangles are congruent, so ∠BAD = ∠CAD (AD is the bisector of ∠A) and ∠ADB = ∠ADC. Being a linear pair, each is 90°, so AD ⊥ BC.

Question: Exercise 9.3, Question 2: ΔABC and ΔDBC are isosceles triangles on the common base BC. Prove that AD is the perpendicular bisector of BC.

Answer: AB = AC (ΔABC is isosceles) and DB = DC (ΔDBC is isosceles). Points equidistant from B and C lie on the perpendicular bisector of BC, so both A and D lie on it. Therefore AD is the perpendicular bisector of BC.

Question: Exercise 9.3, Question 3: PQRS is a square in which X, Y and Z are the midpoints of sides PQ, QR and RS respectively. Prove that ΔPXY ≅ ΔSZY.

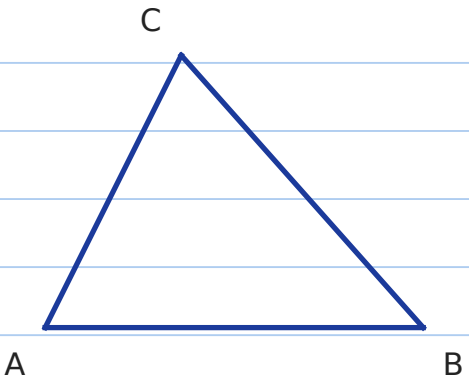
Answer: PX = SZ (halves of equal sides), PY = SY (by Pythagoras in the equal right triangles PQY and SRY), XY = ZY (by Pythagoras in the equal right triangles XQY and ZRY). Therefore by SSS, ΔPXY ≅ ΔSZY.

Question: Exercise 9.3, Question 4: Prove that the medians of an isosceles triangle are equal.

Answer: In isosceles ΔABC with AB = AC, let M be the midpoint of AB and N the midpoint of AC. Then BM = CN (halves of equal sides). In ΔBCM and ΔCBN: BC is common, ∠B = ∠C (base angles), BM = CN. By SAS the two triangles are congruent, so CM = BN, i.e. the medians to the equal sides are equal.

Question: Exercise 9.4, Question 1: Prove that in an isosceles triangle, the perpendiculars drawn from the base vertices to the opposite sides are equal to each other.

Answer: In isosceles ΔABC with AB = AC, let BD ⊥ AC and CE ⊥ AB. In ΔBCE and ΔCBD: ∠BEC = ∠CDB = 90°, ∠EBC = ∠DCB (base angles), BC is common. By AAS the two triangles are congruent, so CE = BD.



Question: Exercise 9.4, Question 2: Prove that if the bisector of an angle of a triangle is perpendicular to the opposite side, then the triangle is an isosceles triangle.

Answer: Let AD be the bisector of ∠A meeting BC at D, with AD ⊥ BC. In ΔABD and ΔACD: ∠BAD = ∠CAD, AD is common, ∠ADB = ∠ADC = 90°. By ASA the two triangles are congruent, so AB = AC, i.e. the triangle is isosceles.

Question: Exercise 9.4, Question 3: Prove that in an isosceles triangle the median to the base, the bisector of the vertex angle and the altitude to the base are the same line.

Answer: In isosceles ΔABC with AB = AC, let D be the midpoint of BC. From ΔABD ≅ ΔACD (SSS), ∠BAD = ∠CAD (AD is the bisector of the vertex angle) and ∠ADB = ∠ADC = 90° (AD is perpendicular to the base). So AD

is at once median, bisector and altitude.

Question: Exercise 9.4, Question 4: Prove that if the three altitudes of a triangle are equal, then the triangle is an isosceles triangle.

Answer: Area of a triangle = (1/2)×base×altitude. If all three altitudes are equal then all three bases (sides) are also equal, i.e. the triangle is equilateral, which is also isosceles.

Question: Review Exercise 9, Question 1: If ΔABC ≅ ΔDEF, m∠F equals: (A) 90° (B) 60° (C) 30° (D) 20°

Answer: (C) 30°, because in ΔABC, ∠C = 180° − 120° − 30° = 30° and by correspondence ∠F = ∠C.

Question: Review Exercise 9, Question 2: Identify the true and false statements from the following. (i) The sum of the measures of all angles of a quadrilateral is 360°.

Answer: True.

Question: Review Exercise 9, Question 2: (ii) The sum of the measures of all angles of a triangle is 270°.

Answer: False, the sum is 180°.

Question: Review Exercise 9, Question 2: (iii) In an isosceles triangle the angles are of the same measure.

Answer: False, only the two base angles are equal.

Question: Review Exercise 9, Question 2: (iv) In a triangle there are two obtuse angles.

Answer: False, a triangle has at most one obtuse angle.

Question: Review Exercise 9, Question 2: (v) In an isosceles triangle, corresponding angles and corresponding sides are equal in measure.

Answer: True (corresponding parts of the congruent halves are equal).

Review Exercise 9, Question 3: Fill in the blanks to complete the sentences. (i) ΔABC ↔ ΔDEF, then the correspondence of AC is _____.

Answer: DF.

Review Exercise 9, Question 3: (ii) ΔKLM ↔ ΔPQR, then the correspondence of ∠MKL is _____.

Answer: ∠QPR.

Review Exercise 9, Question 3: (iii) In an isosceles triangle the base angles are _____.

Answer: Equal.

Review Exercise 9, Question 3: (iv) If each angle of a triangle measures 60°, the triangle is _____.

Answer: Equilateral.

Review Exercise 9, Question 3: (v) In an obtuse triangle, the side opposite the obtuse angle is called the _____.

Answer: Longest side.

Review Exercise 9, Question 3: (vi) In an obtuse triangle, the sum of the measures of the acute angles is _____.

Answer: Less than 90°.

Review Exercise 9, Question 4: Circle a, b, c or d for the correct answer. (i) Which of the following is NOT a suitable form for the congruence of two triangles:

(a) A.S.A ≅ A.S.A

(b) H.S ≅ H.S

(c) S.A.A ≅ S.A.A

✓ (d) A.A.A ≅ A.A.A

Review Exercise 9, Question 4: (ii) In triangle ΔABC, if ∠A ≅ ∠B, then the bisector of angle _____ divides the triangle into two congruent triangles:

(a) ∠A

(b) ∠B

✓ (c) ∠C

(d) none of these

Review Exercise 9, Question 4: (iii) The diagonal of _____, does not divide it into two congruent triangles:

(a) rectangle

✓ (b) trapezium

(c) parallelogram

(d) square

Review Exercise 9, Question 4: (iv) In an acute triangle, how many angles are acute:

(a) 1

(b) 2

(c) 3

(d) more than two



10. Parallelograms and Triangles

Question: Exercise 10.1, Question 1: The line joining the midpoints of any two opposite sides of a parallelogram is parallel to its other two sides.

Answer: In parallelogram ABCD, let M be the midpoint of AD and N the midpoint of BC. AM = BN (halves of equal opposite sides) and AM ∥ BN, so AMNB is a parallelogram, hence MN ∥ AB and MN ∥ CD.

Question: Exercise 10.1, Question 2: The interior angles on any side of a parallelogram are supplementary.

Answer: In parallelogram ABCD, AD ∥ BC and AB is a transversal, so ∠A + ∠B = 180° (consecutive interior angles). Similarly the angles on every side are supplementary.

Question: Exercise 10.1, Question 3: Prove that the bisectors of two angles on the same side of a parallelogram intersect each other at a right angle.

Answer: ∠A + ∠B = 180° (supplementary). At the intersection P of the bisectors, in ΔAPB: ∠PAB + ∠PBA = (1/2)(∠A + ∠B) = 90°, so ∠APB = 90°, i.e. the bisectors are perpendicular to each other.

Question: Exercise 10.1, Question 4: If the diagonals of a quadrilateral bisect each other, then it is a parallelogram.

Answer: In quadrilateral ABCD the diagonals AC and BD meet at O with AO = OC and BO = OD. ∠AOB = ∠COD (vertical angles) so ΔAOB ≅ ΔCOD (SAS) ⇒ AB = CD and ∠OAB = ∠OCD ⇒ AB ∥ CD (alternate angles). Similarly AD ∥ BC. Therefore ABCD is a parallelogram.

Question: Exercise 10.1, Question 5: The opposite angles of any parallelogram are equal.

Answer: From $\angle A + \angle B = 180°$ and $\angle B + \angle C = 180°$ (supplementary), $\angle A = \angle C$. Similarly $\angle B = \angle D$. Therefore opposite angles are equal.

Question: Exercise 10.2, Question 1: Prove that a quadrilateral is a parallelogram if its opposite angles are equal.

Answer: Let $\angle A = \angle C$ and $\angle B = \angle D$. Sum of the quadrilateral's angles: $2\angle A + 2\angle B = 360° \Rightarrow \angle A + \angle B = 180° \Rightarrow AD \parallel BC$. Similarly $\angle B + \angle C = 180° \Rightarrow AB \parallel CD$. Therefore ABCD is a parallelogram.

Question: Exercise 10.2, Question 2: Prove that a quadrilateral is a parallelogram if its diagonals bisect each other.

Answer: The diagonals AC and BD meet at O with $AO = OC$ and $BO = OD$. $\Delta AOB \cong \Delta COD$ (SAS) $\Rightarrow AB = CD$ and $AB \parallel CD$ (alternate angles). Similarly $AD \parallel BC$. Therefore ABCD is a parallelogram.

Question: Exercise 10.2, Question 3: If a quadrilateral is a parallelogram, then its diagonals also form congruent triangles.

Answer: In parallelogram ABCD, diagonal AC divides it into ΔABC and ΔCDA, which are congruent by SSS ($AB = CD$, $BC = AD$, AC common). Similarly diagonal BD gives $\Delta ABD \cong \Delta CDB$.

Question: Exercise 10.2, Question 4: If the angles on every side of a quadrilateral are supplementary, then it is a parallelogram.

Answer: $\angle A + \angle B = 180° \Rightarrow AD \parallel BC$ (consecutive interior angles supplementary). $\angle B + \angle C = 180° \Rightarrow AB \parallel CD$. Therefore ABCD is a parallelogram.

Question: Exercise 10.3, Question 1: If the midpoints of two sides of any triangle are joined, then it is parallel to the third side, and if its length is 4 cm, what will be the length of the third side?

Answer: By the midsegment theorem the joining segment is half the third side, so the third side = 2×4 = 8 cm.

Question: Exercise 10.3, Question 2: Prove that the line segments joining the midpoints of opposite sides of a quadrilateral bisect each other.

Answer: In quadrilateral ABCD, let P, Q, R, S be the midpoints of AB, BC, CD, DA respectively. In ΔABC, PQ ∥ AC and PQ = (1/2)AC; in ΔCDA, SR ∥ AC and SR = (1/2)AC. So PQRS is a parallelogram, hence its diagonals PR and QS bisect each other.

Question: Exercise 10.3, Question 3: Joining the midpoints of the four sides of a quadrilateral in order forms a new quadrilateral which is a parallelogram.

Answer: In quadrilateral ABCD, let P, Q, R, S be the midpoints of AB, BC, CD, DA respectively. From ΔABC and ΔCDA, PQ ∥ SR and PQ = SR = (1/2)AC, and from ΔBCD and ΔDAB, QR ∥ PS and QR = PS = (1/2)BD. Therefore PQRS is a parallelogram.

Question: Exercise 10.3, Question 4: Prove that a line through the midpoint of one side of a triangle parallel to the second side bisects the third side.

Answer: In ΔABC, let D be the midpoint of AB and DE ∥ BC meeting AC at E. From similar triangles, AD/AB = AE/AC ⇒ 1/2 = AE/AC ⇒ AE = (1/2)AC, i.e. E is the midpoint of AC.

Question: Exercise 10.3, Question 5: Prove that joining the midpoints of

the three sides of a triangle forms four small triangles which are congruent to each other.

Answer: By the midsegment theorem, the sides of each small triangle are halves of the corresponding sides of the big triangle, so all four small triangles have the same three sides. Therefore by SSS all four triangles are congruent to each other.

Question: Exercise 10.4, Question 1: Prove that if the three medians of a triangle are equal, then it is an isosceles triangle.

Answer: From the median-length formula (Apollonius): $m_a^2 = (2b^2+2c^2-a^2)/4$ and $m_b^2 = (2a^2+2c^2-b^2)/4$.If $m_a =$

$= m_b$ then $2b^2+2c^2-a^2$

$= 2a^2+2c^2-b^2 \Rightarrow 3b^2$

$= 3a^2 \Rightarrow a$

$= b$. Therefore the triangle is isosceles.

Question: Exercise 10.4, Question 2: If the three medians PB, QC and RA of triangle PQR meet at point G, prove that G is the centroid.

Answer: The three medians of a triangle meet at a single point, which is called the centroid. By the centroid theorem, G divides each median in the ratio 2:1, so G is the centroid.

Question: Exercise 10.4, Question 3: In the given figure, the length of GR is 2 cm. Find the length of AG.

Answer: G is the centroid, which divides the median RA in the ratio 2:1: RG:GA = 2:1. Since GR = 2 cm, AG = 1 cm.

Question: Exercise 10.5, Question 1: The angles of the new triangle formed by joining the midpoints of the sides of any triangle are equal to those of the original triangle.

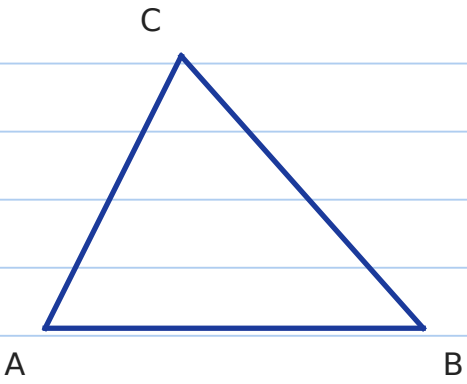
Answer: The four small triangles formed by joining the midpoints are congruent to each other and similar to the original triangle (SSS similarity, ratio 1:2), so the angles of the new triangle equal the angles of the original triangle.

Question: Exercise 10.5, Question 2: The line segment joining the midpoints of the non-parallel sides of a trapezium is also parallel to the other two sides and is equal to half the sum of those two sides.

Answer: In trapezium ABCD with AB ∥ CD, let M be the midpoint of AD and N the midpoint of BC. Diagonal AC meets MN at P. In ΔACD, MP = (1/2)CD and MP ∥ CD; in ΔABC, PN = (1/2)AB and PN ∥ AB. Hence MN ∥ AB ∥ CD and MN = MP+PN = (1/2)(AB+CD).

Question: Exercise 10.5, Question 3: Every line segment drawn from the vertex of any triangle to the base is bisected by the line segment joining the midpoints of the other two sides.

Answer: In ΔABC, let M be the midpoint of AB and N the midpoint of AC; MN is the midsegment. A line AD from A to any point D on BC meets MN at P. In ΔABD, M is the midpoint of AB and MP ∥ BD, so P is the midpoint of AD, i.e. MN bisects AD.



Review Exercise 10, Question 1: Fill in the blanks. (i) In a parallelogram opposite sides are _____.

Answer: Equal.

Review Exercise 10, Question 1: (ii) In a parallelogram opposite angles are _____.

Answer: Equal.

Review Exercise 10, Question 1: (iii) In a triangle the medians are _____.

Answer: Three.

Review Exercise 10, Question 1: (iv) The diagonals of a parallelogram _____ each other.

Answer: Bisect.

Review Exercise 10, Question 1: (v) The consecutive angles of a parallelogram are _____.

Answer: Supplementary.

Review Exercise 10, Question 1: (vi) The medians of a triangle are _____.

Answer: Concurrent at the centroid.

Review Exercise 10, Question 1: (vii) The sum of the measures of the interior angles of a quadrilateral is equal to _____.

Answer: 360°.

Review Exercise 10, Question 2: Tick the correct answer. (i) The diagonals of a square are _____ with each other: (a) perpendicular (b) unequal (c) equal (d) both (a) and (c)

Review Exercise 10, Question 2: (ii) The sum of the measures of the interior angles of a quadrilateral is _____:

(a) 2 right angles

✓ (b) 4 right angles

(c) 3 right angles

(d) none

Review Exercise 10, Question 2: (iii) In triangle ABC, the segment joining the midpoints of AB and AC is 3.5 cm, then mBC is:

(a) 4.5 cm

(b) 5.5 cm

(c) 6 cm

✓ (d) 7 cm

Review Exercise 10, Question 2: (iv) The two medians AD and BE meet at point G. If mGD = 1.7 cm, then mAG is:

(a) 2.7 cm

(b) 8.85 cm

✓ (c) 3.4 cm

(d) 5.1 cm

Review Exercise 10, Question 2: (v) In parallelogram ABCD, the sum of the measures of angles ∠A and ∠C is 130, then m∠C is:

(a) 25°

(b) 115°

✓ (c) 65°

(d) none

Review Exercise 10, Question 2: (vi) If the opposite angles of a quadrilateral are equal in measure and none of them is an obtuse angle, then the quadrilateral is _____:

(a) square

(b) parallelogram

(c) trapezium

✓ (d) rectangle

Review Exercise 10, Question 2: (vii) The centroid is a common point of _____:

✓ (a) the medians of the triangle

(b) the diagonals of the parallelogram

(c) the bisectors of the triangle's angles

(d) the perpendicular bisector of the triangle

Review Exercise 10, Question 2: (viii) The median of any triangle is the

point _____:

(a) at equal distance from the vertex

(b) at equal distance from the midpoint of the side

(c) at equal distance from the altitude

✓ (d) none

11. Line Bisectors and Angles Bisectors

Question: Exercise 11.1, Question 1: Prove that the perpendicular bisectors of any two sides of any triangle meet at a point which is equidistant from all three vertices.

Answer: Let the perpendicular bisectors of AB and AC meet at O. O is on the perpendicular bisector of AB ⇒ OA = OB. O is on the perpendicular bisector of AC ⇒ OA = OC. So OA = OB = OC, i.e. O is equidistant from all three vertices.

Question: Exercise 11.1, Question 2: Prove that the centre of a circle lies on the perpendicular bisector of a chord.

Answer: Let AB be a chord of a circle, and drop perpendicular OM from centre O to AB. In ΔOAM and ΔOBM: OA = OB (radii), OM is common, ∠OMA = ∠OMB = 90°. By RHS the two triangles are congruent ⇒ AM = MB. So OM is the perpendicular bisector of AB, i.e. O lies on the perpendicular bisector of the chord.

Question: Exercise 11.1, Question 3: Where will be the centre of the circle passing through three non-collinear points, and why?

Answer: The centre will be at the point where the perpendicular bisectors of the segments joining the three points meet (the circumcentre), because that point is equidistant from all three points.

Question: Exercise 11.1, Question 4: If two circles intersect each other at points A and B, then prove that the line passing through their centres is the perpendicular bisector of AB.

Answer: Let O1 and O2 be the centres. O1A = O1B (radii of the first circle) and O2A = O2B (radii of the second). So O1 and O2 are both equidistant from A and B, i.e. both lie on the perpendicular bisector of AB. Therefore the line O1O2 is the perpendicular bisector of AB.

Question: Exercise 11.1, Question 5: Three markets A, B and C are not on a line. A businessman wants to build a mosque at a place equidistant from these markets. After fixing the location of the mosque, prove that the place is equidistant from all three markets.

Answer: Build the mosque at the circumcentre — the point where the perpendicular bisectors of AB and AC meet. By the perpendicular bisector property, that point is equidistant from A and B and also from A and C, so it is equidistant from all three markets.

Question: Exercise 11.2, Question 1: Prove that the circumcentre of an acute triangle lies inside the triangle.

Answer: For any side BC, the circumcentre O lies on the same side of BC as the opposite vertex A, when $\angle A < 90°$ (if $\angle A = 90°$, O would lie on BC, and if $\angle A > 90°$, O would lie on the opposite side). Since all three angles are acute, O lies inside the triangle.

Question: Exercise 11.2, Question 2: Prove that the perpendicular bisectors of the four sides of an isosceles trapezium meet at the same point.

Answer: In isosceles trapezium ABCD with AB ∥ CD and AD = BC, the axis of symmetry is the perpendicular bisector of both AB and CD. The perpendicular bisectors of AD and BC meet this axis at the same point by symmetry, so the perpendicular bisectors of all four sides meet at one

point (the circumcentre).

Question: Exercise 11.2, Question 3: Prove that the altitudes of a triangle meet at the same point.

Answer: Place A at (0, 0), B at (c, 0) and C at (u, v). The altitude from C is x = u. The altitude from B is y = (−u/v)(x−c). At x = u: y = u(c−u)/v. The altitude from A is y = (−(u−c)/v)x. At x = u: y = u(c−u)/v. All three altitudes pass through the same point (u, u(c−u)/v), i.e. they meet at one point (the orthocentre).

Question: Exercise 11.3, Question 1: Two isosceles triangles have a common base. Prove that the line joining their vertices bisects the common base at a right angle.

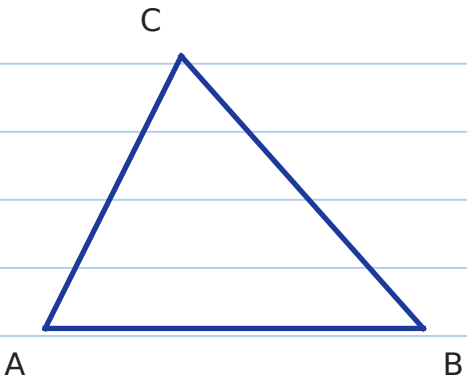
Answer: Let ΔABC and ΔDBC be isosceles triangles with common base BC, AB = AC and DB = DC. Both A and D are equidistant from B and C, so AD is the perpendicular bisector of BC, i.e. AD bisects BC at a right angle.

Question: Exercise 11.3, Question 2: The bisectors of the angles of a triangle bisect the opposite sides. Prove that the triangle is an isosceles triangle.

Answer: The bisector of ∠A bisects BC at D. By the angle bisector theorem, BD/DC = AB/AC, and since BD = DC, AB = AC. Similarly the other angle bisectors give all sides equal, so the triangle is isosceles (equilateral).

Question: Exercise 11.3, Question 3: In isosceles triangle ABC with mAB = mAC, prove that the perpendiculars drawn from vertices B and C to the opposite sides are equal.

Answer: Let BD ⊥ AC and CE ⊥ AB. In ΔBCE and ΔCBD: ∠BEC = ∠CDB = 90°, ∠EBC = ∠DCB (base angles), BC is common. By AAS the two triangles are congruent, so CE = BD.



Question: Review Exercise 11, Question 1: Prove that if two altitudes of a triangle are equal, then the triangle is isosceles.

Answer: Area of a triangle = (1/2)×base×altitude. If the altitudes to two sides are equal, then those sides must be equal, so the triangle is isosceles.

Question: Review Exercise 11, Question 2: Prove that the incentre of a triangle is equidistant from all three sides and lies on the bisectors of all three angles.

Answer: Let I be the intersection of the bisectors of ∠B and ∠C. I is on the bisector of ∠B ⇒ equidistant from BA and BC. I is on the bisector of ∠C ⇒ equidistant from CB and CA. So I is equidistant from all three sides, and being equidistant from BA and CA, I also lies on the bisector of ∠A.

Question: Review Exercise 11, Question 3: Write (T) for true and (F) for false against the following statements. (i) Bisecting a side means we divide the given side into two parts.

Answer: T (into two equal parts).

Question: Review Exercise 11, Question 3: (ii) Each base angle of an isosceles obtuse triangle is 45.

Answer: F (the base angles can take any value less than 90°, e.g. vertex 100° gives base angles 40°).

Question: Review Exercise 11, Question 3: (iii) Triangles with equal sides also have equal angles.

Answer: T (by SSS congruence, corresponding angles are equal).

Review Exercise 11, Question 4: Choose the correct answer. (i) In an acute triangle there are _____ acute angles:

(a) one

(b) two

✓ (c) three

(d) none

Review Exercise 11, Question 4: (ii) A point on a line segment which is equidistant from its two ends is its _____:

✓ (a) perpendicular bisector

(b) perpendicular

(c) centre

(d) midpoint

Review Exercise 11, Question 4: (iii) The _____ of the sides of any acute triangle intersect each other inside the triangle:

(a) altitudes

(b) perpendicular bisectors

(c) wide

(d) acute

Review Exercise 11, Question 4: (iv) The bisectors of the angles of a triangle are _____:

(a) concurrent

(b) parallel

(c) do not intersect

(d) equal

12. Sides and Angles of a Triangle

Question: Exercise 12.1, Question 1: O is an interior point of triangle ABC. Prove that mOA + mOB + mOC > (1/2)(mAB + mBC + mCA).

Answer: In ΔOAB: OA + OB > AB; in ΔOBC: OB + OC > BC; in ΔOCA: OC + OA > CA (triangle inequality). Adding the three: 2(OA+OB+OC) > AB+BC+CA ⇒ OA+OB+OC > (1/2)(AB+BC+CA).

Question: Exercise 12.1, Question 2: In triangle ABC, m∠B = 70° and m∠C = 45°. Which side of the triangle will be the largest?

Answer: ∠A =

= 180° − 70° − 45°

= 65°. The largest angle is ∠B = 70°, so the side opposite it, AC, is the largest.

Question: Exercise 12.1, Question 3: In triangle ABC, m∠A = 58° and m∠B = 65°. Which side of the triangle will be the smallest?

Answer: ∠C =

= 180° − 58° − 65°

= 57°. The smallest angle is ∠C = 57°, so the side opposite it, AB, is the smallest.

Question: Review Exercise 12, Question 1: Tick (✓) the true and false

statements. (i) The sum of two sides of a triangle is greater than the third side.

Answer: True.

Question: Review Exercise 12, Question 1: (ii) The difference of two sides of a triangle is greater than the third side.

Answer: False, the difference is less than the third side.

Question: Review Exercise 12, Question 1: (iii) The perpendicular distance is the shortest distance from a point to a line.

Answer: True.

Question: Review Exercise 12, Question 1: (iv) The largest angle of an obtuse triangle is 100°.

Answer: False, an obtuse angle can take any value between 90° and 180°.

Question: Review Exercise 12, Question 1: (v) A perpendicular on a line always makes a 90° angle.

Answer: True.

Review Exercise 12, Question 2: Fill in the blanks to complete the sentences. (i) The _____ of an obtuse triangle is the largest side.

Review Exercise 12, Question 2: (ii) In any triangle, the sum of the sides forming the obtuse angle is _____.

Answer: third side.

Review Exercise 12, Question 2: (iii) In triangle ABC, if m∠A = 50° and

m∠B = 30°, then the _____ side will be larger than the other two sides.

Answer: 100°).

Review Exercise 12, Question 2: (iv) The length of the diagonal of a quadrilateral is _____ than the sum of its surrounding sides.

Answer: Less.

Review Exercise 12, Question 3: Tick () the correct answer. (i) If one side of an isosceles triangle measures 6 cm, then its median will be _____ than 9 cm:

✓ (a) less

(b) greater

(c) equal

(d) none

Review Exercise 12, Question 3: (ii) If the perimeter of a rectangle is 22 cm, then the length of its diagonal will be _____ than 11 cm:

(a) equal

(b) greater

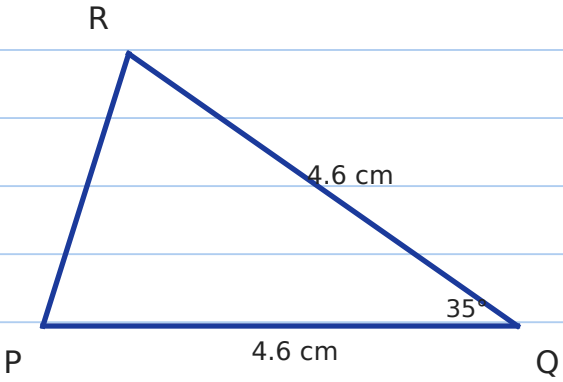
✓ (c) less

(d) none

13. Practical Geometry - Triangles

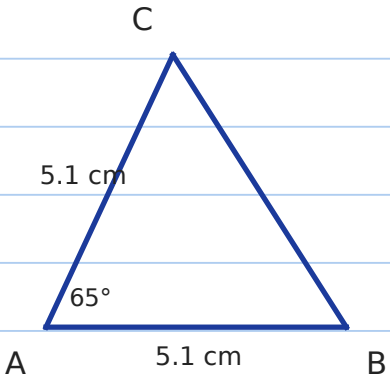
Question: Exercise 13.1, Question 1: Construct triangle PQR with m∠Q = 35° and mPQ = mQR = 4.6 cm.

Answer: (i) Draw QR = 4.6 cm. (ii) At Q construct an angle of 35°. (iii) Mark QP = 4.6 cm on the ray. (iv) Join PR. ΔPQR is the required triangle.



Question: Exercise 13.1, Question 2: Construct triangle ABC with m∠A = 65° and mAB = mAC = 5.1 cm.

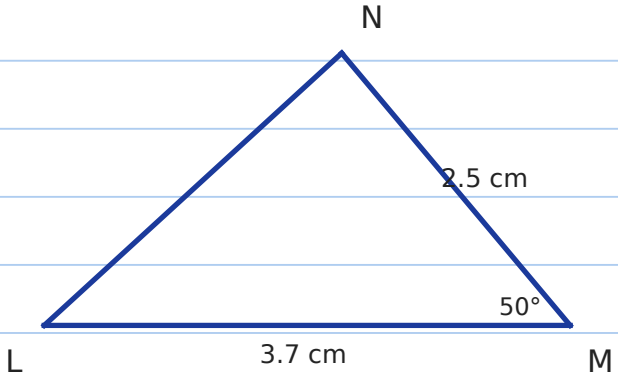
Answer: (i) Draw AB = 5.1 cm. (ii) At A construct an angle of 65°. (iii) Mark AC = 5.1 cm on the ray. (iv) Join BC. ΔABC is the required triangle.



Question: Exercise 13.1, Question 3: Construct triangle LMN with ∠M =

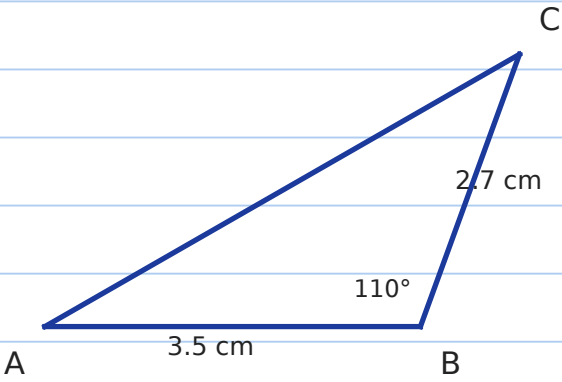
50° and mLM = 3.7 cm, mMN = 2.5 cm.

Answer: (i) Draw MN = 2.5 cm. (ii) At M construct an angle of 50°. (iii) Mark ML = 3.7 cm on the ray. (iv) Join LN. ΔLMN is the required triangle.



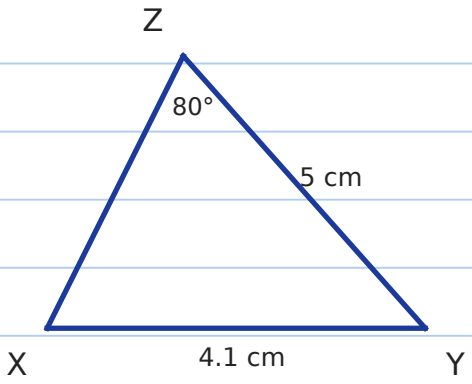
Question: Exercise 13.1, Question 4: Construct triangle ABC with ∠B = 110° and mAB = 3.5 cm, mBC = 2.7 cm.

Answer: (i) Draw BC = 2.7 cm. (ii) At B construct an angle of 110°. (iii) Mark BA = 3.5 cm on the ray. (iv) Join AC. ΔABC is the required triangle.



Question: Exercise 13.1, Question 5: Construct triangle XYZ with ∠Z = 80° and mXY = 4.1 cm, mYZ = 5 cm.

Answer: (i) Draw YZ = 5 cm. (ii) At Z construct an angle of 80°. (iii) With centre Y and radius 4.1 cm draw an arc cutting the other ray at X. (iv) Join XY. ΔXYZ is the required triangle.

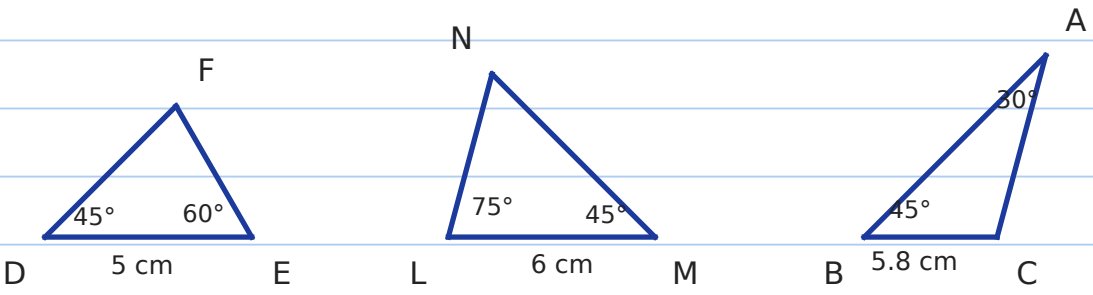


mXY = 4.1 cm, mYZ = 5 cm, ∠Z = 80°

Question: Exercise 13.1, Question 6: Construct the following triangles: (i) mDE = 5 cm, m∠D = 45° and m∠E = 60° (ii) mLM = 6 cm, m∠L = 75° and m∠M = 45° (iii) mBC = 5.8 cm, m∠A = 30° and m∠B = 45°.

Answer: (i) Draw DE = 5 cm, construct angles 45° at D and 60° at E; the rays meet at F. (ii) Draw LM = 6 cm, construct angles 75° at L and 45° at M; the rays meet at N. (iii) ∠C = 180° − 30° − 45° = 105°; draw BC = 5.8 cm, construct 45° at B and 105° at C; the rays meet at A.

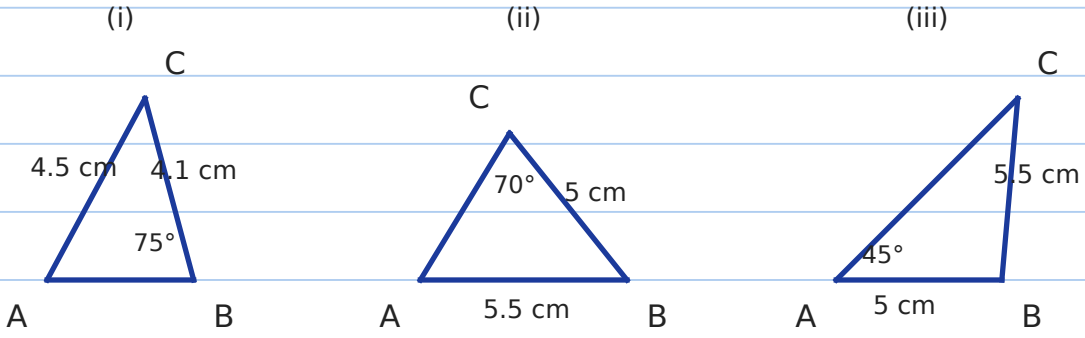
(i) (ii) (iii)



Question: Exercise 13.1, Question 7: Construct triangle ABC when the measures of two of its sides and the angle opposite one side are given: (i) mBC = 4.1 cm, mAC = 4.5 cm and m∠B = 75° (ii) mBC = 5 cm, mAB = 5.5 cm and m∠C = 70° (iii) mAB = 5 cm, mBC = 5.5 cm and m∠A = 45°.

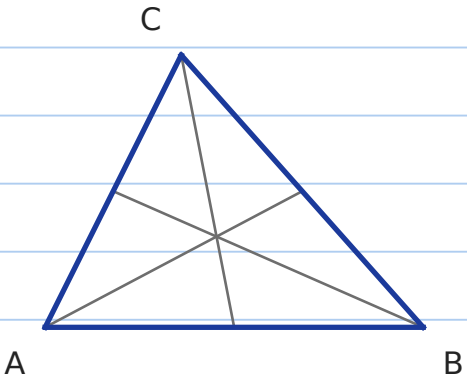
Answer: (i) Construct ∠B = 75°, mark BC = 4.1 cm on one arm, with centre C and radius 4.5 cm draw an arc cutting the other arm at A, join AC.

(ii) Construct ∠C = 70°, mark CB = 5 cm on one arm, with centre B and radius 5.5 cm draw an arc cutting the other arm at A, join AB. (iii) Construct ∠A = 45°, mark AB = 5 cm on one arm, with centre B and radius 5.5 cm draw an arc cutting the other arm at C, join BC.



Question: Exercise 13.2, Question 1: Draw a triangle and draw its medians. Prove that they meet at the same point.

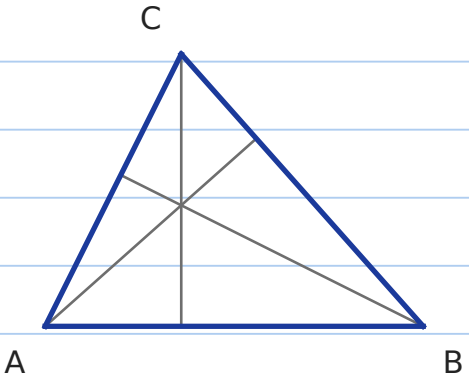
Answer: Draw ΔABC, bisect each side to get the midpoints, join the vertices to the opposite midpoints. All three medians meet at the same point G (the centroid), which divides each median in the ratio 2:1.



Question: Exercise 13.2, Question 2: Draw a triangle and draw its altitudes. Prove that they meet at the same point.

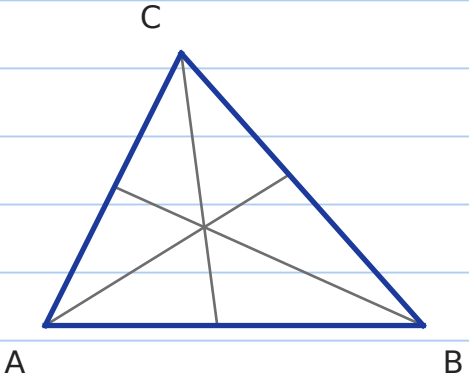
Answer: Draw ΔABC, draw the perpendicular from each vertex to the opposite side. All three altitudes meet at the same point H (the

orthocentre).



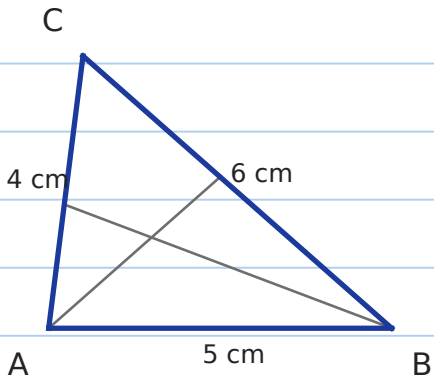
Question: Exercise 13.2, Question 3: Draw a triangle, draw the bisectors of its interior angles and prove that they meet at the same point.

Answer: Draw ΔABC, bisect the three interior angles. All three bisectors meet at the same point I (the incentre), which is equidistant from all three sides.



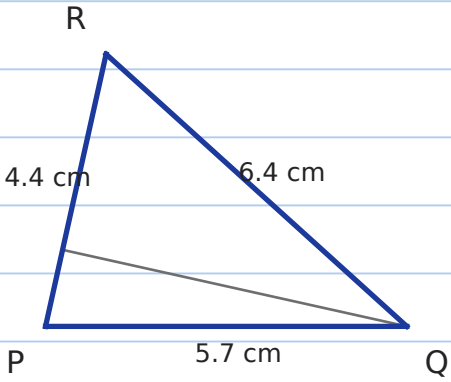
Question: Exercise 13.2, Question 4: Construct triangle ABC with mBC = 6 cm, mCA = 4 cm, mAB = 5 cm and draw the bisectors of A and B.

Answer: Draw BC = 6 cm, get A from arcs of 5 cm from B and 4 cm from C, join ABC. Bisect ∠A and ∠B; the two bisectors meet at the incentre I.



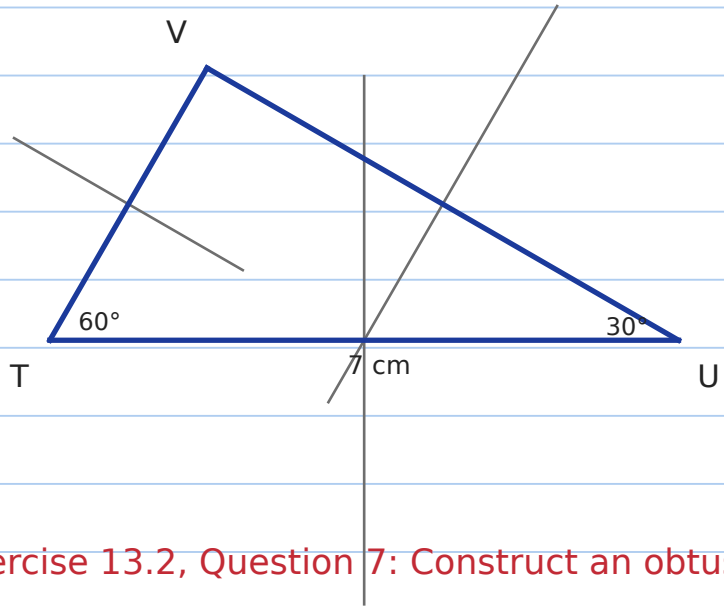
Question: Exercise 13.2, Question 5: Construct triangle PQR with mPQ = 5.7 cm, mQR = 6.4 cm, mPR = 4.4 cm; draw the altitude from vertex Q.

Answer: Draw PR = 4.4 cm, get Q from arcs of 5.7 cm from P and 6.4 cm from R, join PQR. Draw the perpendicular from Q to PR.



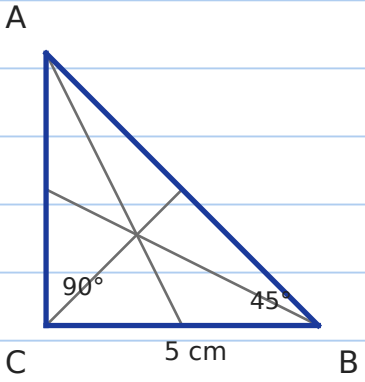
Question: Exercise 13.2, Question 6: Construct triangle TUV with mTU = 7 cm, ∠T = 60°, ∠U = 30°, draw the perpendicular bisectors of the triangle's sides and prove they meet at the same point.

Answer: Draw TU = 7 cm, construct 60° at T and 30° at U; the rays meet at V (∠V = 90°). Bisect the three sides and draw the perpendicular bisectors; all three meet at the same point O (the circumcentre), which is the midpoint of the hypotenuse TU.



Question: Exercise 13.2, Question 7: Construct an obtuse triangle with ∠B = 45°, ∠C = 90° and mCB = 5 cm and draw its medians. [note: ∠C = 90° as printed makes it a right triangle, not obtuse]

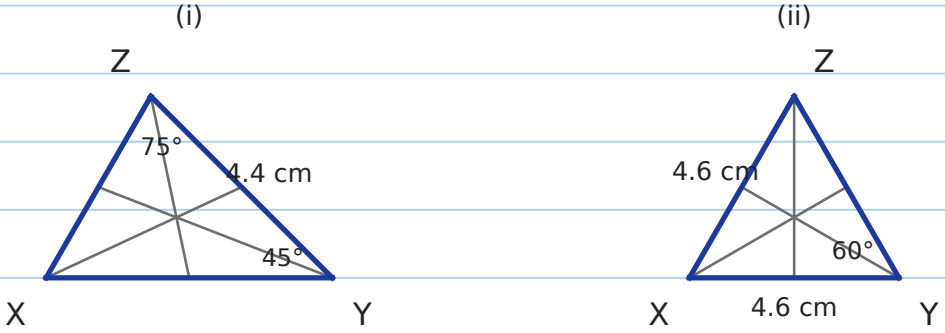
Answer: Draw CB = 5 cm, construct 45° at B and 90° at C; the rays meet at A (∠A = 45°). Bisect the three sides to get the midpoints and join the vertices; the three medians meet at the centroid.



Question: Exercise 13.2, Question 8: Construct ΔXYZ, draw its three medians. Prove that they are concurrent. (i) mYZ = 4.4 cm, m∠Y = 45° and m∠Z = 75° (ii) mXY = 4.6 cm, mXZ = 4.6 cm and m∠Y = 60°.

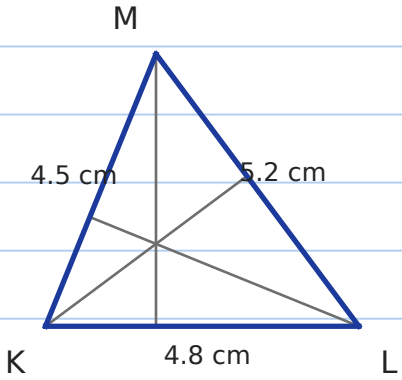
Answer: (i) Draw YZ = 4.4 cm, construct 45° at Y and 75° at Z; the rays meet at X. (ii) Since XY = XZ the base angles are equal, ∠Y = ∠Z = 60°, so ∠X = 60° — the triangle is equilateral. Draw XY = 4.6 cm, construct 60°

at Y and X; the rays meet at Z. In both cases bisect the sides to draw the medians; all three medians meet at one point.



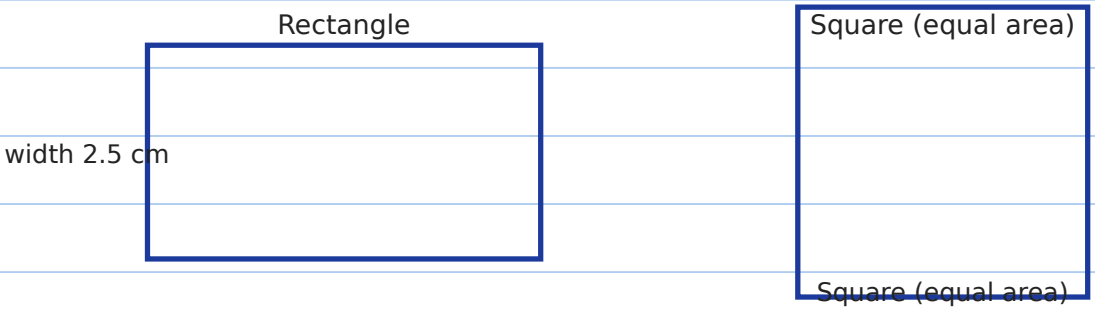
Question: Exercise 13.2, Question 9: Construct ΔKLM with mKL = 4.8 cm, mLM = 5.2 cm, mMK = 4.5 cm and draw its altitudes and verify that they are concurrent.

Answer: Draw KL = 4.8 cm, get M from arcs of 5.2 cm from L and 4.5 cm from K, join KLM. Draw the altitudes from the three vertices to the opposite sides; all three altitudes meet at one point (the orthocentre).



Question: Exercise 13.3, Question 1: Construct a rectangle whose width side is 2.5 cm. Construct a square whose area equals the area of the given rectangle. [note: print gives only the 2.5 cm side with 'respectively', suggesting a second side measure was dropped; construction method below uses the rectangle's two sides as given]

Answer: Draw the rectangle. Equal-area square: extend one side of the rectangle by the length of the other side, bisect the extended line and draw a semicircle on it, erect a perpendicular at the end of the first side meeting the semicircle. That perpendicular is the mean proportional of the two sides and is the side of the required square; construct the square on it. Area of square = area of rectangle.

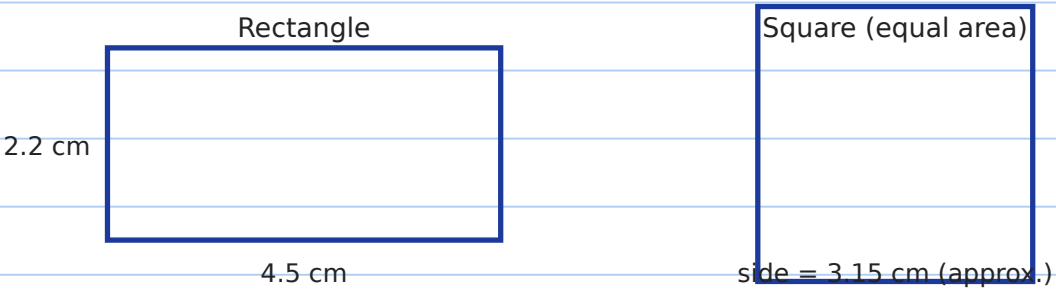


Question: Exercise 13.3, Question 2: Construct a rectangle whose adjacent sides are 4.5 cm and 2.2 cm respectively, measure the sides of the square and find its area, compare with the area of the rectangle and verify that the perimeter of the square is less than the perimeter of the rectangle.

Answer: Area of rectangle =

= 4.5 × 2.2

= 9.9 cm². Side of the equal-area square = √9.9 ≈ 3.15 cm, its area = 9.9 cm² = area of rectangle. Perimeter of square = 4 × 3.15 = 12.6 cm, perimeter of rectangle = 2(4.5 + 2.2) = 13.4 cm. 12.6 < 13.4, so the square's perimeter is less.

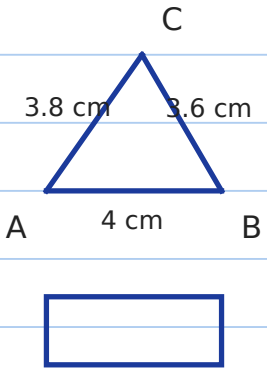


Question: Exercise 13.3, Question 3: Construct a triangle whose base is 4 cm and the other two sides are 3.6 cm and 3.8 cm. Convert it into a rectangle of equal area.

Answer: Construct the triangle (base 4 cm, arcs 3.6 and 3.8 cm). Draw the altitude from the vertex to the base (height h ≈ 3.11 cm, area ≈ 6.22 cm²). Bisect the altitude and draw a line through its midpoint parallel to the base; complete the rectangle with base 4 cm and height h/2 ≈ 1.55 cm.Area of rectangle =

= 4 × 1.55 ≈ 6.22 cm²

= area of triangle.



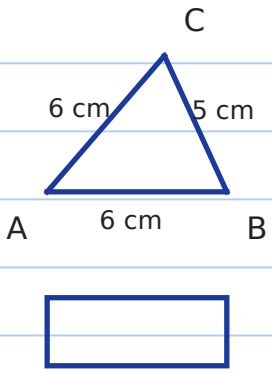
Rectangle of equal area (base 4 cm, height = half the altitude)

Question: Exercise 13.3, Question 4: Construct a triangle whose base is 6 cm and the other two sides are 5 cm and 6 cm. Construct a rectangle of area equal to the given triangle.

Answer: Construct the triangle (base 6 cm, arcs 5 and 6 cm). Draw the altitude from the vertex to the base (height h ≈ 4.55 cm, area ≈ 13.64 cm²). Bisect the altitude and draw a line through its midpoint parallel to the base; complete the rectangle with base 6 cm and height h/2 ≈ 2.27 cm.Area of rectangle =

= 6 × 2.27 ≈ 13.64 cm²

= area of triangle.



Rectangle of equal area (base 6 cm, height = half the altitude)

Review Exercise 13, Question 1: Fill in the blanks. (i) The side opposite the largest angle of a triangle is _____.

Answer: largest.

Review Exercise 13, Question 1: (ii) A line segment joining a vertex of a triangle and perpendicular to the opposite side is called _____.

Answer: Altitude.

Review Exercise 13, Question 1: (iii) A line segment joining a vertex of a triangle with the midpoint of the opposite side is called _____.

Answer: Median.

Review Exercise 13, Question 1: (iv) The perpendicular bisectors of the sides of a triangle are _____.

Answer: Concurrent.

Review Exercise 13, Question 1: (v) Two or more triangles are said to be _____ to each other if their angles are equal and the measures of their sides are _____.

Answer: similar).

Review Exercise 13, Question 2: Tick () the correct answer. (i) A triangle whose three sides are equal is _____:

(a) scalene

(b) obtuse



(c) equilateral

(d) isosceles

Review Exercise 13, Question 2: (ii) A quadrilateral each angle of which is 90° and all sides are equal. It is called _____:

(a) rhombus

(b) rectangle

(c) trapezium



(d) square

Review Exercise 13, Question 2: (iii) The medians of a triangle are _____:

(a) collinear

(b) equal

✓ (c) concurrent

(d) rhombus

Review Exercise 13, Question 2: (iv) The _____ altitudes of an equilateral triangle are equal to each other:

(a) two

✓ (b) three

(c) four

(d) none

Review Exercise 13, Question 2: (v) The diagonals of a rectangle are _____ to each other:

✓ (a) equal

(b) divided into three parts

(c) cut at an obtuse angle

(d) none

Review Exercise 13, Question 2: (vi) The _____ of a triangle cut each other in the ratio 2:1:

(a) altitudes

(b) angle bisectors

(c) perpendicular bisectors

✓ (d) medians

Review Exercise 13, Question 2: (vii) If each base angle of an isosceles triangle is 45, then the [vertex] angle measures _____:

(a) 30°

(b) 60°

✓ (c) 90°

(d) 120°

Review Exercise 13, Question 2: (viii) If all three medians of a triangle are equal, then the triangle is _____:

(a) obtuse

✓ (b) equilateral

(c) isosceles

(d) acute

Review Exercise 13, Question 2: (ix) If two _____ of a triangle are equal, then the triangle will be isosceles:

(a) altitudes

(b) medians

(c) perpendicular bisectors

✓ (d) sides

14. Theorems Related with Area

Question: Exercise 14.1, Question 1: In the given figure ABCD is a parallelogram and EFCD is a rectangle with AL ⊥ DC. Prove that (i) area ABCD = area EFCD (ii) area ABCD = mDC × mAL.

Answer: (i) Parallelogram ABCD and rectangle EFCD are on the same base DC and between the same parallels EB and DC, so their areas are equal. (ii) Area of rectangle = DC × FC, and FC = AL (perpendiculars between the same parallels), so area ABCD = DC × AL.

Question: Exercise 14.1, Question 2: In the given figure, if the diagonals of a quadrilateral divide it into four equal triangles, show that it is a rhombus.

Answer: Since the four triangles are equal (congruent), the four sides of the quadrilateral are corresponding sides of those triangles, so all four sides are equal, i.e. the quadrilateral is a rhombus.

Question: Exercise 14.1, Question 3: In the given figure AD ∥ BC, ABC is an obtuse triangle, from vertex B, with mBC = 7 cm. mAC = 11 cm; find the area of ΔABC. [flagged: as printed the height is not given; BC = 7 and diagonal AC = 11 alone do not determine the area]

Answer: [flagged: the question as printed cannot be solved — the height is not given, and mBC = 7 cm and diagonal mAC = 11 cm alone do not determine the area.]

Question: Exercise 14.1, Question 4: Prove that the medians of a triangle divide it into two triangles of equal area.

Answer: A median joins a vertex to the midpoint of the opposite side, so

the two triangles have equal bases and the same altitude, hence equal areas.

Question: Exercise 14.1, Question 5: Prove that the line joining the midpoints of the opposite sides of a rectangle divides it into two equal rectangles.

Answer: The joining line is parallel to the other two sides and bisects the length (or width) of the rectangle, so both parts are rectangles of equal dimensions, i.e. equal rectangles.

Question: Exercise 14.1, Question 6: If two equal-area parallelograms have the same base, then their altitudes are also equal.

Answer: Area of a parallelogram = base × altitude. Since the areas are equal and the base is the same, the altitudes are also equal.

Question: Exercise 14.1, Question 7: Prove that the bisector of an angle of an equilateral triangle divides it into two triangles of equal area.

Answer: In an equilateral triangle the angle bisector is also the median to the opposite side, so it bisects the opposite side; the two triangles have equal bases and the same altitude, so their areas are equal.

Question: Exercise 14.1, Question 8: Prove that a rhombus is divided by its diagonals into four triangles of equal area.

Answer: The diagonals of a rhombus bisect each other, so the four triangles have equal bases (half-diagonals) and equal altitudes (the other half-diagonal), so all four triangles have equal areas.

Question: Review Exercise 14, Question 1: Tick T for true and F for false

against the following. (i) The area of a closed figure means the part enclosed by its shape.

Answer: T.

Question: Review Exercise 14, Question 1: (ii) The diagonal of a rectangle divides it into two congruent triangles.

Answer: T.

Question: Review Exercise 14, Question 1: (iii) Identical shapes have different areas.

Answer: F (identical shapes have equal areas).

Question: Review Exercise 14, Question 1: (iv) The area of a parallelogram equals base × altitude.

Answer: T.

Question: Review Exercise 14, Question 1: (v) The median of a triangle means the perpendicular from its vertex to the opposite side.

Answer: F (that is the altitude; the median joins the vertex to the midpoint).

Question: Review Exercise 14, Question 1: (vi) The perpendicular distance of two parallelograms can sometimes be different.

Answer: T.

Question: Review Exercise 14, Question 1: (vii) The perpendicular drawn from the vertex point always bisects the opposite side.

Answer: F (only in an isosceles or equilateral triangle).

Question: Review Exercise 14, Question 1: (viii) Two triangles are equal in area if they have the same base and equal altitudes.

Answer: T.

Review Exercise 14, Question 2: Tick () the correct answer. (i) If the perpendicular distance between two lines is the same, then those lines are _____:

(a) perpendicular to each other

✓ (b) parallel to each other

(c) intersecting each other

(d) none

Review Exercise 14, Question 2: (ii) If the areas of two triangles are equal, then _____ will also be congruent:

✓ (a) not necessary

(b) it is necessary that

(c) absolutely

(d) none

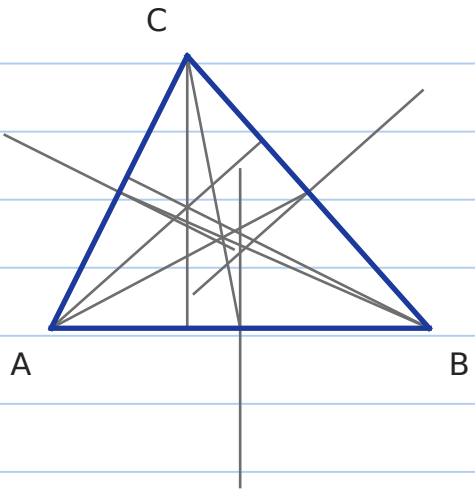
Review Exercise 14, Question 2: (iii) The perpendicular drawn from the vertex of a triangle to the opposite side is called _____:

(a) median

(b) perpendicular bisector

✓ (c) altitude

(d) angle bisector



Review Exercise 14, Question 2: (iv) Parallelograms with the same base and same altitude are _____:

(a) congruent

✓ (b) equal in area

(c) alike

(d) all of these

Review Exercise 14, Question 2: (v) If parallelograms have equal bases, then they have the same area if:

✓ (a) their altitudes are equal

(b) their altitude is the same

(c) they lie between the same parallels

(d) all of these

Review Exercise 14, Question 2: (vi) If ΔABC and ΔDEF have equal bases and equal altitudes, then these triangles are _____:

✓ (a) equal in area

(b) congruent

(c) alike

(d) none of these

15. Projection of a side of a Triangle

Question: Exercise 15.1, Question 1: Find the length AB in triangle ABC when (i) mAC = 3 cm, mBC = 6 cm and m∠C = 120° (ii) mCD = mBC cos(180° − m∠C).

Answer: Let D be the foot of the perpendicular from B to AC extended.

From (ii), CD = mBC cos(180° − 120°) = 6 × cos 60° = 6 × 0.5 = 3 cm.

Obtuse-angle theorem: $(mAB)^2 = (mAC)^2 + (mBC)^2 + 2(mAC)(mCD)$

$= 9 + 36 + 2 \times 3 \times 3$

$= 63$, so $mAB = \sqrt{63} \approx 7.94$ cm.

Question: Exercise 15.2, Question 1: In triangle ΔABC with m∠A = 30°, prove that $(mBC)^2 = (mAC)^2 + (mAB)^2 - \sqrt{3}(mAB)(mAC)$.

Answer:

Acute-angle theorem: $(mBC)^2 = (mAB)^2 + (mAC)^2 - 2(mAB)(AD)$, where AD is the projection of AC on AB

AD =

$= (mAC) \cos 30°$

= (mAC)(√3/2).

Substituting in the theorem: (mBC)² = (mAB)² + (mAC)² − 2(mAB)(mAC)(√3/2)

= (mAC)² + (mAB)² − √3(mAB)(mAC).

Question: Exercise 15.2, Question 2: Solve triangle ΔABC when mBC = __ cm, mAC = 5 cm and m∠A = 60°. [flagged: the mBC number is blank in print, so the question cannot be solved]

Answer: [flagged: the mBC number is not given in print (blank space between the equals sign and cm), so the triangle is not determined and no solution is possible.]

Question: Exercise 15.2, Question 3: Given a triangle with sides 3 cm, 4 cm, 5 cm, determine whether the triangle will be acute, right, or obtuse.

Answer:

Pythagoras rule: $a^2 + b^2 = c^2$ (c = longest side) — the triangle is right-angled

$3^2 + 4^2 = 9 + 16 = 25 = 5^2$, so the triangle is right-angled.

Question: Exercise 15.2, Question 4: Given a triangle with sides 4 cm, 7 cm, 8 cm, determine whether the triangle will be acute, right, or obtuse.

Answer:

Rule: $a^2 + b^2 > c^2$ — acute-angled; $a^2 + b^2 < c^2$ — obtuse-angled (c = longest side)

The largest side is 8 cm. $4^2 + 7^2 = 16 + 49 = 65 > 64 = 8^2$, so the triangle is acute-angled.

Review Exercise 15, Question 1: Fill in the blanks. (i) In obtuse triangle ΔABC, $(mBC)^2 = (mAC)^2 + (mAB)^2 +$ _____.

Answer: projection of AC on AB extended.

Review Exercise 15, Question 1: (ii) In triangle ΔABC, every two sides are equal to 4 cm; it is called a _____ triangle.

Answer: Equilateral.

Review Exercise 15, Question 1: (iii) In triangle ΔABC with $m\angle B = 90°$, $(mAB)^2 + (mBC)^2 =$ _____.

Answer: $(mAC)^2$ (Pythagoras).

Review Exercise 15, Question 1: (iv) 8 cm, 15 cm and 17 cm are the sides of _____.

Answer: A right triangle ($8^2 + 15^2 = 64 + 225 = 289 = 17^2$).

Question: Review Exercise 15, Question 2: In triangle ΔABC with $mAC = 3$ cm, $mBC = 6$ cm and $\angle C = 120°$, find mAB.

Answer:

Obtuse-angle theorem: $(mAB)^2 = (mAC)^2 + (mBC)^2 + 2(mAC)(mCD)$, where D is the foot of the perpendicular from B on AC produced

CD =

= mBC cos(180° − 120°)

= 6 × 0.5

= 3 cm. (mAB)² = (mAC)² + (mBC)² + 2(mAC)(mCD) = 9 + 36 + 18 = 63, so mAB = √63 ≈ 7.94 cm.

16. Introduction to Coordinate Geometry/ Analytical Geometry

Question: Exercise 16.1, Question 1: Using the distance formula, find the distance between the following pairs of points. (i) (6,6) and (−4,5)

Answer:

Distance formula: $d = \sqrt{((x_2 - x_1)^2 + (y_2 - y_1)^2)}$

$d =$

$= \sqrt{((-4-6)^2 + (5-6)^2)}$

$= \sqrt{(100+1)}$

$= \sqrt{101}$ units.

Question: Exercise 16.1, Question 1: (ii) (2,3) and (2,2)

Answer:

Distance formula: $d = \sqrt{((x_2 - x_1)^2 + (y_2 - y_1)^2)}$

$d =$

$= \sqrt{((2-2)^2 + (2-3)^2)}$

= √1

= 1 unit.

Question: Exercise 16.1, Question 1: (iii) (2,3) and (0,1)

Answer:

Distance formula: d = √((x₂ − x₁)² + (y₂ − y₁)²)

d =

= √((0−2)² + (1−3)²)

= √(4+4)

= √8

= 2√2 units.

Question: Exercise 16.1, Question 1: (iv) (2,3) and (0,1) [note: (iv) repeats (iii) in print]

Answer:

Distance formula: d = √((x₂ − x₁)² + (y₂ − y₁)²)

d =

$= \sqrt{((0-2)^2 + (1-3)^2)}$

$= \sqrt{8}$

$= 2\sqrt{2}$ units.

Question: Exercise 16.1, Question 2: A(a,0) and B(0,b) are points; find the distance between A and B when (i) a = −3, b = −4

Answer:

Distance formula: $d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$

AB =

$= \sqrt{(a^2 + b^2)}$

$= \sqrt{(9+16)}$

$= \sqrt{25}$

= 5 units.

Question: Exercise 16.1, Question 2: (ii) a = −9, b = 6

Answer:

Distance formula: $d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$

AB =

$= \sqrt{(a^2 + b^2)}$

$= \sqrt{(81+36)}$

$= \sqrt{117}$

$= 3\sqrt{13}$ units.

Question: Exercise 16.1, Question 2: (iii) a = 3, b = 4

Answer:

Distance formula: $d = \sqrt{((x_2 - x_1)^2 + (y_2 - y_1)^2)}$

AB =

$= \sqrt{(a^2 + b^2)}$

$= \sqrt{(9+16)}$

$= \sqrt{25}$

= 5 units.

Question: Exercise 16.1, Question 2: (iv) $a = \sqrt{2}$, $b = -2\sqrt{2}$

Answer:

Distance formula: $d = \sqrt{((x_2 - x_1)^2 + (y_2 - y_1)^2)}$

AB =

= √(a² + b²)

= √(2+8)

= √10 units.

Question: Exercise 16.1, Question 3: Find the perimeter of the triangle with vertices A(0,0), B(4,0), C(2,2√3).

Answer:

Distance formula: d = √((x₂ − x₁)² + (y₂ − y₁)²)

AB = 4, BC = √((2−4)² + (2√3)²) = √(4+12) = 4, CA = √((2)² + (2√3)²) = √(4+12) = 4.Perimeter =

= 4 + 4 + 4

= 12 units (the triangle is equilateral).

Question: Exercise 16.2, Question 1: Show that P(−3,−4), Q(2,6), R(0,2) are collinear.

Answer:

Distance formula: d = √((x₂ − x₁)² + (y₂ − y₁)²)

PQ = √(25+100) = √125 = 5√5, QR = √(4+16) = √20 = 2√5, PR = √(9+36) = √45 = 3√5. PR + QR = 3√5 + 2√5 = 5√5 = PQ, so the points are collinear (R lies between P and Q).

Question: Exercise 16.2, Question 2: Show that A(−1,0), B(1,0), C(0,√3) are non-collinear.

Answer:

Distance formula: $d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$

AB = 2, BC = √(1+3) = 2, AC = √(1+3) = 2. No sum of two distances equals the third (2 + 2 ≠ 2), so the points are non-collinear (they form an equilateral triangle).

Question: Exercise 16.2, Question 3: Show that L(0,√3), M(−1,0), N(1,0) form an equilateral triangle.

Answer:

Distance formula: $d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$

LM = √(1+3) = 2, MN = 2, NL = √(1+3) = 2. All three sides are equal, so the triangle is equilateral.

Question: Exercise 16.2, Question 4: Show whether A(2,3), B(8,11), C(0,17) form an isosceles triangle or not.

Answer:

Distance formula: $d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$

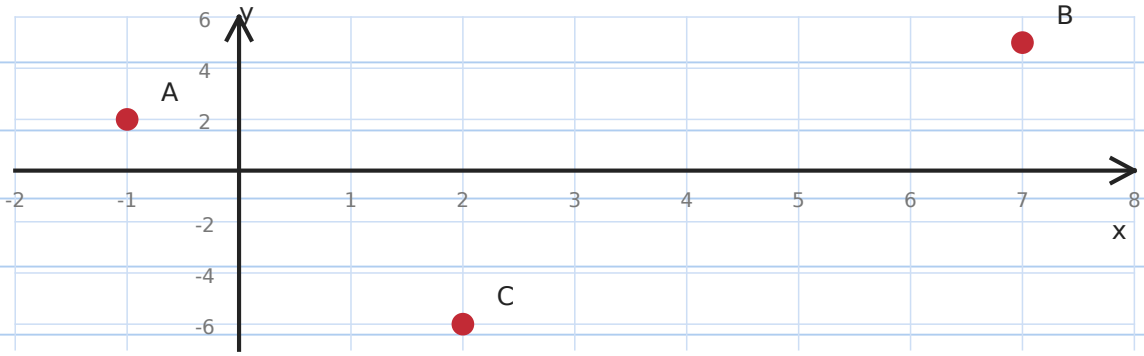
AB = √(36+64) = 10, BC = √(64+36) = 10, AC = √(4+196) = √200 = 10√2. AB = BC = 10, so the triangle is isosceles (and since $AB^2 + BC^2 = AC^2$, it is right isosceles).

Question: Exercise 16.2, Question 5: Do the points A(−1,2), B(7,5), C(2,−6) form an obtuse triangle?

Answer:

Distance formula: $d = \sqrt{((x_2 - x_1)^2 + (y_2 - y_1)^2)}$

No. AB = √(64+9) = √73, AC = √(9+64) = √73, BC = √(25+121) = √146. $AB^2 + AC^2 = 73 + 73 = 146 = BC^2$, so the triangle is right-angled (isosceles), not obtuse.



Question: Exercise 16.2, Question 6: If A(3,1), B(9,1), C(6,k), find the value of k for which they form an equilateral triangle.

Answer:

Distance formula: $d = \sqrt{((x_2 - x_1)^2 + (y_2 - y_1)^2)}$

AB = 6, AC = √(9 + (k−1)²). For equilateral, AC = 6: 9 + (k−1)² = 36 ⇒ (k−1)² = 27 ⇒ k = 1 ± 3√3.

Question: Exercise 16.2, Question 7: Show that P(1,2), Q(3,4), R(0,−1) are vertices of a scalene triangle.

Answer:

Distance formula: d = √((x₂ − x₁)² + (y₂ − y₁)²)

PQ = √(4+4) = √8 = 2√2, QR = √(9+25) = √34, RP = √(1+9) = √10. All three sides are different, so the triangle is scalene.

Question: Exercise 16.2, Question 8: Show that A(2,3), B(8,11), C(0,17), D(−6,9) are vertices of a rhombus.

Answer:

Distance formula: d = √((x₂ − x₁)² + (y₂ − y₁)²)

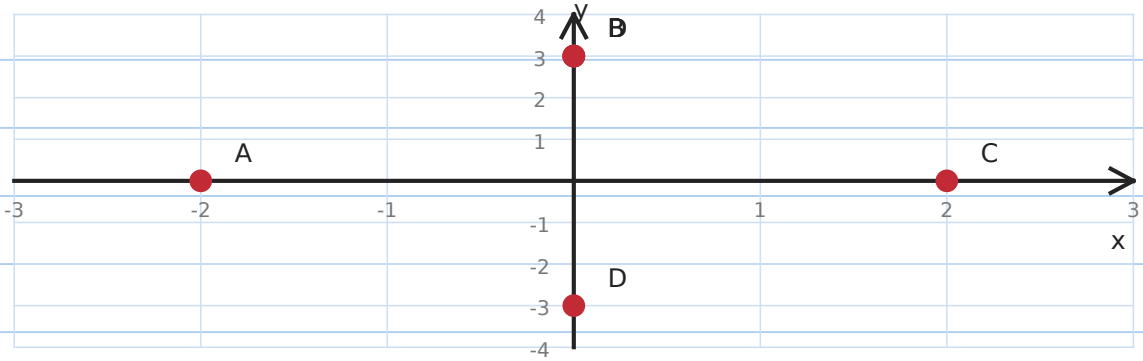
AB = 10, BC = 10, CD = √(36+64) = 10, DA = √(64+36) = 10. All four sides are equal, so the quadrilateral is a rhombus (the diagonals are also equal, AC = BD = 10√2, so it is in fact a square).

Question: Exercise 16.2, Question 9: Show that A(−2,0), B(0,3), C(2,0), D(0,3) form a square. [flagged: print shows D(0,3) duplicating B(0,3); read as D(0,−3); the points then form a rhombus, not a square — diagonals 4 ≠ 6]

Answer:

Distance formula: d = √((x₂ − x₁)² + (y₂ − y₁)²)

[flagged: in print D(0,3) duplicates B(0,3), so no quadrilateral is formed. Reading D as (0,−3), all four sides equal √13, so a rhombus is formed, but since the diagonals AC = 4 and BD = 6 are not equal, it is not a square.]



Question: Exercise 16.2, Question 10: Show that A(3,2), B(4,1), C(5,4), D(6,3) are vertices of a rectangle.

Answer:

Distance formula: $d = \sqrt{((x_2 - x_1)^2 + (y_2 - y_1)^2)}$

AB = √2, BC = √10, CD = √2, DA = √10. Opposite sides are equal (a parallelogram), and the diagonals AC = √8 = 2√2 and BD = √8 = 2√2 are equal, so the quadrilateral is a rectangle.

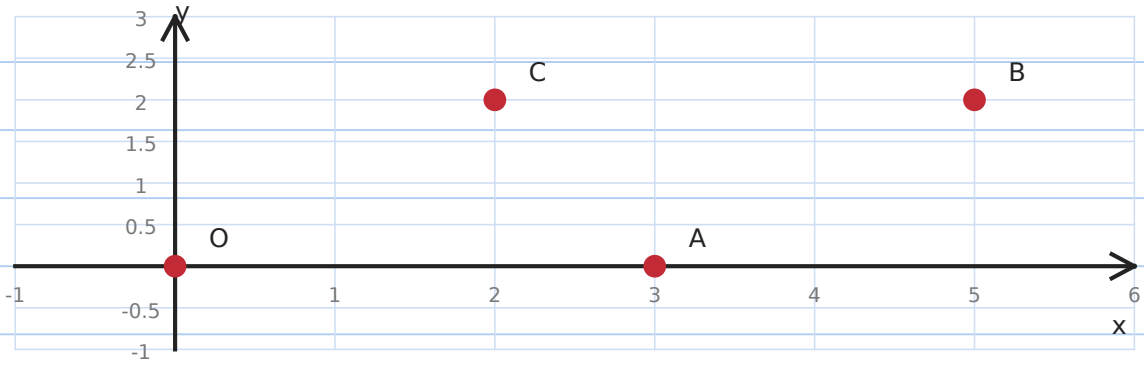
Question: Exercise 16.2, Question 11: Using the distance formula, show that O(0,0), A(3,0), B(5,2), C(2,2) are not vertices of a parallelogram.

[flagged: as printed the points DO form a parallelogram — opposite sides equal and diagonals bisect each other — so the book's 'are not' claim is incorrect]

Answer:

Distance formula: $d = \sqrt{((x_2 - x_1)^2 + (y_2 - y_1)^2)}$

[flagged: the printed claim is incorrect. OA = 3, AB = √8 = 2√2, BC = 3, CO = √8 = 2√2, i.e. opposite sides are equal. Both diagonals have midpoint (2.5,1), so the diagonals bisect each other, i.e. OABC is a parallelogram, not the opposite.]

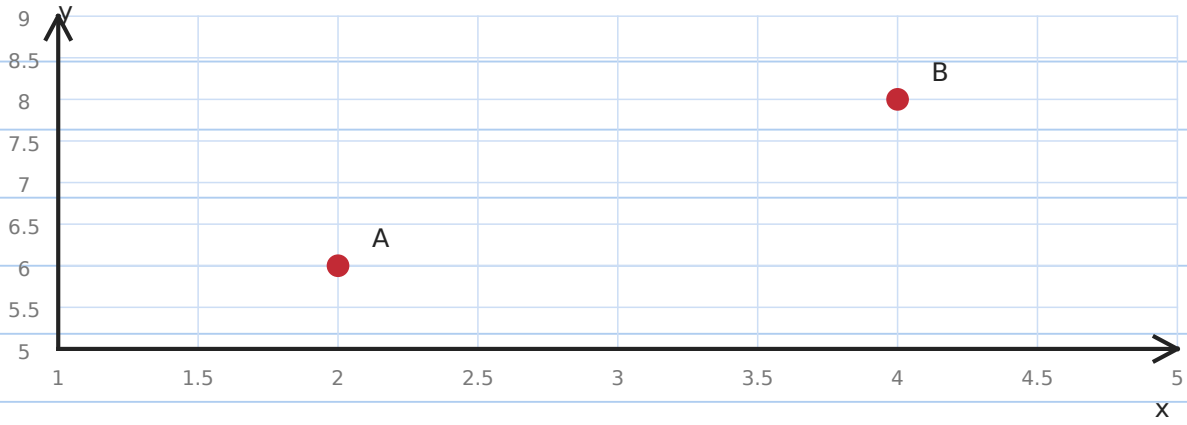


Question: Exercise 16.3, Question 1: Find the midpoint of the following pairs of points. (i) B(4,8) and A(2,6)

Answer:

Midpoint formula: $M = ((x_1 + x_2) / 2, (y_1 + y_2) / 2)$

Midpoint = ((4+2)/2, (8+6)/2) = (3,7).



Question: Exercise 16.3, Question 1: (ii) P(−3,−1) and Q(5,2)

Answer:

Midpoint formula: M = $((x_1 + x_2) / 2, (y_1 + y_2) / 2)$

Midpoint = ((−3+5)/2, (−1+2)/2) = (1, 1/2).

Question: Exercise 16.3, Question 1: (iii) L(0,6) and M(−8,0)

Answer:

Midpoint formula: M = $((x_1 + x_2) / 2, (y_1 + y_2) / 2)$

Midpoint = ((0−8)/2, (6+0)/2) = (−4,3).

Question: Exercise 16.3, Question 1: (iv) C(0,0) and D(2√3,4√3)

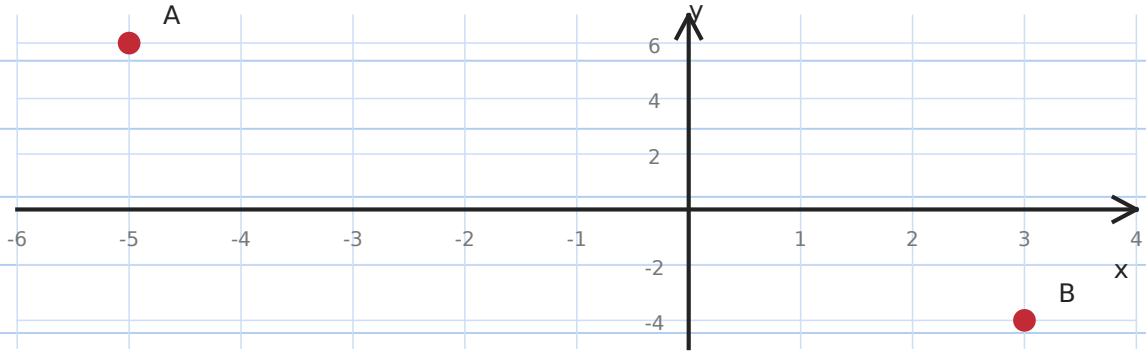
Answer:

Midpoint formula: M = $((x_1 + x_2) / 2, (y_1 + y_2) / 2)$

Midpoint = ((0+2√3)/2, (0+4√3)/2) = (√3, 2√3).

Question: Exercise 16.3, Question 2: Find the centre of a circle when the endpoints of the diameter are A(−5,6) and B(3,−4).

Answer: Centre = midpoint = ((−5+3)/2, (6−4)/2) = (−1,1).

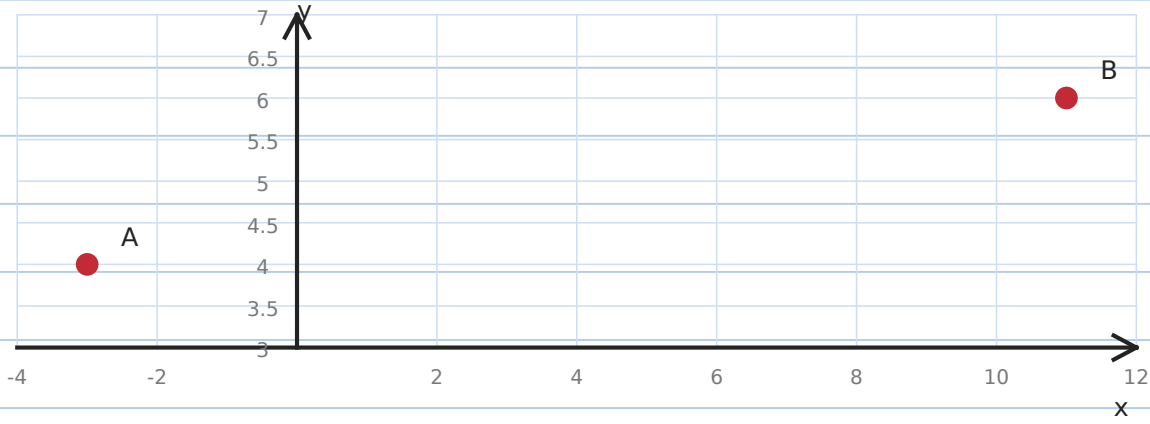


Question: Exercise 16.3, Question 3: The centre of a circle is (3,4) and one endpoint of the diameter is (4,6); find the other endpoint.

Answer: Let the other endpoint be (x,y). (4+x)/2 = 3 ⇒ x = 2, (6+y)/2 = 4 ⇒ y = 2. The other endpoint is (2,2).

Question: Exercise 16.3, Question 4: The endpoints of the diameter of a circle are A(−3,4) and B(11,6). Find the centre and radius of the circle.

Answer: Centre = ((−3+11)/2, (4+6)/2) = (4,5). Diameter = √((11+3)² + (6−4)²) = √200 = 10√2, so radius = 5√2 units.



Question: Exercise 16.3, Question 5: Prove that a triangle is isosceles if it has two equal medians.

Answer: Let the equal medians be AD and BE (D midpoint of BC, E midpoint of AC). They meet at the centroid G, AG = (2/3)AD = (2/3)BE = BG, so ΔAGB is isosceles and ∠BAG = ∠ABG. In ΔABD and ΔBAE: AB is common, AD = BE, ∠BAD = ∠ABE, so by SAS they are congruent, BD = AE ⇒ (1/2)BC = (1/2)AC ⇒ BC = AC, i.e. the triangle is isosceles.

Question: Exercise 16.3, Question 6: Prove that the diagonals of a parallelogram always bisect each other.

Answer:

Midpoint formula: $M = ((x_1 + x_2) / 2, (y_1 + y_2) / 2)$

Place parallelogram OABC with O(0,0), A(a,0), C(b,d), B(a+b,d). Midpoint of diagonal OB = ((a+b)/2, d/2), midpoint of diagonal AC = ((a+b)/2, d/2). Both midpoints are the same, so the diagonals bisect each other.

Question: Review Exercise 16, Question 1: Read the following sentences carefully, circle "T" for a true statement and "F" for a false one. (i) Point R lies on PQ if R exists between P and Q.

Answer: T.

Question: Review Exercise 16, Question 1: (ii) All sides of a scalene triangle are equal.

Answer: F (no two sides of a scalene triangle are equal).

Question: Review Exercise 16, Question 1: (iii) Perpendicular lines meet at

an angle of 135°.

Answer: F (perpendicular lines meet at 90°).

Question: Review Exercise 16, Question 1: (iv) Collinear points cannot form a triangle.

Answer: T.

Question: Review Exercise 16, Question 1: (v) Non-collinear points do not form a triangle.

Answer: F (non-collinear points do form a triangle).

Question: Review Exercise 16, Question 1: (vi) In an isosceles triangle, two sides and angles are equal.

Answer: T.

Question: Review Exercise 16, Question 1: (vii) All points lying on the y-axis are collinear.

Answer: T.

Question: Review Exercise 16, Question 1: (viii) The x-axis and y-axis intersect at (0,0).

Answer: T.

Question: Review Exercise 16, Question 1: (ix) The distance of (6,0) from the origin is 36 units.

Answer: F (the distance is 6 units).

Review Exercise 16, Question 2: Fill in the blanks. (i) If A(x1,y1) and

B(x2,y2) are two points, then |AB| = _____.

Answer: √((x2−x1)² + (y2−y1)²).

Review Exercise 16, Question 2: (ii) Collinear points lie on the same _____.

Answer: Line.

Review Exercise 16, Question 2: (iii) In the fourth quadrant, x > 0 and y _____.

Answer: (y < 0).

Review Exercise 16, Question 3: Tick () the correct answer. (i) Two perpendicular lines meet at an angle of _____:

(a) 45°

(b) 60°

✓ (c) 90°

(d) 180°

Review Exercise 16, Question 3: (ii) ((x1+x2)/2, (y1+y2)/2) is called:

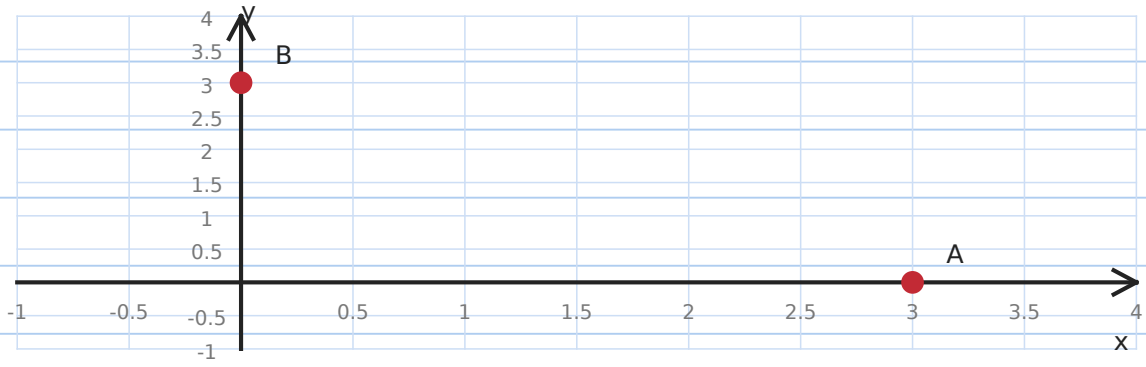
✓ (a) the midpoint formula

(b) the distance formula

(c) the section formula

(d) the ratio formula

Review Exercise 16, Question 3: (iii) On the plane, A(3,0) and B(0,3) are two points, then |AB| = _____: (a) 6 units (b) 6√2 (c) 3√2 (d) 3√2



Review Exercise 16, Question 3: (iv) Three points are collinear when:

(c) Both

✓ (d) mAB = mBC + mAC